

Mathematical continuation audit

Actual cohomology over the tau base

Complete continuation proofs

13 September 2026

The complete proof texts in this volume retain their original objects, cochain signs, arithmetic actions, measures, nilpotent jets and topologies. Equation numbers are local to each chapter; original tags are preserved. The chapter source manifest records the exact source bytes and every layout-only change made during assembly.

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Chapter 1

Typed source maps and the full arithmetic lift

Verified typed maps in the original τ -base construction

This is an independent source-map verification for the transcript continuation audit. It reads the entire primary note and the entire canonical source route. It does not claim to audit the whole shared transcript or the analytic arithmetic estimates. It does not duplicate the homotopy-fibre/right-adjoint continuation assigned elsewhere.

Source pins and exact scope

1. output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md, 25,512 bytes, SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. All 611 lines read. Equation locators below refer to this body.
2. work/fl_geometry_current_source_route_20260913.md, SHA-256 a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74. Entire body read. Its dependencies were not thereby deemed independently reread.
3. output/split_zero_rh_tandem_2026-09-12/tex/support_diagrams.tex, SHA-256 7586ac17247456a55bdacfcf8b46c196c29eafd3b57846a71792e918704e069b. Entire body read to check the kernel types and support-index transport. Its labels D1–D8 are used below.

The primary note explicitly works in the algebraic category

$$\mathcal{A} = \text{Fun}(P, \text{Vect}_E), \quad E = \mathbb{C}, \quad + < \eta < \sigma, \quad - < \eta < \sigma.$$

The topology is the upper-set topology; diagrams are covariant. The calculations below keep that category, its arrows, and its cohomological degrees. The source does not replace the Schwartz topology by a theorem about its continuous dual: it calculates an algebraic right adjoint, and says so.

1. The structural map does not perform Boolean localization

Primary equations (2)–(3), lines 33–45, use the blueprint map

$$j_R : \mathbf{F}_{1,\tau} = \{\tau, 1\} \longrightarrow B_R, \quad j_R(\tau) = \tau, \quad j_R(1) = 1_R^\bullet.$$

The only additional additive relation imposed by the source blueprint is that its empty sum represents τ . The target retains that relation, its own addition relations, and its original multiplicative monoid, so

the displayed map preserves all source relations. Every proper prime contains τ and excludes the unit. Its inverse image is consequently exactly $\{\tau\}$. This proves the structural map to the one-point base.

Primary equations (6)–(7), lines 64–80, concern a separate map $\chi_R : G(R) \rightarrow \mathbb{B}$, with $\chi_R(\tau) = 0$, $\chi_R(r^\bullet) = 1$, and localization at $e = 0_R^\bullet$. Inverting the idempotent e gives $e=1$; then $er^\bullet = e$ gives $r^\bullet = 1$. No cancellation with τ occurs. The source already proves this localization, and does not use it as the definition of the structural base pushforward.

The canonical route supplies the exact comparison for every $G(R)$ -semimodule M :

$$\Phi : \mathbb{B} \otimes_{G(R)} M \longrightarrow eM, \quad b \otimes m \longmapsto b \cdot em.$$

Here eM is a Boolean semimodule under its original addition. For a supported scalar a , $ea = e$, so both sides of the balancing relation map to $b \cdot em$; for $a=\tau$ both map to the global zero. Additivity in m follows from distributivity of e . Additivity in b uses $em + em = em$. Its inverse is

$$\Psi : eM \longrightarrow \mathbb{B} \otimes_{G(R)} M, \quad l \longmapsto 1 \otimes l.$$

Indeed $\Phi\Psi(l) = el = l$, and $\Psi\Phi(1 \otimes m) = 1 \otimes em = \chi_R(e) \otimes m = 1 \otimes m$; tensors with Boolean coefficient 0 vanish. Both formulas commute with every split-linear map. The comparison is therefore a natural isomorphism, not a claimed lack of relation between the two constructions.

2. The global pushforward and its canonical derived restriction

Primary equations (17)–(20), lines 200–242, are correct with their displayed domains and signs.

For x in P , $P_x(y) = E$ when $x \leq y$, and 0 otherwise, with identity maps on nonzero arrows. A natural transformation $P_x \rightarrow F$ is determined by the image v of 1 at x ; at y above x its value must be $r_{xy}v$. Conversely those values are natural. Thus

$$\mathrm{Hom}_{\mathcal{A}}(P_x, F) = F_x.$$

Evaluation is exact because kernels and cokernels in the diagram category are computed at each point. This proves projectivity of P_x .

The source resolution is

$$0 \longrightarrow P_\eta \xrightarrow{\delta} P_+ \oplus P_- \xrightarrow{\varepsilon} \mathbf{E} \longrightarrow 0, \quad \delta(v) = (v, -v), \quad \varepsilon(u, v) = u + v.$$

At $+$ its sequence is $0 \rightarrow 0 \rightarrow E \xrightarrow{1} E \rightarrow 0$, and at $-$ it is the same sequence in the second summand. At η and σ its sequence is $0 \rightarrow E \rightarrow E^2 \rightarrow E \rightarrow 0$, with kernel of addition exactly $\{(v, -v) : v \in E\}$. Restriction maps commute with both displayed maps because each nonzero component is an identity. This checks exactness and naturality on every point and arrow.

Compatible global sections of F are determined by $(u, v) \in F_+ \oplus F_-$ with $r_+u = r_-v$. Their η -value is this common vector and their σ -value is its image under $r_{\eta\sigma}$. Hence $\Gamma(P, F) = \mathrm{Hom}(\mathbf{E}, F)$, and the resolution computes

$$\mathbf{A}_\tau(F) = R\Gamma(P, F) \simeq C_F, \quad C_F^0 = F_+ \oplus F_-, \quad C_F^1 = F_\eta, \quad d_F(u, v) = r_+u - r_-v.$$

The convention is the displayed resolution coboundary by precomposition. The degrees and the differential are those of the source, not an identification with F_σ .

Put $r = r_{\eta\sigma}$. The two maps $c_+, c_- : C_F \rightarrow F_\sigma[0]$ have degree-zero parts

$$c_+^0(u, v) = rr_+u, \quad c_-^0(u, v) = rr_-v,$$

and degree-one parts 0. They are chain maps because the target has no term in degree 1. With degree -1 homotopy h defined by $h^1 = r : F_\eta \rightarrow F_\sigma$,

$$(c_+ - c_-)^0 = h^1 d_F, \quad (c_+ - c_-)^1 = 0 = d_{F_\sigma} h^1.$$

They therefore give a single canonical derived morphism $\mathrm{res}_\sigma : \mathbf{A}_\tau(F) \rightarrow F_\sigma[0]$. Its action on a global compatible section is its actual σ -value. This proves the comparison, including its sign, without substituting the target for the source.

3. The actual arithmetic cohomology and its three support fibres

Primary equations (10)–(15), (21)–(25), retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \hat{\phi}(0) = 0\}, & \hat{\phi}(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \\ \mathcal{B} &= \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0 \right\}, \\ \Theta\phi(x) &= \sum_{n \neq 0} \phi(nx), & JF(x) &= x^{-1}F(1/x), & J\Theta &= \Theta\mathcal{F}. \end{aligned}$$

Its joint diagram is $(V, V, \mathcal{B}, 0)$, with differential

$$d(\phi, \psi) = \Theta\phi - J\Theta\psi = \Theta(\phi - \hat{\psi}).$$

Consequently its image is exactly ΘV : one inclusion follows from Fourier invariance of V , and the reverse inclusion follows by using $(\phi, 0)$. Fourier invariance follows from evenness and $\mathcal{F}^2\phi(x) = \phi(-x)$, which exchanges the two vanishing moments and gives $\mathcal{F}^2 = 1$ on V . Using the source's established injectivity of Θ gives

$$\ker d = \{(\hat{\psi}, \psi) : \psi \in V\}, \quad H^1 C_{\mathcal{T}} = Q = \mathcal{B}/\Theta V.$$

This verifier does not replace the source's Mellin/Poisson proof of injectivity by an independent analytic claim. The algebraic identification just made uses that named source result at its exact place.

For the $+$ singleton mask the differential is Θ . For the $-$ singleton mask it is $-J\Theta = -\Theta\mathcal{F}$. Each is injective, and each has image ΘV , so its H^0 is 0 and its H^1 is Q . The inclusions from either singleton to the joint mask are coordinate inclusions in degree 0 and identity in degree 1; they are chain maps since the joint differential restricts to the indicated signed singleton differential. Their induced H^1 maps are the identity of Q . The empty support diagram is zero.

Thus the entire H^1 support diagram has zero bottom fibre and Q at all three nonempty masks, with identity transitions among nonempty comparable masks. Its reconstruction is exactly

$$M = \widetilde{H}_\tau^1 = \{\tau\} \sqcup \coprod_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q.$$

The linear map induced by res_σ on H^1 is $Q \rightarrow H^1(0[0]) = 0$ at each nonempty mask. Reconstructing that same-mask diagram map gives

$$\epsilon_M : M \longrightarrow eM, \quad \tau \longmapsto \tau, \quad (A, q) \longmapsto (A, 0).$$

This is the precise typed reading of the source's statement immediately after (25), at lines 283–288. It is not a degree-one map to the unshifted linear stalk treated as if that stalk had nonzero H^1 . The supported skeleton is obtained by the explicit support reconstruction. Its fibres are

$$\epsilon_M^{-1}(\tau) = \{\tau\}, \quad \epsilon_M^{-1}(A, 0) = \{A\} \times Q \quad (A \neq \emptyset).$$

Every arithmetic amplitude remains in its recorded fibre. The projection and its section $(A, 0) \mapsto (A, 0)$ prove the relation to the boundary skeleton exactly.

4. The right adjoint is computed algebraically on that same chart

Primary equations (26)–(30), lines 294–347, give the correct right-adjoint object.

Define $I_x(W)(y) = W$ if $y \leq x$ and 0 otherwise. A map $F \rightarrow I_x(W)$ is uniquely determined by $\ell : F_x \rightarrow W$: its y -component for $y \leq x$ is ℓr_{yx} , and its other components are 0. These components are natural on every arrow. Therefore

$$\text{Hom}_{\mathcal{A}}(F, I_x(W)) = \text{Hom}_E(F_x, W).$$

Every vector space is injective: a linear map defined on a subspace extends after extending a basis of that subspace to a basis of the whole space. Evaluation at x is exact, so $I_x(W)$ is injective in $\mathcal{A}$.

The source complex is

$$K_\tau(W)^{-1} = I_\eta(W), \quad K_\tau(W)^0 = I_+(W) \oplus I_-(W), \quad d_K = (+\text{res}, -\text{res}).$$

A bounded complex of injectives calculates the derived Hom from a sheaf here. Evaluation of its Hom complex gives

$$R\text{Hom}_{\mathcal{A}}(F, K_\tau(W)) = \left[\text{Hom}_E(F_\eta, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \text{Hom}_E(F_+, W) \oplus \text{Hom}_E(F_-, W) \right],$$

in degrees -1 and 0 . The ordinary Hom complex from C_F to $W[0]$ has exactly these terms. For a degree -1 map ℓ its differential is $-(-1)^{-1}\ell d_F = \ell d_F$, giving exactly the displayed pair. This proves the chain isomorphism in (28), naturally in F and W , and hence the stated adjunction.

For bounded coefficient complexes the same result follows by taking the finite total complexes, preserving the original differential of W . An explicit convention is

$$K_\tau(W)^n = I_\eta(W^{n+1}) \oplus I_+(W^n) \oplus I_-(W^n), \quad d(s, u, v) = (-d_W s, \text{res}_+ s + d_W u, -\text{res}_- s + d_W v).$$

Its square is zero: the first component is $d_W^2 s$, and each other component cancels its two restriction-of- $d_W s$ terms, leaving $d_W^2 u$ or $d_W^2 v$. This is the signed totalization of the displayed two-term adjoint.

At $+$, $-$, η , σ , respectively, the W -vector-space stalk complexes are

$$[W \xrightarrow{1} W], \quad [W \xrightarrow{-1} W], \quad W[1], \quad 0.$$

The inclusion $S_\eta W[1] \rightarrow K_\tau(W)$, identity at η and zero at the other points, is natural: at each arrow entering η its source is zero; at arrows leaving η its target is zero; all other squares have a zero entry on the source side. It induces an isomorphism in cohomology at every stalk, since the $\pm$ complexes have invertible differentials. Hence

$$K_\tau(W) \simeq S_\eta W[1]$$

by an explicit quasi-isomorphism. In particular $K_\tau(W)_\sigma = 0$ does not imply that the right adjoint is zero, nor that its Hom from the arithmetic sheaf is zero. Its degree -1 Hom cohomology is

$$\{\ell : \mathcal{B} \rightarrow W : \ell \Theta V = 0\} \cong \text{Hom}_E(Q, W).$$

The inverse correspondence sends a quotient functional $\lambda : Q \rightarrow W$ to $\lambda \circ q$; the forward correspondence sends ℓ to $[F] \mapsto \ell(F)$. Equality of values on representatives is exactly $\ell \Theta V = 0$.

This verifies the target of the full finite-jet/residue morphism (43), whose source has been retained as $\overline{A_Z}$ and whose codomain is $H^{-1}R\text{Hom}_{\mathcal{A}}(\mathcal{T}, K_\tau(\mathcal{L}_1))$. The source's surjective $J_Z : Q \rightarrow A_Z$ and perfect finite residue pairing are the specified reasons its precomposition map is injective. Neither a continuous-dual identification nor involutive Verdier duality is supplied by this algebraic adjunction.

5. Exact categorical and all-support kernels

Use D1–D8 with $M = \mathcal{T}(L, V)$, $N = \mathcal{T}(L', W)$, and a split-linear map

$$F = \mathcal{T}(\alpha, f), \quad F(l, x) = (\alpha(l), f_l x).$$

The target's global zero is $(\perp', 0)$, while its support skeleton consists of all $(k, 0)$. Consequently

$$F(l, x) = 0_N \iff \alpha(l) = \perp' \text{ and } f_l x = 0,$$

$$F(l, x) \in eN \iff f_l x = 0.$$

Writing $L_0 = \alpha^{-1}(\perp')$, these equalities give

$$\ker_{G(R)} F = \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}),$$

$$F^{-1}(eN) = \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN.$$

The original transition maps restrict to these kernels: naturality gives $f_k \rho_{lk} x = \rho'_{\alpha(l), \alpha(k)} f_l x = 0$. Also L_0 is closed under joins because α preserves joins. Therefore both displayed carriers are subsemimodules. Any map with the corresponding vanishing property has its image in the indicated carrier and hence factors uniquely through the inclusion. This proves the two universal properties, not just set identities. Their exact comparison is the inclusion induced by $L_0 \hookrightarrow L$.

For ϵ_M above, α is the identity and every fibre map is the zero map $Q \rightarrow 0$. Its bottom fibre is zero. Thus

$$\ker_{G(E)} \epsilon_M = \{\tau\}, \quad M \times_{eM} eM = M.$$

A trivial categorical kernel here cannot establish disappearance of supported arithmetic cohomology: the second displayed object is the entire original arithmetic carrier. The formula states the scope of the trivial-kernel result exactly.

The quotient q at a fixed nonempty mask has source $\mathcal{B}$, boundary subspace ΘV , and target Q . Its supported form sends (A, F) to $(A, [F])$. In the all-support sense its kernel fibre is ΘV , while every boundary maps to $(A, 0)$, never τ . This follows from $[F] = 0 \iff F \in \Theta V$. More generally, a cochain map has

$$\ker H(f) = \{z \in \ker d_C : fz \in \operatorname{im} d_D\} / \operatorname{im} d_C.$$

The map sends the class of z to its cohomology class. It is well-defined because boundaries map to boundaries, onto because every cohomology class has a cycle representative, and injective because the zero class is precisely a boundary. This is D7, and reconstructing it at every label gives the all-support object in D8.

6. Dual support arrows: the exact reversal and its limit

Primary equation (31), lines 349–356, retains the correct map for every inclusion $A \subseteq B$:

$$\rho_{AB}^\vee : \operatorname{Hom}_E(H_B, W) \longrightarrow \operatorname{Hom}_E(H_A, W), \quad \ell \mapsto \ell \rho_{AB}.$$

For $A \subseteq B \subseteq C$, $\rho_{AB}^\vee(\rho_{BC}^\vee \ell) = \ell \rho_{BC} \rho_{AB} = \ell \rho_{AC}$, proving functoriality on L^{op} . The identity arrow acts as the identity. Thus the exact dual-index diagram exists.

It is not a covariant diagram on the original inclusion-ordered L with the same arrow directions. The source does not assert that it is. A further reconstruction into split semimodules must declare its index semilattice. For the finite Boolean lattice in this source, L^{op} itself has join given by intersection and bottom $\{+, -\}$. Applying D1–D5 to this opposite lattice is valid and keeps the dual arrows, but its bottom fibre is then $\operatorname{Hom}_E(H_{\{+, -\}}, W)$, not the original empty-mask fibre. In degree 1 this is $\operatorname{Hom}_E(Q, W)$. D1–D5 explicitly allow such nonzero bottom fibres. Thus an argument that applies a theorem requiring a singleton bottom fibre to this reconstructed dual has changed a hypothesis and must calculate the actual bottom fibre. The original dual map (43), formed label by label with precomposition, does not require that unsupported identification.

7. A completed exact jet map that makes the eigenline wording precise

Primary (40), (44)–(45), and (49) retain full nilpotent jets. The phrase “residue-value eigenvector” in (49) is read through the following exact maps, so it does not discard the nilpotents of (40).

At an actual zero ρ of multiplicity m , keep the original coordinate $z = s - \rho$ and factor

$$g(\rho + z) = u(z)z^m, \quad u(0) \neq 0.$$

Then $A_\rho = \mathcal{O}_\rho/(g) = \mathcal{O}_\rho/(z^m)$, with the identity on germs giving the latter equality of ideals because u is a unit. Its reduction map is

$$q_\rho : A_\rho \longrightarrow E, \quad [h] \longmapsto h(\rho).$$

The endomorphism J_g of (44) factors as an actual quotient followed by an injection:

$$A_\rho \xrightarrow{q_\rho} E \xrightarrow{t_\rho} A_\rho, \quad t_\rho(c) = c[g'], \quad J_g = t_\rho q_\rho.$$

Indeed

$$g' = u'(z)z^m + mu(z)z^{m-1} \equiv mu(0)z^{m-1} \pmod{z^m}.$$

Writing $h = h(0) + z\tilde{h}$, multiplication by $[g']$ kills the second summand, so $[g'h] = h(0)[g']$, proving factorization. Since $mu(0) \neq 0$, $[g'] \neq 0$; therefore t_ρ is injective. Its image is exactly

$$\text{Ann}_{A_\rho}(z) = Ez^{m-1}.$$

To verify this equality, multiply $\sum_{j=0}^{m-1} c_j z^j$ by z . The product is zero modulo z^m exactly when $c_0 = \dots = c_{m-2} = 0$. This includes $m=1$, when the annihilator is all of $A_\rho = E$. Consequently

$$\ker J_g = \ker q_\rho = zA_\rho, \quad \text{im } J_g = \text{Ann}_{A_\rho}(z).$$

The actual scaling operator remains

$$U_a = a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N^j, \quad N[z^j] = [z^{j+1}], \quad a > 0.$$

Reduction gives $q_\rho U_a = a^\rho q_\rho$, since every $j > 0$ term has zero constant coefficient. Multiplication of $t_\rho(c)$, which lies in Ez^{m-1} , by a $j > 0$ term is zero, so $U_a t_\rho(c) = t_\rho(a^\rho c)$. Both maps are equivariant. Thus the residue-value representation and its socle image inside the full jet are linked by the explicit nonzero map t_ρ . When $m > 1$, the constant lift 1 of a residue value is not generally an eigenvector of U_a for $a \neq 1$, because its z coefficient is $a^\rho \log a \neq 0$. The socle image $t_\rho(1)$ is an actual eigenvector. No full jet was replaced by its reduction in proving that fact.

The tensor of r copies of this explicit eigenvector is nonzero in $A_\rho^{\otimes_{E^r}}$: the coefficient of $(z^{m-1})^{\otimes r}$ is $(mu(0))^r \neq 0$. Its action on that finite tensor quotient is exactly $a^{r\rho}$. Its precise relation to primary (47) is the equivariant surjection

$$Q^{\otimes_{E^r}} \xrightarrow{(\text{pr}_\rho J_Z)^{\otimes r}} A_\rho^{\otimes_{E^r}}.$$

Here Z contains ρ , $J_Z : Q \rightarrow A_Z$ is the source's full jet quotient, and pr_ρ is its block projection. Each map is equivariant by the full jet scaling formula (40). The first is surjective by the source's jet-surjectivity result, the block projection is surjective, and their tensor is surjective because each simple target tensor has a tensor of preimages and such tensors span the target. This proves the exact relation to $Q^{\otimes r}$.

The quotient-and-socle argument of this subsection, taken alone, does not establish an equivariant lift to Q . The original arithmetic section supplies that additional map: Subsection 1 below retains its full Taylor unit and proves its scaling equivariance, giving the actual eigenline inside Q and its tensor inside $Q^{\otimes r}$. Thus this is a completed source lift, with the quotient-only argument and the additional original section both specified. On the finite quotient the tensor modulus expression is

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2 \text{Re } \rho - 1) \log a \quad (a > 1),$$

with its original action and coefficient line. This is an identity on the actual eigenline embedded in $A_\rho^{\otimes r}$, which is a quotient of the tensor cohomology in (47); it proves no upper bound on that identity. At every nonempty supported mask, q_ρ , t_ρ , and J_g lift by retaining the mask, so a killed nilpotent maps to the supported zero of its mask.

Audit uses

The source already contains the structural base, global pushforward, derived restriction, right adjoint, full supported Q , dual-index arrows, and the finite-jet morphism into that right adjoint. A passage that calls any of these missing must be tested against the equation locators above. The algebraic calculation does not supply analytic bounds; that limitation has the exact scope stated in the primary note. A boundary-stalk vanishing or a trivial categorical kernel does not reach the original supported Q , because the explicit maps above retain the whole Q fibre. A duality argument must retain the dual arrow direction, its coefficient line, and its actual algebraic category. A tensor argument can use the exact socle embedding into the finite tensor quotient above while retaining every nilpotent in that ambient jet object; it must retain the displayed quotient map from the full tensor cohomology. Subsection 1 constructs the source eigenvector using the original arithmetic section and its proved finite scaling cocycle, rather than inferring it from the quotient alone.

8. The original arithmetic section completes the source eigenline lift

The section used here is already constructed in the retained arithmetic source, equations (A7)–(A11) of output/split_zero_rh_tandem_2026-09-12/tex/arithmetic_input.tex, SHA-256 a50b0fb587b644c4ea94dba45b2b423d038f2bfbe7fef188543939ba850e58d2. The group-level source integral is proved in Sections 2–3, equations (9)–(19), of output/tau_f1_transcript_audit_2026-09-13/proofs/theta_scaling_cocycle.md, SHA-256 5aae80114cbf3cb0c3bec25e4e655b37117a932c1a92132161c162393b7ea377. We retain their maps and give the source-lift proof explicitly. The earlier quotient calculation did not prove this additional map, but the map is present in the original construction and is completed here at the full scaling-group level.

The original packet and its unit. Let Z be a nonempty finite set of actual nontrivial zeros of the original $g = 2\xi$, with each complete multiplicity m_ρ , and set

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[X]/(h), \quad A = M_X.$$

The letter X here is the original polynomial variable denoted t in (A10)–(A11); the notation E_h distinguishes that source's finite packet E from the ground field $E = \mathbb{C}$ used earlier in this verification. The complex Mellin coordinate remains s . At each selected zero, the quotient

$$v(s) = g(s)/h(s), \quad \varepsilon = j_h(h/g) \in E_h$$

is holomorphic and nonzero. Indeed, in the original coordinate $z = s - \rho$, both g and h have order exactly m_ρ , so their quotient has a nonzero constant coefficient. A truncated power series with nonzero constant coefficient is invertible by recursively solving for its inverse coefficients. The Chinese-remainder decomposition therefore makes ε a unit, with $j_h(v)\varepsilon = 1$. No coefficient of this unit is replaced by 1. For $u \in E_h$, let $R_Z(u)$ be the unique polynomial of degree less than d representing εu . Polynomial division by the monic h proves this existence and uniqueness.

The retained source is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP.$$

Its moments and this Mellin identity are the proved original (A3), with the same Fourier convention and both factors of two retained. Write

$$D = -x\partial_x, \quad p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)|, \quad b \in \mathbb{Z}, \quad k \geq 0.$$

These are the defining seminorms of the original $\mathcal{B}$. The operator inverse is the actual inverse

$$S_h^{\mathcal{B}} : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B} \quad \text{of} \quad h(D) : \mathcal{B} \longrightarrow \ker(j_h \mathcal{M}).$$

Equations (A5)–(A7), or equation (9) in the finite-cocycle chapter, prove its range, injectivity, and continuity. The explicit one-factor inverse used in that proof is

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad \mathcal{M}F(\rho) = 0.$$

It satisfies $(D - \rho)S_\rho F = F$. The zero moment writes its upper integral as the negative lower integral; these two expressions give, respectively, all decay bounds at infinity and zero. More explicitly, for $N > \operatorname{Re} \rho$ and $N + \operatorname{Re} \rho > 0$,

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \operatorname{Re} \rho} x^{-N}, \quad |S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \operatorname{Re} \rho} x^N.$$

Using the first bound for $x \geq 1$, the second for $x \leq 1$, and $DS_\rho F = \rho S_\rho F + F$ proves every defining derivative seminorm bound. The homogeneous solution $cx^{-\rho}$ belongs to $\mathcal{B}$ only when $c = 0$. Integration by parts gives $\mathcal{M}S_\rho F(s) = \mathcal{M}F(s)/(s - \rho)$, with the removable value filled in. Iteration removes one of the prescribed zero factors at each step and preserves every other vanishing order. This proves the stated inverse on the complete jet kernel and injectivity of $h(D)$, without assuming an unproved operator range.

Define the original section exactly as in (A10):

$$R_{\text{ref}}u = S_h^{\mathcal{B}}\Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad q : \mathcal{B} \longrightarrow Q = \mathcal{B}/\Theta V, \quad \sigma = qR_{\text{ref}}.$$

The input to $S_h^{\mathcal{B}}$ has Mellin transform $gR_Z(u)$, which has all required zero jets, so this definition has the stated domain for every u . The inverse formula gives

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}}u = j_h(v)\varepsilon u = u.$$

For $\phi \in V$, the original Mellin identity $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ holomorphic at the selected zeros, gives $J_Z\Theta\phi = 0$. Thus J_Z factors as $\bar{J}_Z q$, and

$$\bar{J}_Z \sigma = 1_{E_h}, \quad h(D)\sigma(u) = q\Theta(R_Z(u)(D)\phi_*) = 0.$$

The displayed left inverse proves that $\sigma : E_h \hookrightarrow Q[h(D)]$ is injective. This is an actual section inside the arithmetic cohomology, not a section inferred from finite-dimensional quotient data alone.

All positive dilations act equivariantly after the original boundary. Set $\ell(u) = [X^{d-1}]R_Z(u)$. Since multiplication by X commutes with the unit ε , the polynomial $XR_Z(u) - R_Z(Au)$ is divisible by h . Its degree is at most d ; its leading coefficient is $\ell(u)$. Hence the exact polynomial identity is

$$XR_Z(u) - R_Z(Au) = h(X)\ell(u).$$

Applying the defining inverse, or multiplying by the injective $h(D)$, gives the original boundary coefficient (A11):

$$DR_{\text{ref}} - R_{\text{ref}}A = \Theta\phi_*\ell.$$

For a single simple zero this coefficient is $\ell(u) = u/g'(\rho) = u/(2\xi'(\rho))$; its arithmetic unit and factor of two remain present.

For $a > 0$ let $\lambda = \log a$, $T_r = e^{rA}$, and retain $U_a F(x) = F(x/a)$. Define

$$c_a(u) = \int_0^\lambda U_{e^{\lambda-r}} \phi_* \ell(T_r u) dr \in V.$$

When $\lambda < 0$ this is the oriented integral $\int_0^\lambda = -\int_\lambda^0$. No orientation changes are hidden. Fix any norm on the finite-dimensional E_h , only for this convergence estimate, with the induced operator norms for A and ℓ . For the original Schwartz seminorms $q_{N,k}(\phi) = \sup_x (1 + |x|)^N |\partial_x^k \phi(x)|$,

$$q_{N,k}(U_b \phi) \leq b^{-k} \max(1, b)^N q_{N,k}(\phi).$$

On the interval between 0 and λ , this inequality and $\|T_r\| \leq e^{r\|A\|}$ give

$$q_{N,k}(c_a(u)) \leq |\lambda| e^{(N+k+\|A\|)|\lambda|} q_{N,k}(\phi_*) \|\ell\| \|u\|.$$

The integrand is continuous in every seminorm. Its Riemann sums converge in the complete closed Schwartz subspace V : uniform continuity on the compact parameter interval makes them Cauchy in each seminorm, and their limits retain evenness and both zero moments. Differentiating dilation is legitimate in every seminorm, as the difference quotient's integral remainder is bounded by finitely many seminorms of $D^2\phi$ on that compact interval. The same argument applies in $\mathcal{B}$, using $p_{b,k}(U_a F) = a^b p_{b,k}(F)$. Consequently

$$\frac{d}{dr}(U_{e^{\lambda-r}} R_{\text{ref}} T_r u) = -U_{e^{\lambda-r}} (D R_{\text{ref}} - R_{\text{ref}} A) T_r u = -\Theta U_{e^{\lambda-r}} \phi_* \ell(T_r u).$$

Integrating in the actual $\mathcal{B}$ seminorms and using continuity of Θ proves, for every $a > 0$,

$$U_a R_{\text{ref}} - R_{\text{ref}} a^A = \Theta c_a, \quad a^A = e^{(\log a)A}.$$

This is precisely equation (15) of the finite-cocycle calculation. The original relation space is invariant because $U_a \Theta = \Theta U_a$, so U_a induces $\bar{U}_a$ on Q . Applying q to the identity gives the required full intertwining

$$\boxed{\bar{U}_a \sigma = \sigma a^A \quad \text{for every } a > 0.}$$

Inserting $U_a R_{\text{ref}} b^A$ into the defect for ab and using injectivity of Θ also gives the retained cocycle law $c_{ab} = U_a \circ c_b + c_a \circ b^A$. This proves group equivariance of the section in Q ; it does not assert strict equivariance of R_{ref} in $\mathcal{B}$. Indeed the displayed boundary is exactly what connects those two maps.

The exact full-jet insertion and the source eigenline. The identity on germs gives the Chinese-remainder isomorphism

$$E_h \xrightarrow{\sim} \bigoplus_{\nu \in Z} A_\nu, \quad A_\nu = \mathcal{O}_\nu / (g) = \mathcal{O}_\nu / ((s - \nu)^{m_\nu}).$$

At each centre the ideals generated by h and g both equal the displayed power ideal, since their remaining factors are units. For a fixed $\rho \in Z$, let $\iota_\rho : A_\rho \hookrightarrow E_h$ be the inverse Chinese-remainder map with all other blocks zero. An explicit formula, retaining all Taylor coefficients, uses $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$: for a germ representative f ,

$$\iota_\rho[f] = \left[H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{1}{j!} \frac{d^j}{dX^j} \left(\frac{f(X)}{H_\rho(X)} \right) \right]_{X=\rho} (X - \rho)^j \Big|_h.$$

At ρ its jet is f ; at every other selected centre the factor H_ρ gives all required zero jets. A change of representative f by a germ of order at least m_ρ leaves the displayed derivatives unchanged. These facts prove that this is the exact well-defined insertion, inverse to its block projection. In each block,

$$A|_\nu = \nu I + N_\nu, \quad a^A|_\nu = a^\nu \sum_{j=0}^{m_\nu-1} \frac{(\log a)^j}{j!} N_\nu^j.$$

Thus ι_ρ intertwines the full block scaling; it does not replace that block by its residue-value quotient. Retain the preceding subsection's actual factorization $g(\rho + z) = u_\rho(z) z^m$, $m = m_\rho$, and put

$$t_\rho : \mathbb{C} \longrightarrow A_\rho, \quad t_\rho(c) = c[g'] = c m u_\rho(0) z^{m-1}.$$

Its domain has action $c \mapsto a^\rho c$. The previous calculation proves that t_ρ is an injective intertwiner, since $m u_\rho(0) \neq 0$ and $N_\rho z^{m-1} = 0$. Therefore the complete original source lift is

$$\boxed{L_\rho = \sigma \iota_\rho t_\rho : (\mathbb{C}, a^\rho) \hookrightarrow (Q, \bar{U}_a), \quad w_\rho = L_\rho(1).}$$

Its equivariance follows by composing the three proved intertwiners. Its nonvanishing follows without any choice of a quotient representative: $\bar{J}_Z w_\rho = \iota_\rho[g'] \neq 0$. For an explicit left inverse, let pr_ρ denote the block projection and set

$$\Lambda_\rho(q_0) = \frac{[z^{m-1}] \text{pr}_\rho \bar{J}_Z(q_0)}{mu_\rho(0)}.$$

Then $\Lambda_\rho L_\rho = 1_{\mathbb{C}}$. This linear left inverse is used to certify injectivity; no equivariance of Λ_ρ on the whole Q is asserted. In particular, for $m > 1$ it is not residue evaluation, which would vanish on the socle. The construction keeps precisely the nilpotent coefficient required for a nonzero lift.

The full tensor lift in the original top cohomology. For every integer $r \geq 1$, tensor over $\mathbb{C}$ to obtain

$$L_\rho^{\otimes r} = \sigma^{\otimes r} \iota_\rho^{\otimes r} t_\rho^{\otimes r} : \mathbb{C}^{\otimes r} \hookrightarrow Q^{\otimes r}, \quad w_\rho^{\otimes r} = L_\rho^{\otimes r}(1^{\otimes r}).$$

The map $\Lambda_\rho^{\otimes r}$ is a left inverse, since the tensor of the identities $\Lambda_\rho L_\rho = 1$ is the identity. Thus this is a nonzero tensor in the source cohomology, not only in its finite quotient. Composing the exact scaling intertwiners gives

$$\bar{U}_a^{\otimes r} w_\rho^{\otimes r} = a^{r\rho} w_\rho^{\otimes r} \quad (a > 0).$$

The top-degree comparison in primary (47) identifies this tensor with an actual class of $H^r \mathbf{A}_{\tau,r}(\mathcal{T}^{\otimes r})$. Explicitly, put

$$b_\rho = R_{\text{ref}} \iota_\rho[g'] \in \mathcal{B}.$$

Its representative in the original tensor complex is $b_\rho^{\otimes r} \in \mathcal{B}^{\otimes r}$. Top boundaries are sums of tensors with a ΘV factor. The map $J_Z^{\otimes r}$ kills every such tensor, whereas it sends this representative to $(\iota_\rho[g'])^{\otimes r} \neq 0$. This also proves its nonzero top cohomology class directly in the original complex.

The group action on this representative has the original source boundary and tensor signs. Put $c = c_a(\iota_\rho[g'])$. Then $U_a b_\rho - a^\rho b_\rho = \Theta c$. In the joint two-leg complex, $d(c, 0) = \Theta c$. With each $\mathcal{B}$ factor in degree one, form the degree $r - 1$ tensor

$$H_{a,r} = \sum_{j=1}^r (-1)^{j-1} (U_a b_\rho)^{\otimes(j-1)} \otimes (c, 0) \otimes (a^\rho b_\rho)^{\otimes(r-j)}.$$

Only the degree-zero factor contributes to its differential, with sign $(-1)^{j-1}$ from its preceding degree-one factors. It cancels the displayed coefficient. The resulting sum telescopes, giving

$$dH_{a,r} = (U_a b_\rho)^{\otimes r} - (a^\rho b_\rho)^{\otimes r}.$$

This is the cochain-level proof of the stated tensor eigencharacter on the original complex, retaining its source boundary rather than postulating a strict invariant representative.

Finally, the source character now has the exact original modulus identity

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2 \operatorname{Re} \rho - 1) \log a \quad (a > 1)$$

on the nonzero class $w_\rho^{\otimes r}$ itself. The actual moment line still has character a ; no norm or mass has been changed. This identity supplies the source eigenline requested by the tensor construction. It does not supply an upper estimate for its modulus. At each original nonempty support mask, σ , ι_ρ , and t_ρ lift by keeping that mask and its labelled zero. Their cohomological lifts commute with the identity transports among the nonempty Q -fibres; external τ remains external absence. The joint tensor class above therefore belongs to the retained supported arithmetic cohomology over the same absolute base.

Chapter 2

Restriction fibre, adjoint extension and tensor signs

1. Objects, provenance, and cochain conventions

The complete successor *A global retraction of the original theta complex over the tau base*, retained as `Tau_Global_Comparison_2026-09-12/RESEARCH_NOTE.md`, was also read completely. Its SHA256 is `496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859`. Equations (6.12a)–(6.12i) below reproduce its constructive retraction and evaluate the previously general closure kernel for this exact source.

This calculation continues equations (18)–(48) of *Arithmetic cohomology over the tau-base*. The complete source `NOTE.md` has SHA256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`. Its complete retained dependency `sources/Split_Support_Actual_Comparison__CONTINUATION.tex` has SHA256 `98e8aaa63b1f7ee4661c043cd985ca274646b749f36d23d2ec46cbc4b0737dc5`. Both were read completely for this calculation. The category, original theta source, masks, analytic topology, multiplicities, and actions below are those sources' objects.

Let $E = \mathbb{C}$, let P have order $+$ $< \eta < \sigma$ and $-$ $< \eta < \sigma$, and let $\mathcal{A} = \text{Fun}(P, \text{Vect}_E)$. A diagram $F \in \mathcal{A}$ is specified by

$$U = F_+, \quad V_- = F_-, \quad M = F_\eta, \quad S = F_\sigma, \quad r_+ : U \rightarrow M, \quad r_- : V_- \rightarrow M, \quad t : M \rightarrow S. \quad (1.1)$$

Its remaining two arrows are the composites tr_+ and tr_- . The symbol V_- here denotes a general stalk; V in Section 5 denotes the original even Schwartz source.

Use cohomological complexes. For a chain map $f : C \rightarrow D$, define

$$\text{Cone}(f)^n = D^n \oplus C^{n+1}, \quad d(y, x) = (d_D y + f x, -d_C x), \quad d_{C[k]} = (-1)^k d_C. \quad (1.2)$$

After ordering its summands as $C^n \oplus D^{n-1}$, the homotopy fibre is

$$\text{Fib}(f) = \text{Cone}(f)[-1], \quad d(c, y) = (d_C c, -d_D y - f c). \quad (1.3)$$

For internal Hom use

$$d_{\text{Hom}} h = d_W h - (-1)^{|h|} h d_C. \quad (1.4)$$

These conventions, including the minus sign in the second component of (1.3), remain fixed throughout.

2. The original pushforward and the complete restriction fibre

Define $P_x(y) = E$ for $x \leq y$, and zero otherwise, with identity arrows between nonzero stalks. A natural map $P_x \rightarrow F$ is uniquely specified by the image of $1 \in P_x(x)$, so $\text{Hom}(P_x, F) = F_x$. Evaluation

is exact, hence P_x is projective. The sequence

$$0 \longrightarrow P_\eta \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \mathbf{E} \longrightarrow 0 \quad (2.1)$$

is exact: at $+$ and $-$ its last arrow is identity; at η, σ its kernel is the displayed anti-diagonal; all restriction squares commute. Consequently a representative of the source functor is

$$C(F) = A_\tau(F) = [U \oplus V_- \xrightarrow{\delta} M], \quad \delta(u, v) = r_+u - r_-v, \quad (2.2)$$

in degrees 0, 1. With (1.4), applying internal Hom to (2.1) initially gives differential $-\delta$; the explicit isomorphism to (2.2) is identity in degree zero and minus identity in degree one. This is a change between two stated cochain presentations, not a modification of the arithmetic differential in (2.2).

The two representatives of derived restriction are

$$f_+(u, v) = tr_+u, \quad f_-(u, v) = tr_-v, \quad C(F) \longrightarrow S[0]. \quad (2.3)$$

They are chain maps because the target has only degree zero. Their difference is $t\delta$. Thus $h^1 = t : M \rightarrow S$ satisfies $f_+ - f_- = d_S h + h d_C$. In particular both represent the same natural derived morphism.

Using f_+ , the complete fibre is

$$\boxed{B_\sigma(F) = [U \oplus V_- \xrightarrow{b} M \oplus S], \quad b(u, v) = (r_+u - r_-v, -tr_+u).} \quad (2.4)$$

The cone itself has degrees $-1, 0$ and differential

$$U \oplus V_- \longrightarrow S \oplus M, \quad (u, v) \mapsto (tr_+u, -r_+u + r_-v). \quad (2.5)$$

No stalk or quotient has been dropped in either presentation.

Put

$$I = \text{im } r_+ \cap \text{im } r_- \subset M. \quad (2.6)$$

Then the complete cohomology of (2.4) is

$$H^0 B_\sigma(F) = \{(u, v) : r_+u = r_-v \in \ker t\}, \quad (2.7)$$

$$H^1 B_\sigma(F) = \frac{M \oplus S}{\{(r_+u - r_-v, -tr_+u) : u \in U, v \in V_-\}}, \quad (2.8)$$

and all its other cohomology groups vanish. There are exact sequences

$$0 \longrightarrow \ker r_+ \oplus \ker r_- \longrightarrow H^0 B_\sigma(F) \xrightarrow{(u, v) \mapsto r_+u} I \cap \ker t \longrightarrow 0, \quad (2.9)$$

$$0 \longrightarrow S/t(I) \xrightarrow{[s] \mapsto [(0, s)]} H^1 B_\sigma(F) \xrightarrow{[(m, s)] \mapsto [m]} M/(\text{im } r_+ + \text{im } r_-) \longrightarrow 0. \quad (2.10)$$

For (2.9), every element of the displayed intersection has a preimage under each leg, and the kernel is exactly the indicated direct sum. For (2.10), the last map is well defined and surjective. A class in its kernel has $m = \delta(u, v)$; subtracting $b(u, v)$ replaces it by $(0, s + tr_+u)$. The class of $(0, s)$ vanishes exactly when there exist u, v with $r_+u = r_-v$ and $s = -tr_+u$, which is precisely $s \in t(I)$. This proves the sequence, including the sign in its presentation. These are exact sequences of specified objects, without a chosen splitting.

Equivalently the restriction long exact sequence is

$$0 \rightarrow H^0 B_\sigma(F) \rightarrow \ker \delta \xrightarrow{f_+} S \xrightarrow{s \mapsto [(0, -s)]} H^1 B_\sigma(F) \rightarrow \text{coker } \delta \rightarrow 0. \quad (2.11)$$

Its signs agree with (1.2): when the fibre-to-source arrow is the positive projection, the rotated distinguished triangle has boundary arrow minus the inclusion of S in $\text{Cone}(f) = B_\sigma(F)[1]$. This accounts for the minus in (2.11); the independently specified exact presentation (2.10) uses the

positive inclusion. The only nonzero possible cone cohomology is $H^{-1} \text{Cone}(f) = H^0 B_\sigma(F)$ and $H^0 \text{Cone}(f) = H^1 B_\sigma(F)$.

The two representative fibres are related by the chain isomorphism

$$B_{\sigma,+}(F) \longrightarrow B_{\sigma,-}(F), \quad \Phi^0 = 1, \quad \Phi^1(m, s) = (m, s + tm), \quad (2.12)$$

whose inverse subtracts tm . Indeed $-f_+ + t\delta = -f_-$. Thus the fibre calculation does not depend on a privileged choice of the plus leg.

3. Naturality and the right-adjoint comparison

A morphism $a : F \rightarrow F'$ consists of maps $a_+, a_-, a_\eta, a_\sigma$ commuting with r_+, r_-, t . The fibre map is exactly

$$B_\sigma(a)^0 = a_+ \oplus a_-, \quad B_\sigma(a)^1 = a_\eta \oplus a_\sigma. \quad (3.1)$$

For example its second differential component is $-a_\sigma t r_+ u = -t' r'_+ a_+ u$; the first follows from the two other commutative squares. Formula (3.1) also commutes with (2.12), because $a_\sigma t = t' a_\eta$. This proves the naturality of every kernel, quotient, exact sequence, and representative comparison above.

For a vector space W , let $I_x(W)(y) = W$ for $y \leq x$, and zero otherwise. A natural map $F \rightarrow I_x(W)$ is uniquely determined by its component $F_x \rightarrow W$; the other nonzero components are precomposition with $F_y \rightarrow F_x$. Hence

$$\text{Hom}_{\mathcal{A}}(F, I_x(W)) = \text{Hom}_E(F_x, W). \quad (3.2)$$

Every vector space is injective over E : a linear map on a subspace extends after extending a basis. Therefore the right side is exact contravariantly in F , and $I_x(W)$ is injective.

Let

$$K_\tau(W) = [I_\eta(W) \xrightarrow{(+\text{res}, -\text{res})} I_+(W) \oplus I_-(W)] \quad (3.3)$$

have degrees $-1, 0$. It is a bounded injective complex. Applying (3.2) gives

$$R \text{Hom}_{\mathcal{A}}(F, K_\tau(W)) = [\text{Hom}(M, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \text{Hom}(U, W) \oplus \text{Hom}(V_-, W)]. \quad (3.4)$$

The internal Hom of (2.2) into $W[0]$ has exactly this differential: the element ℓ has degree -1 , so (1.4) gives $d\ell = +\ell\delta$. In particular the signs in the original single-factor adjoint and residue arrow require no repair.

For completeness the adjunction holds for bounded complexes, with all totalization signs specified. Define

$$C(F)^n = (F_+^n \oplus F_-^n) \oplus F_\eta^{n-1}, \quad d(b, m) = (d_F b, \delta b - d_F m), \quad (3.5)$$

and $K_\tau(W) = K_\tau(E) \otimes W$, with tensor differential. A degree n map $F \rightarrow K_\tau(W)$ consists of $b_p : F_+^p \oplus F_-^p \rightarrow W^{p+n}$ and $a_p : F_\eta^p \rightarrow W^{p+n+1}$. The isomorphism to $\text{Hom}^n(C(F), W)$ keeps b_p and takes

$$\ell_p = (-1)^{n+1} a_p. \quad (3.6)$$

On the source differential these components become

$$a'_p = -d_W a_p - (-1)^n a_{p+1} d_F, \quad b'_p = d_W b_p + a_p \delta - (-1)^n b_{p+1} d_F. \quad (3.7)$$

On the target they become

$$\ell'_p = d_W \ell_p + (-1)^n \ell_{p+1} d_F, \quad b'_p = d_W b_p - (-1)^n (b_{p+1} d_F + \ell_p \delta). \quad (3.8)$$

Substitution of (3.6) gives equality in both components. Bounded injective targets compute derived Hom, and (2.1), with the degree n multiplier $(-1)^n$ on its eta component, gives (3.5) from its projective resolution. This proves the bounded derived adjunction as an actual natural chain isomorphism.

The stalk complexes of (3.3) are $[W \xrightarrow{1} W]$ at $+$, $[W \xrightarrow{-1} W]$ at $-$, $W[1]$ at η , and zero at σ . The stalkwise kernel diagram is $S_\eta W$, supported only at η . Its inclusion in degree -1 is a natural quasi-isomorphism

$$S_\eta W[1] \longrightarrow K_\tau(W). \quad (3.9)$$

Since every point of P lies below σ , $I_\sigma(W)$ is the constant diagram W . It represents the dual of the sigma stalk. The actual adjoint of restriction is

$$\alpha_+ : I_\sigma(W) \longrightarrow K_\tau(W), \quad (3.10)$$

given in degree zero by restriction into $I_+(W)$, and zero into $I_-(W)$. The alternative α_- restricts into $I_-(W)$. Their difference is $d_K h$, where $h : I_\sigma(W) \rightarrow I_\eta(W)$ is restriction in degree -1 . Applying Hom to (3.10) sends $w : S \rightarrow W$ to $(wtr_+, 0)$, which is exactly precomposition with f_+ . This identifies the morphism, rather than only comparing its endpoints.

4. The dual of the restriction fibre and its nonsplit extension

Define the bounded injective complex

$$R_\tau(W) = [I_\eta(W) \oplus I_\sigma(W) \xrightarrow{d_R} I_+(W) \oplus I_-(W)], \quad (4.1)$$

in degrees $-1, 0$, where

$$d_R(\ell, w) = (\ell \text{res}_+ - w \text{res}_{\sigma,+}, -\ell \text{res}_-). \quad (4.2)$$

Here restriction symbols denote the natural maps of injective diagrams. Formula (3.2) makes its Hom differential

$$(\ell, w) \mapsto (\ell r_+ - wtr_+, -\ell r_-). \quad (4.3)$$

This is precisely precomposition with the two components $(\delta, -f_+)$ of (2.4), with the positive sign supplied by degree -1 in (1.4). Therefore

$$\boxed{R \text{Hom}_{\mathcal{A}}(F, R_\tau(W)) \cong R \text{Hom}_E(B_\sigma(F), W).} \quad (4.4)$$

Equivalently $R_\tau(W) = \text{Cone}(-\alpha_+)$. Multiplying its $I_\sigma(W)$ cone summand by -1 identifies this presentation with $\text{Cone}(\alpha_+)$; (4.2) fixes which presentation is being used in (4.4).

At $+$, the differential is $(\ell, w) \mapsto \ell - w$, with kernel (w, w) . At $-$, it is $(\ell, w) \mapsto -\ell$, with kernel $(0, w)$. At η it has domain $W \oplus W$ and zero target, and at σ it has domain W and zero target. All degree-zero cokernels vanish. Consequently, if $D(W)$ has stalks

$$D(W)_+ = W, \quad D(W)_- = W, \quad D(W)_\eta = W \oplus W, \quad D(W)_\sigma = W, \quad (4.5)$$

and arrows

$$w \mapsto (w, w) \quad (+ \rightarrow \eta), \quad w \mapsto (0, w) \quad (- \rightarrow \eta), \quad (u, w) \mapsto w \quad (\eta \rightarrow \sigma), \quad (4.6)$$

the kernel inclusion is a natural quasi-isomorphism $D(W)[1] \rightarrow R_\tau(W)$.

There is an exact sequence of diagrams

$$\boxed{0 \longrightarrow S_\eta W \xrightarrow{w \mapsto (u, 0)} D(W) \xrightarrow{\text{second coordinate}} I_\sigma(W) \longrightarrow 0.} \quad (4.7)$$

At $+$, $-$, σ the last map is identity under (4.5); at η its kernel is the first coordinate. Naturality follows directly from (4.6). For $W \neq 0$, the extension has no natural splitting. Indeed a splitting is identity at the plus and minus stalks, since the quotient there is identity. Commutation along $+$ $\rightarrow \eta$ forces its η component to be $w \mapsto (w, w)$, whereas commutation along $- \rightarrow \eta$ forces $w \mapsto (0, w)$. They disagree on every nonzero $w \in W$. This proves nonsplitting explicitly.

In the original case $F_\sigma = 0$, every term $\text{Hom}(F, I_\sigma(W)) = \text{Hom}(0, W)$ is zero. Thus the inclusion $K_\tau(W) \rightarrow R_\tau(W)$ induces an actual equality of the represented Hom complexes, as well as an isomorphism in the derived category. The nonzero original global arithmetic classes are therefore all in the relative fibre, and their original adjoint is precisely the restriction of (4.4) to these objects.

5. Calculation on the original theta source and every mask

Retain the exact original spaces and maps

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad (5.1)$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad (5.2)$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}. \quad (5.3)$$

The last identity is the retained Poisson identity. Explicitly, the full Poisson equation is $\sum_n \phi(nx) = x^{-1} \sum_n \hat{\phi}(n/x)$; the two zero terms vanish by (5.1), giving (5.3). At infinity every differentiated summand is bounded by a constant times $(|n|x)^{-N}$ for any chosen $N > 1$; summing $2 \sum_{n \geq 1} n^{-N}$ proves rapid decay. Applying (5.3) proves arbitrary power decay at zero, with all derivatives, so $\Theta : V \rightarrow \mathcal{B}$. The Fourier transform preserves (5.1) because it exchanges the value and integral conditions, and its square is reflection, which is identity on even functions.

The map Θ is injective. For $\Re s = c > 1$, absolute convergence gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s), \quad \mathcal{M}F(s) = \int_0^\infty F(x)x^s \frac{dx}{x}. \quad (5.4)$$

For absolute convergence, the integral of $|\phi(nx)|x^{c-1}$ is $n^{-c} \int_0^\infty |\phi(u)|u^{c-1}du$, and both the integral and $\sum n^{-c}$ are finite. The Euler product is nonzero for $c > 1$, as follows from absolute convergence of its logarithm. If $\Theta\phi = 0$, (5.4) makes $\mathcal{M}\phi(c+it) = 0$ for every real t . This is the Fourier transform, with its frequency sign explicitly reversed, of the integrable function $y \mapsto e^{cy}\phi(e^y)$. Fourier uniqueness gives zero almost everywhere and continuity gives zero everywhere. Evenness then gives $\phi = 0$ on the full real line.

The joint sheaf is

$$\mathcal{T} = (V, V, \mathcal{B}, 0; \Theta, \Theta\mathcal{F}, 0), \quad C(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}]. \quad (5.5)$$

Its restriction map is zero on both chain representatives, so

$$\boxed{B_\sigma(\mathcal{T}) = C(\mathcal{T}), \quad \text{Cone}(\text{res}_\sigma) = C(\mathcal{T})[1].} \quad (5.6)$$

Injectivity just proved gives

$$H^0 B_\sigma(\mathcal{T}) = \{(\hat{\psi}, \psi) : \psi \in V\}, \quad H^1 B_\sigma(\mathcal{T}) = Q := \mathcal{B}/\Theta V. \quad (5.7)$$

The quotient here is the algebraic quotient with its original quotient locally convex topology. The later global retraction source proves that this image is closed; its exact retraction and the resulting zero closure kernel are included after (6.12) below.

Let $L = \mathcal{P}(\{+, -\})$. For each nonempty mask $A \in L$, retain V at each admitted leg and the zero coefficient space at a missing leg, retain $\mathcal{B}$ at eta and zero at sigma, and retain the restrictions in (5.5). The empty mask has the zero diagram. Every nonempty A has restriction fibre exactly $C(\mathcal{T}_A)$; its differential image is ΘV , since both Θ and $\Theta\mathcal{F}$ have that image. A one-leg differential is injective, with its literal plus or minus sign retained. It follows that

$$H^0 B_\sigma(\mathcal{T}_A) = \begin{cases} V, & A = \{+, -\}, \\ 0, & |A| = 1, \end{cases} \quad H^1 B_\sigma(\mathcal{T}_A) = Q \quad (A \neq \emptyset). \quad (5.8)$$

For mask inclusions the eta map is identity and leg maps are coordinate inclusions, hence the induced H^1 map is exactly identity on Q . This explicitly proves compatibility with every original mask transport.

Reconstruction gives

$$\widetilde{H}^1 B_\sigma = \{\tau\} \sqcup \bigsqcup_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q. \quad (5.9)$$

The scalar $e = 0_E^\bullet \in G(E)$ sends (A, q) to $(A, 0)$. The scalar τ sends it to the absent element. Linear restriction to the zero coefficient at sigma sends q to zero, but its supported reconstruction is $(A, q) \mapsto (A, 0)$, preserving A . No class of (5.9) becomes absent merely because its coefficient restricts to zero. The same statement applies to (2.4), (3.4), and (4.4) fibrewise, with dual label arrows given by actual precomposition and hence indexed by L^{op} .

6. Finite jets, their full kernel, and the retained topology

Put $g = 2\xi$, where

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (6.1)$$

The source's full-plane multiplier identity is

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_\phi = \frac{\mathcal{M}\phi}{\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)} \in \mathcal{O}(\mathbb{C}). \quad (6.2)$$

Here is the full continuation used in this identity. Write the Taylor expansion of the even Schwartz function at zero as $\phi(x) = \sum_{j=0}^N a_{2j}x^{2j} + O(x^{2N+2})$. Subtract this finite sum in its Mellin integral over $(0, 1)$. The remainder is holomorphic for $\Re s > -2N - 2$, while the subtracted terms have integrals $a_{2j}/(s+2j)$. The integral over $(1, \infty)$ is entire by Schwartz decay. Since N is arbitrary, $\mathcal{M}\phi$ has meromorphic continuation with only possible simple poles at the nonpositive even integers. The reciprocal $1/\Gamma(s/2)$ has a simple zero at each such integer and cancels each possible pole. At $s = 0$, the condition $a_0 = \phi(0) = 0$ removes the Mellin pole, and $1/\Gamma(s/2)$ cancels the explicit factor s in the denominator of H_ϕ . At $s = 1$, evenness and the integral condition give $\mathcal{M}\phi(1) = \frac{1}{2} \int_{\mathbb{R}} \phi = 0$, cancelling $s - 1$. No other denominator zero occurs, since the gamma function has no zeros. Thus the displayed H_ϕ is entire. Formula (5.4) proves (6.2) first on $\Re s > 1$, and analytic continuation proves it everywhere, with the original factor $g = 2\xi$ retained.

Let Z be a finite set of actual zeros of g with their actual orders m_ρ , and write

$$z_\rho = s - \rho, \quad g(s) = u_\rho(z_\rho)z_\rho^{m_\rho}, \quad u_\rho(0) \neq 0, \quad A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho/(g). \quad (6.3)$$

The germ coordinate z_ρ , the full jet length, and the actual unit u_ρ are retained. The jet arrow is

$$J_Z : \mathcal{B} \longrightarrow A_Z, \quad (J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j. \quad (6.4)$$

Equation (6.2) proves $J_Z \Theta = 0$, so it induces $q_Z : Q \rightarrow A_Z$.

Here is a full finite interpolation proof, with the original dx Riesz density preserved. Index jets by $\alpha = (\rho, j)$, and fix any $R > 0$ and $0 < R' < R$. Let

$$\eta_{R'}(y) = \begin{cases} \exp[-1/((R')^2 - y^2)], & |y| < R', \\ 0, & |y| \geq R', \end{cases} \quad r_\alpha(x) = x^{\bar{\rho}-1}(\log x)^j/j!. \quad (6.5)$$

Set

$$B_{\alpha\beta} = \frac{1}{j!k!} \int_{-R'}^{R'} \eta_{R'}(y) e^{(\rho+\bar{\sigma}-1)y} y^{j+k} dy, \quad \beta = (\sigma, k). \quad (6.6)$$

This is $\ell_\alpha(\eta_{R'}(\log x)r_\beta(x))$, where ℓ_α is the corresponding component of (6.4). It is Hermitian. For a column c , its quadratic form is

$$\bar{c}^\top B c = \int_{-R'}^{R'} \eta_{R'}(y) e^{-y} \left| \sum_\alpha \bar{c}_\alpha e^{\rho y} y^j / j! \right|^2 dy. \quad (6.7)$$

The exponential polynomials are independent: for a fixed exponent ρ , apply $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{m_\sigma}$ to a vanishing linear combination. It kills every other exponent block, and acts on the remaining block by $e^{\rho y}$ times $\prod_{\sigma \neq \rho} (\partial_y + \rho - \sigma)^{m_\sigma}$. Each factor on finite-degree polynomials is invertible, since it is a nonzero constant plus a nilpotent differentiation operator, with inverse given by the terminating geometric series. Hence that polynomial block is zero. This holds for every ρ , proving independence. Therefore (6.7) is strictly positive for $c \neq 0$, so B is invertible.

For $v \in A_Z$, the function

$$s_Z(v)(x) = \eta_{R'}(\log x) \sum_{\beta} (B^{-1}v)_{\beta} r_{\beta}(x) \quad (6.8)$$

is smooth, belongs to $\mathcal{B}$, has support in $[e^{-R'}, e^{R'}] \subset (e^{-R}, e^R)$, and satisfies $J_Z s_Z(v) = v$ by matrix multiplication. Thus (6.4) and q_Z are surjective, with an explicitly calculated finite-dimensional continuous section of (6.4).

For continuity, choose an integer $N > |\Re \rho|$ and put $p_N(F) = \max(\sup x^N |F(x)|, \sup x^{-N} |F(x)|)$. Splitting the integral at one gives the exact bound

$$|\ell_{\rho,j}(F)| \leq p_N(F) ((N + \Re \rho)^{-j-1} + (N - \Re \rho)^{-j-1}). \quad (6.9)$$

Indeed the two integrals of $|\log x|^j/j!$ against $x^{N+\Re \rho} dx/x$ and $x^{-N+\Re \rho} dx/x$ are the two displayed terms. This also proves that the Mellin jet derivatives used here are well-defined by dominated differentiation.

Define the actual retained kernel spaces, using $\mathcal{K}_Z$ to avoid changing the different space named $\mathcal{B}_Z$ in the source's finite arithmetic extension:

$$\mathcal{K}_Z = \ker J_Z, \quad Q_Z^0 = \mathcal{K}_Z / \Theta V. \quad (6.10)$$

They give exact sequences

$$0 \longrightarrow \mathcal{K}_Z \longrightarrow \mathcal{B} \xrightarrow{J_Z} A_Z \longrightarrow 0, \quad \boxed{0 \longrightarrow Q_Z^0 \longrightarrow Q \xrightarrow{q_Z} A_Z \longrightarrow 0}. \quad (6.11)$$

The first is split by (6.8); the second is split as a topological vector-space sequence by the induced section. No equivariance of this section is asserted: only the arrows (6.11) are canonical scaling morphisms.

To verify the topology assertion, the map $(F_0, v) \mapsto F_0 + s_Z(v)$ is a continuous linear bijection $\mathcal{K}_Z \oplus A_Z \rightarrow \mathcal{B}$, with continuous inverse $F \mapsto (F - s_Z J_Z F, J_Z F)$. Since $\Theta V \subset \mathcal{K}_Z$, passing to the quotient gives the claimed topological direct sum. By (6.9), $\mathcal{K}_Z$ is closed, so $\overline{\Theta V} \subset \mathcal{K}_Z$. The same proof gives

$$0 \rightarrow \mathcal{K}_Z / \overline{\Theta V} \rightarrow Q^H := \mathcal{B} / \overline{\Theta V} \rightarrow A_Z \rightarrow 0, \quad \ker(Q \rightarrow Q^H) = \overline{\Theta V} / \Theta V. \quad (6.12)$$

Thus every finite jet survives both specified quotients, with the closure kernel written explicitly. No new global topology has been substituted.

For this actual source the kernel in (6.12) is zero. We include the later source's constructive reason, retaining its auxiliary cutoffs and contraction constant. Let μ be the arithmetic Möbius function, and define on $x \neq 0$

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|). \quad (6.12a)$$

For $j \geq 0$, write $x^j \partial_x^j = P_j(D)$, where $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, and let $C_{N,j}(F)$ be the sum of $p_{N,k}(F) = \sup x^N |D^k F|$ weighted by the absolute coefficients of P_j . For each integer $N > 1$, termwise differentiation gives

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j} \quad (x > 0). \quad (6.12b)$$

Uniform convergence away from zero justifies each derivative. On the original theta image the double sum is absolutely convergent, bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which converges by Schwartz decay and $d(k) \leq k$. The identity $\sum_{n|k} \mu(n) = 1$ for $k = 1$, zero otherwise, therefore proves

$$\mathcal{U}\Theta\phi = \phi, \quad \mathcal{U}J\Theta\phi = \hat{\phi} \quad (x \neq 0). \quad (6.12c)$$

Choose the original type of real even compactly supported smooth cutoffs $0 \leq \alpha, \beta \leq 1$, each equal to one near zero, and set on $L^2(\mathbb{R}, dx)$

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad c = \|T\|, \quad YF = (1 - \alpha)UF + \alpha \mathcal{F}^{-1}((1 - \beta)UJF). \quad (6.12d)$$

The kernel of T is $\alpha(x)\check{\beta}(x - y)$, with squared L^2 norm $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Both its factors are contractions. If $c = 1$, compactness supplies a norm-attaining unit vector f ; equality in both contraction inequalities forces $\hat{f}$ to have compact support where $\beta = 1$ and f to have compact support where $\alpha = 1$. Compact Fourier support makes the inverse Fourier integral entire, because the compactly supported L^2 transform is L^1 and allows all complex derivatives under its integral. This entire function vanishes on a real interval outside the support of f , hence vanishes identically, a contradiction. Therefore $c < 1$, and

$$(1 - T)^{-1} = \sum_{n \geq 0} T^n, \quad \|(1 - T)^{-1}\| \leq (1 - c)^{-1}. \quad (6.12e)$$

The cut-off exterior maps in Y are continuous into even Schwartz functions by (6.12b); their values are extended by zero where the cutoff vanishes near zero. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: compact Fourier support bounds every derivative of $\mathcal{F}^{-1} M_\beta \mathcal{F} f$ by Cauchy–Schwarz, and multiplication by smooth compactly supported α controls every Schwartz seminorm. Thus $\Phi = (1 - T)^{-1}Y$ is Schwartz-valued and continuous: the equation $\Phi F = YF + T\Phi F$ bounds each Schwartz seminorm p by

$$p(\Phi F) \leq p(YF) + C_p(1 - c)^{-1} \|YF\|_2. \quad (6.12f)$$

It preserves evenness. With $\gamma_0 = e^{-\pi x^2}$ and $\gamma_2 = 2\pi x^2 e^{-\pi x^2}$, whose value/integral columns are $(1, 1)$, $(0, 1)$, define $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ by

$$P_V \phi = \phi - \phi(0)\gamma_0 - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) \gamma_2, \quad \Lambda = P_V(1 - T)^{-1}Y. \quad (6.12g)$$

This continuous projection has image V and is identity on V . On $\Theta\phi$, (6.12c) gives $Y\Theta\phi = (1 - T)\phi$, so

$$\Lambda\Theta = 1_V, \quad K = 1_{\mathcal{B}} - \Theta\Lambda, \quad K^2 = K, \quad \ker K = \Theta V. \quad (6.12h)$$

The last equality follows in both directions by direct substitution. Since K is continuous, its kernel is closed. Therefore $\Theta\bar{V} = \Theta V$, $Q = Q^H$, and the closure kernel in (6.12) is exactly zero. The section

$$s : Q \rightarrow \mathcal{B}, \quad s[F] = KF \quad (6.12i)$$

is well-defined and continuous in the quotient topology, satisfies $qs = 1$ and $\Lambda s = 0$, and gives the topological decomposition $\mathcal{B} \cong V \oplus Q$ by $F \mapsto (\Lambda F, qF)$. Every cutoff and the factor $(1 - c)^{-1}$ remain in this construction. It establishes closedness of the theta image, and makes no assertion that the different kernel Q_Z^0 in (6.11) vanishes.

Let $\mathcal{T}_Z$ have the same source legs and sigma stalk as $\mathcal{T}$, eta stalk $\mathcal{K}_Z$, and the original restrictions. Then

$$0 \rightarrow \mathcal{T}_Z \rightarrow \mathcal{T} \xrightarrow{J_Z} S_\eta A_Z \rightarrow 0 \quad (6.13)$$

is exact at each stalk; its eta surjectivity is (6.8). Applying the explicit complexes gives a termwise exact sequence

$$0 \rightarrow C(\mathcal{T}_Z) \rightarrow C(\mathcal{T}) \rightarrow A_Z[-1] \rightarrow 0. \quad (6.14)$$

The kernel inclusion into the homotopy fibre of the last map is a quasi-isomorphism. This follows directly by quotienting that inclusion: the quotient fibre is the fibre of the identity of $A_Z[-1]$, whose two identical terms have an invertible differential and zero cohomology. Therefore the full cohomology of this jet-comparison fibre is

$$H^0 = V \quad \text{via } \psi \mapsto (\hat{\psi}, \psi), \quad H^1 = Q_Z^0, \quad (6.15)$$

and nothing else. On a nonempty one-leg mask $H^0 = 0$ and $H^1 = Q_Z^0$; the same proof applies to each exact mask diagram. On $S_\eta A_Z$ itself, sigma restriction is zero, its restriction fibre is exactly $A_Z[-1]$, and its restriction cone is $A_Z[0]$.

For disjoint finite packets Z, W of actual zeros, simultaneous application of (6.8) to $Z \cup W$ with prescribed coordinates $(0, w)$ proves the further exact surjection

$$Q_Z^0 \longrightarrow A_W. \quad (6.16)$$

Consequently the kernel in (6.11) retains all these explicitly reachable omitted packets. It has not been set to zero.

7. Original scaling, generator, and dual actions

For $a > 0$, retain $U_a F(x) = F(x/a)$. Substitution $x = ay$ gives

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = aU_{1/a} \hat{\phi}. \quad (7.1)$$

These operators preserve the original test spaces: dilation rescales each defining Schwartz or $\mathcal{B}$ seminorm by finite constants, preserves evenness and the value-zero condition, and multiplies the integral by a . The source sheaf action is exactly

$$U_a^{\mathcal{T}} : (\phi, \psi, F) \mapsto (U_a \phi, aU_{1/a} \psi, U_a F). \quad (7.2)$$

The identities $\Theta U_a = U_a \Theta$ and (7.1) give

$$d(U_a \phi, aU_{1/a} \psi) = U_a \Theta(\phi - \hat{\psi}). \quad (7.3)$$

Thus (7.2) is a diagram automorphism and (3.1) proves equivariance of every restriction-fibre map. On $H^0 = V$ in the parameter ψ of (5.7), its action is $\psi \mapsto aU_{1/a} \psi$; on Q it is $[F] \mapsto [U_a F]$. Every support mask is fixed.

The infinitesimal generator for $a = e^t$ at $t = 0$ is $D = -x\partial_x$ on the plus and eta stalks, and $1 - D$ on the minus stalk. Differentiating (7.1) gives the original relation $\mathcal{F}D = (1 - D)\mathcal{F}$. Integration by parts, with zero boundary terms because $F \in \mathcal{B}$, gives $\mathcal{M}(DF) = s\mathcal{M}F$.

On the full local jet algebra,

$$U_a|_\rho = a^\rho \exp((\log a)N_\rho) = a^\rho \sum_{k=0}^{m_\rho-1} \frac{(\log a)^k}{k!} N_\rho^k, \quad N_\rho(z_\rho^j) = z_\rho^{j+1}, \quad (7.4)$$

where $N_\rho(z_\rho^{m_\rho-1}) = 0$. The generator is $\rho I + N_\rho$. This follows by multiplying the jet by the exact germ $a^{\rho+z_\rho}$, so (6.4), (6.11), (6.13), and (6.14) intertwine it; in particular $\mathcal{K}_Z$ and Q_Z^0 are stable under scaling. Nilpotent directions have not been replaced by their trace.

Let $\mathcal{L}_1 = E$ with action $w \mapsto aw$, the original integral-moment line. On every dual object the action is the induced Hom action

$$(U_a \ell)(v) = a\ell(U_{1/a} v). \quad (7.5)$$

The formula is forced by the evaluation map $\ell \otimes v \mapsto \ell(v) \in \mathcal{L}_1$: applying (7.5) and U_a to both inputs multiplies the result by a . Since all maps in Sections 3 and 4 are defined by precomposition and restriction, this proves their equivariance with (7.5), including every contravariant mask transport.

The known continuous section (6.12i) has an exact, retained scaling defect. Denote the induced action on Q by $\overline{U}_a$, and define

$$k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a. \quad (7.6)$$

To prove the second equality, use $1_{\mathcal{B}} = sq + \Theta\Lambda$ on $U_a s$, and $qU_a s = \overline{U}_a$. The group identity then gives, without dropping a boundary term,

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (7.7)$$

Indeed expand $U_{ab}s - s\overline{U}_{ab}$ by first applying the equation for b , then that for a , and cancel the injective Θ . These maps are continuous by their formulas. Thus the known topological splitting is accompanied by the full original source-valued action defect.

8. Full finite residue arrow and its chain degree

The symmetry of the original $g = 2\xi$ used here follows directly from the original Gaussian Poisson formula, with no assumption on its zeros. Put $\psi_0(t) = \sum_{n \geq 1} e^{-\pi n^2 t}$. Poisson gives $\psi_0(t) = t^{-1/2} \psi_0(1/t) + \frac{1}{2}(t^{-1/2} - 1)$. For $\Re s > 1$, termwise integration gives $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^\infty \psi_0(t) t^{s/2} dt/t$. Splitting at one and substituting $t = 1/v$ in the reflected term therefore yields the entire continuation

$$g(s) = 1 + s(s-1) \int_1^\infty \psi_0(t) (t^{s/2} + t^{(1-s)/2}) \frac{dt}{t}. \quad (8.0)$$

The constant is exactly one: the remaining elementary integral is $1/(s-1) - 1/s = 1/[s(s-1)]$. The bound $\psi_0(t) \leq e^{-\pi t}/(1 - e^{-\pi t})$ for $t \geq 1$ proves locally uniform convergence with all s -derivatives. The expression is invariant under $s \mapsto 1-s$, and its real coefficients give conjugation symmetry. Consequently $g^\dagger = g$, and g is not the zero function (for example its continuation has $g(0) = 1$).

Assume Z is reflection-stable, meaning the finite chosen set satisfies $\iota Z = Z$ for $\iota\rho = 1 - \bar{\rho}$; this selects an actual finite divisor, without a hypothesis on the location of its points. The completed function has $g^\dagger = g$ for $f^\dagger(s) = \overline{f(1-\bar{s})}$. In original coordinates,

$$f_\rho^\dagger(z_\rho) = \sum_j (-1)^j \overline{f_{\iota\rho, j}} z_\rho^j, \quad u_\rho(z) = (-1)^{m_\rho} \overline{u_{\iota\rho}(-\bar{z})}. \quad (8.1)$$

The residue pairing, with its raw orientation, is

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger(s) h(s)}{g(s)} ds. \quad (8.2)$$

Changing any representative by a multiple of g adds a holomorphic differential, so this is well-defined. At ρ , if $u_\rho(z)^{-1} = \sum_{k \geq 0} c_{\rho, k} z^k$, its literal formula is

$$R_Z(f, h) = \sum_\rho \sum_{i, j=0}^{m_\rho-1} (-1)^i \overline{f_{\iota\rho, i}} h_{\rho, j} c_{\rho, m_\rho-1-i-j}, \quad c_{\rho, k} = 0 \ (k < 0). \quad (8.3)$$

The matrix $G_{ij} = c_{\rho, m_\rho-1-i-j}$ has determinant $(-1)^{m_\rho(m_\rho-1)/2} u_\rho(0)^{-m_\rho} \neq 0$: reversing the columns makes it triangular with diagonal $u_\rho(0)^{-1}$, and reversal has the displayed sign. Multiplication of the i -th row by $(-1)^i$, together with the invertible reflection-permutation of blocks, preserves invertibility. Hence R_Z is perfect, including all nilpotent jets and actual analytic units.

For a meromorphic function k , expansion in z_ρ gives

$$\text{Res}_\rho k^\dagger(s) ds = -\overline{\text{Res}_{\iota\rho} k(s)} ds. \quad (8.4)$$

Only its coefficient of z^{-1} contributes, and reflection multiplies that coefficient by $(-1)^{-1} = -1$. Apply (8.4) to $k = h^\dagger f/g$, sum over the permuted divisor, and use $g^\dagger = g$. The result is

$$R_Z(f, h) = -\overline{R_Z(h, f)}. \quad (8.5)$$

The raw pairing is therefore skew-Hermitian. No phase has been inserted.

The identity $(a^s)^\dagger = a^{1-s}$ gives

$$R_Z(U_a f, U_a h) = a R_Z(f, h). \quad (8.6)$$

Thus the following is an injective linear equivariant map from the conjugate vector space:

$$\overline{A_Z} \longrightarrow H^{-1} R \operatorname{Hom}_{\mathcal{A}}(\mathcal{T}, K_{\tau}(\mathcal{L}_1)), \quad \bar{f} \mapsto \ell_f, \quad \ell_f(F) = R_Z(f, J_Z F). \quad (8.7)$$

Here ℓ_f is literally the degree -1 cochain on $\mathcal{B}$ in (3.4). It is closed, since (6.2) gives $\ell_f \Theta = 0$, and $\ell_f \Theta \mathcal{F} = 0$. There is no degree -2 term, so this cocycle represents zero exactly when it is the zero functional. Surjectivity of J_Z and perfectness of (8.2) then prove injectivity. Formula (8.6), evaluated on $J_Z U_{1/a} F$, proves $\ell_{U_a f}(F) = a \ell_f(U_{1/a} F)$, exactly (7.5). Bound (6.9) also proves that all these functionals are continuous on the original quotient topology, even though the adjunction (3.4) uses the full algebraic dual. Thus (8.7) exhibits a specified subspace of continuous functionals inside that full dual, by an actual injection.

The contraction $J_g : h \mapsto g' h$ retains its local form

$$g' = m_{\rho} u_{\rho}(z) z^{m_{\rho}-1} + u'_{\rho}(z) z^{m_{\rho}} \equiv m_{\rho} u_{\rho}(z) z^{m_{\rho}-1} \pmod{z^{m_{\rho}}}. \quad (8.8)$$

Using $g'/g = m_{\rho}/z + u'_{\rho}/u_{\rho}$ gives

$$R_Z(f, J_g h) = \sum_{\rho} m_{\rho} \overline{f(\iota \rho)} h(\rho) = \operatorname{Tr}(M_{f \dagger h} \mid A_Z). \quad (8.9)$$

The trace equality follows because multiplication by a germ in the ordered basis $1, z, \dots, z^{m-1}$ is triangular with its value repeated m times on the diagonal. The radical in either variable of (8.9) is exactly the subspace of all packets whose constant term is zero: the formula annihilates that subspace, and arbitrary constant terms in the reflected block detect every nonzero remaining value. The differentiated symmetry is $(g')^{\dagger} = -g'$, which together with (8.5) also confirms the Hermitian orientation of (8.9). No positivity of the raw residue form was used or asserted.

Every arrow in (8.7) is reconstructed at its original label A . A zero functional is $(A, 0)$, and the absent input alone is sent to τ . Precomposition with the mask inclusions commutes with the residue formula because H^1 transport in (5.8) is identity. This completes the labelled adjoint arrow with its degree, sign, topology, and action.

9. Tensor compatibility, including the dual Koszul sign

All coefficient tensors in this section are the original algebraic tensors over E ; no unproved completed tensor topology is substituted. For a product P^r , tensoring the stalkwise exact resolution (2.1) produces a finite projective resolution of the constant diagram. To see exactness without assuming a tensor theorem, at each stalk split an exact finite complex into its cohomology and elementary two-term identity complexes by choosing complements to boundaries and cycles. Tensoring an elementary identity complex with any complex is contractible, by its usual contracting homotopy tensored with identity. Only the tensor of the degree-zero constant cohomology remains. Each external product of the P_x 's represents evaluation at the corresponding product point and is projective. This proves the claimed resolution and yields the natural cochain identity

$$A_{\tau, r}(F_1 \boxtimes \cdots \boxtimes F_r) \simeq C(F_1) \otimes \cdots \otimes C(F_r), \quad (9.1)$$

with the full differential

$$d(x_1 \otimes \cdots \otimes x_r) = \sum_{j=1}^r (-1)^{\sum_{k < j} |x_k|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_r. \quad (9.2)$$

The standard cross product of cohomology classes is an isomorphism because the same split-complex argument kills all summands with a contractible factor. The proof uses a splitting to verify that the canonical cross product is bijective; it does not make that map dependent on the splitting.

Product restriction is the tensor $f_1 \otimes \cdots \otimes f_r$ to $S_1 \otimes \cdots \otimes S_r[0]$. Changes from plus to minus representatives are compatible: for $g_i = f_{i,-}$ and $f_i = f_{i,+}$, a homotopy is

$$H = \sum_{j=1}^r g_1 \otimes \cdots \otimes g_{j-1} \otimes h_j \otimes f_{j+1} \otimes \cdots \otimes f_r, \quad (9.3)$$

where tensor maps act with the Koszul sign and $h_j^1 = t_j$. The differential of each tensor summand is the same tensor with $d_{\text{Hom}} h_j = f_j - g_j$; every other factor is a degree-zero chain map. Summing telescopes to $\otimes f_i - \otimes g_i$, proving the homotopy identity.

For the original sheaves and all nonempty masks, every $S_i = 0$. Thus product restriction is zero and

$$B_{\sigma^r}(\mathcal{T}_{A_1} \boxtimes \cdots \boxtimes \mathcal{T}_{A_r}) = \bigotimes_{i=1}^r C(\mathcal{T}_{A_i}) = \bigotimes_{i=1}^r B_{\sigma}(\mathcal{T}_{A_i}). \quad (9.4)$$

The complete cohomology is the graded tensor product of the two groups in (5.8). Explicitly its degree n is the direct sum over subsets $I \subseteq \{1, \dots, r\}$, $|I| = n$, with Q in factors indexed by I and $H^0 C(\mathcal{T}_{A_i})$ in the other factors. In particular

$$\boxed{H^r B_{\sigma^r}(\mathcal{T}^{\boxtimes r}) = Q^{\otimes r}.} \quad (9.5)$$

The numerator is $\mathcal{B}^{\otimes r}$, and its denominator is exactly

$$\sum_{j=1}^r \mathcal{B}^{\otimes(j-1)} \otimes \Theta V \otimes \mathcal{B}^{\otimes(r-j)}. \quad (9.6)$$

Signs in (9.2) change none of these subspaces but remain in the differential itself. The quotient map to $Q^{\otimes r}$ has kernel (9.6), by successive tensoring of the quotient presentation over the field.

For general diagrams one cannot replace the fibre of a tensor product by the tensor of fibres. The exact replacement can itself be computed. For composable chain maps $X \xrightarrow{f} Y \xrightarrow{g} Z$, put $B(f) = \text{Fib}(f)$ with (1.3). There are chain maps

$$u : B(f) \rightarrow B(gf), \quad (x, y) \mapsto (x, gy), \quad v : B(gf) \rightarrow B(g), \quad (x, z) \mapsto (fx, z). \quad (9.7)$$

Their composite has nullhomotopy $H(x, y) = (-y, 0)$, since $dH + Hd = (fx, gy)$. A surjective chain map

$$\text{Cone}(u) \rightarrow B(g), \quad (x, z; x', y') \mapsto (fx - y', z) \quad (9.8)$$

has kernel $z = 0, y' = fx$, with remaining coordinates $x \in X^n, x' \in X^{n+1}$ and differential $(x, x') \mapsto (dx + x', -dx')$. The degree -1 map $(x, x') \mapsto (0, x)$ contracts this kernel. Thus (9.8) is a quasi-isomorphism and proves the fibre triangle explicitly. Apply this with $f = f_1 \otimes 1_{C_2}$ and $g = 1_{S_1} \otimes f_2$. The resulting triangle is

$$B(f_1) \otimes C_2 \rightarrow B(f_1 \otimes f_2) \rightarrow S_1 \otimes B(f_2) \rightarrow (B(f_1) \otimes C_2)[1]. \quad (9.9)$$

The first fibre identification follows immediately from (1.3) and the tensor differential; in the second, S_1 is in degree zero. This proves the general tensor relation and explains precisely why it reduces to (9.4) for the original source.

The product right adjoint is

$$K_{\tau, r}(W) = K_{\tau}(E)^{\boxtimes r} \otimes W, \quad K_{\tau, r}(W) \simeq S_{(\eta, \dots, \eta)} W[r]. \quad (9.10)$$

Its terms are finite direct sums of $I_{(x_1, \dots, x_r)}(W)$, hence injective. Its only cohomology follows by the same stalkwise contraction proof used for (3.9) and (9.1). No assertion that an infinite-dimensional algebraic dual commutes with tensor products is used.

The adjunction on the product has a sign that must be stated. In total degree $-p$, multiply the coefficient functional arising from $K_{\tau}(E)^{\boxtimes r}$ by

$$\epsilon_p = (-1)^{p(p-1)/2} \quad (9.11)$$

to identify it with $\text{Hom}^{-p}(\bigotimes C(F_i), W)$. To check this, an index set I of p eta factors contributes to degree $-p$. Removing $j \in I$ in d_K has sign $(-1)^{|I \cap \{1, \dots, j-1\}|}$. The corresponding incoming cochain differential in (9.2) has the same sign, while its internal Hom differential has the additional factor $(-1)^{p+1}$ from (1.4). The equality $\epsilon_{p-1} = (-1)^{p+1} \epsilon_p$ proves commutation in every component. This verifies the sign on every product differential.

In top degree the identification is therefore

$$H^{-r} R \text{Hom}_{\mathcal{A}(P^r)}(\mathcal{T}^{\boxtimes r}, K_{\tau, r}(W)) \cong \text{Hom}_E(Q^{\boxtimes r}, W), \quad (9.12)$$

with the coefficient multiplier ϵ_r . In particular, for $r = 2$ it is -1 . If a product residue arrow is written on the right of (9.12) as the literal functional

$$([F_1] \otimes \cdots \otimes [F_r]) \mapsto \prod_{i=1}^r R_Z(f_i, J_Z F_i), \quad (9.13)$$

its coefficient representative in the unaltered tensor complex $K_{\tau, r}$ is ϵ_r times that product. This is the cohomological Koszul sign, not a phase inserted into (8.2) and not a modification of its positivity properties.

Tensoring (6.11) gives the exact surjection $Q^{\boxtimes r} \rightarrow A_Z^{\boxtimes r}$, with kernel

$$\sum_{j=1}^r Q^{\boxtimes(j-1)} \otimes Q_Z^0 \otimes Q^{\boxtimes(r-j)}. \quad (9.14)$$

This follows by successively quotienting each factor; a vector-space complement verifies equality of the kernel, without imposing an equivariant complement. The fibre of the cochain jet map

$$C(\mathcal{T})^{\boxtimes r} \longrightarrow A_Z^{\boxtimes r}[-r] \quad (9.15)$$

has the same cohomology as $C(\mathcal{T})^{\boxtimes r}$ in degrees below r , has (9.14) in degree r , and zero above r . Indeed the target has only degree r , its degree- r map is the proved surjection, and the explicit fibre long exact sequence then has exactly these kernels and cokernels.

The action in all tensor expressions is the tensor of the original (7.2) and (7.4). On a specified packet tuple $(\rho_1, \dots, \rho_r)$ it is

$$a^{\rho_1 + \cdots + \rho_r} \exp \left((\log a) \sum_{j=1}^r 1 \otimes \cdots \otimes N_{\rho_j} \otimes \cdots \otimes 1 \right). \quad (9.16)$$

The summands commute, so their exponential is exactly the product of the terminating exponentials in (7.4). On the target line $\mathcal{L}_1^{\boxtimes r}$ the action is a^r ; (8.6) proves covariance of (9.13) by this exact factor. Formula (9.16) retains every nilpotent term, not only its eigenvalue. Support reconstruction is performed on each original admitted tensor fibre and its balancing relations: a zero tensor amplitude remains the supported zero of that fibre, whereas an absent tensor input is absorbing. The proof never identifies the coefficient tensor with a Cartesian product of label sets.

There is also the source's proved passage to completed projective tensor products, now obtained from the retained continuous contraction instead of an assumed exactness of completion. For the joint complex $C = C(\mathcal{T})$, let $H_0 = V[0] \oplus Q[-1]$, with zero differential, and put

$$p^0(\phi, \psi) = \psi, \quad p^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s, \quad H^1(F) = (\Lambda F, 0). \quad (9.17)$$

Then $pi = 1$ and $1 - ip = dH + Hd$: in degree zero the latter equation is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$, obtained from $\Lambda\Theta = 1$; in degree one it is $1 - sq = \Theta\Lambda$. All the maps are continuous in the already specified topologies. If $P = ip$, the tensor homotopy

$$H_r = \sum_{j=1}^r P^{\boxtimes(j-1)} \otimes H \otimes 1^{\boxtimes(r-j)} \quad (9.18)$$

uses the sign $(-1)^{\sum_{k < j} |x_k|}$ on homogeneous inputs. The differential telescopes to $1 - P^{\otimes r}$, since $d_{\text{Hom}} H = 1 - P$, while $P, 1$ are chain maps of degree zero. Every finite tensor map is continuous for the projective tensor topology and therefore extends uniquely to its completion; the identity extends by density. The completed complex is thus continuously homotopy equivalent to $H_0^{\widehat{\otimes} \pi^r}$. Its degree n cohomology is the direct sum of ordered completed tensors with Q in n slots and V in the others, and in particular its top cohomology is $Q^{\widehat{\otimes} \pi^r}$. The comparison from (9.5) is the canonical completion map on these tensors. For an individual one-leg mask replace H_0 by $Q[-1]$, keep $p^1 = q, i^1 = s$, and take $H^1 = \Lambda$ on the plus leg or $H^1 = -\mathcal{F}\Lambda$ on the minus leg. On the latter the differential is $-\Theta\mathcal{F}$, so $Hd = \mathcal{F}\Lambda\Theta\mathcal{F} = 1_V$ and $dH = \Theta\Lambda = 1 - sq$; the plus leg has the same identities with its positive signs. Thus the same tensor proof applies to every mask. This specifies the exact completed continuation while retaining the algebraic tensor calculation and the action defect (7.6).

10. Exact consequences for continuation

The original global arithmetic object, its generic-boundary restriction, and its right adjoint are related by the explicit maps (2.3), (3.10), and (4.4), and the nonsplit extension (4.7). On the original theta sheaf, the restriction fibre is the original cochain complex itself, with every supported copy of Q retained. Finite jets give the exact kernel sequence (6.11), its sheaf and fibre realizations (6.13)–(6.15), and the injective continuous residue arrow into the original adjoint (8.7). Tensor products give (9.5), (9.14), and the full action (9.16), with the independently checked Koszul sign (9.11).

These are completed calculations on the original $\Theta, V, \mathcal{B}, \mathbf{A}_\tau, K_\tau$ and $g = 2\xi$. No vanishing of Q , of the finite-jet kernel Q_Z^0 , or of an arithmetic boundary contribution was used to obtain them. The calculations establish exact comparison morphisms and retain their kernels; they supply no upper estimate for the original tensor action beyond the identities proved here.

Chapter 3

Full reversed-support dual

Scope and source pins

This calculation completes the specific promise in transcript A1545, node bacc895-0427-42ee-b6f6-4c61752dfde2, to compute “the full τ -chart dual, including bottom fibers created by reversed support arrows.” It uses the actual four-point chart, coefficient spaces, differential, and right adjoint of the retained *Arithmetic cohomology over the τ -base*, NOTE.md, SHA-256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3. The complete transcript segment U0041–U0054 has SHA-256 e79b46722d9a2ee95b68ae6512f37a19fd41e4b3ecc4e53583cd357d0442c251. The reconstruction convention allowing a nonzero bottom fibre is the fully read `tex/support_diagrams.tex`, equations (D1)–(D8). No generic-stalk restriction replaces global cohomology here.

1. Original objects, signs, and both index sets

Retain $E = \mathbb{C}$, the even Schwartz space

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

and

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}.$$

Write $\mathcal{F} : V \rightarrow V$ for this Fourier transform; on this even space $\mathcal{F}^{-1} = \mathcal{F}$. Retain

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad J\Theta = \Theta\mathcal{F}, \quad Q = \mathcal{B}/\Theta V,$$

and denote the quotient by $q : \mathcal{B} \rightarrow Q$. The source proves injectivity of Θ from its Mellin identity in $\Re s > 1$; no zero in the critical strip is assumed absent.

The geometric index poset remains $P = \{+, -, \eta, \sigma\}$, with $+$ $< \eta < \sigma$ and $-$ $< \eta < \sigma$. The support index poset is separately

$$L = \{\emptyset, +, -, J_0\}, \quad J_0 = \{+, -\},$$

ordered by inclusion. The symbol J_0 denotes a support mask; J above denotes the original inversion operator. At active mask A , the coefficient diagram $\mathcal{T}_A$ retains precisely the source legs belonging to A , has overlap $\mathcal{B}$, and has zero vector stalk at σ . At the empty mask it is the zero diagram. The source computes

$$\mathbf{A}_\tau(F) = R\Gamma(P, F) \simeq [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta], \quad K_\tau(W) = [I_\eta(W) \longrightarrow I_+(W) \oplus I_-(W)],$$

with degrees 0, 1 and $-1, 0$, respectively, and differential signs $+, -$. Its proved adjunction gives

$$D_A(W) := R\mathrm{Hom}_{\mathrm{Fun}(P, \mathrm{Vect}_E)}(\mathcal{T}_A, K_\tau(W)) \cong R\mathrm{Hom}_E(\mathbf{A}_\tau(\mathcal{T}_A), W).$$

Here W is an ordinary coefficient vector space, and $X^\vee = \text{Hom}_E(X, W)$ throughout the algebraic calculation.

2. The complete complexes and every support arrow

The following are literal representatives, not just their cohomology:

A	D_A^{-1}	D_A^0	δ_A
$\emptyset$	0	0	0
$+$	$\mathcal{B}^\vee$	$V^\vee$	$\ell \mapsto \ell\Theta$
$-$	$\mathcal{B}^\vee$	$V^\vee$	$\ell \mapsto -\ell\Theta\mathcal{F}$
J_0	$\mathcal{B}^\vee$	$V^\vee \oplus V^\vee$	$\ell \mapsto (\ell\Theta, -\ell\Theta\mathcal{F})$.

For $A \subseteq A'$, the arrow $D_{A'} \rightarrow D_A$ is precomposition with the original coefficient inclusion. On degree -1 it is the identity between active masks. On degree zero the arrows $J_0 \rightarrow +$ and $J_0 \rightarrow -$ are respectively

$$(a, b) \mapsto a, \quad (a, b) \mapsto b.$$

Both leaf-to-empty arrows are zero, as is the direct joint-to-empty arrow. Applying either coordinate projection to $\delta_{J_0} \ell$ gives the corresponding displayed leaf differential; the maps therefore commute with the differentials. Both composites from joint to empty are zero. These checks prove all naturality and composition identities in $\text{Fun}(L^{\text{op}}, \text{Ch}(E))$.

3. Cohomology, with an explicit quotient and inverse

For every active mask, the degree- -1 kernel is

$$\{\ell \in \mathcal{B}^\vee : \ell\Theta = 0\} \xleftarrow[\sim]{q^\vee} Q^\vee, \quad q^\vee(\lambda) = \lambda q.$$

The inverse sends ℓ to $[F] \mapsto \ell(F)$. It is well-defined precisely because $\ell\Theta = 0$. The Fourier isomorphism makes the condition on the minus face equivalent to the same equation. All active transition maps on this kernel are the identity in these $Q^\vee$ coordinates.

On either single leg, the degree-zero cohomology is zero. Indeed any linear functional on V extends to $\mathcal{B}$ through the injective map Θ . More concretely, the global retraction constructed in transcript A1694, fully reproved with explicit fixed cutoffs and estimates in Appendix A below, is a continuous linear map $\Lambda : \mathcal{B} \rightarrow V$ with $\Lambda\Theta = 1$. Then $a\Lambda$ lifts $a \in V^\vee$ on the plus face and $-b\mathcal{F}^{-1}\Lambda$ lifts $b \in V^\vee$ on the minus face.

At the joint mask define

$$\pi_0 : V^\vee \oplus V^\vee \longrightarrow V^\vee, \quad \pi_0(a, b) = a\mathcal{F} + b.$$

This evaluates the functional on the original cycle $\psi \mapsto (\mathcal{F}\psi, \psi)$. The composite $\pi_0\delta_{J_0}$ is zero. Conversely, if $a\mathcal{F} + b = 0$, then $b = -a\mathcal{F}$, and

$$\delta_{J_0}(a\Lambda) = (a, -a\mathcal{F}) = (a, b).$$

The quotient map is surjective, with representative $c \mapsto (0, c)$. Its inverse on the quotient sends c to the class of $(0, c)$: subtracting this representative from (a, b) gives

$$(a, b) - (0, a\mathcal{F} + b) = (a, -a\mathcal{F}) = \delta_{J_0}(a\Lambda).$$

This proves both inverse identities. Consequently the entire cohomology diagram is

A	J_0	$+$	$-$	$\emptyset$
$H^{-1}D_A$	$Q^\vee$	$Q^\vee$	$Q^\vee$	0
H^0D_A	$V^\vee$	0	0	0 .

The arrows in degree -1 are identity from joint to either leaf and zero to the empty mask. Every arrow in degree zero is zero. No degree-zero joint class is discarded.

In particular $V^\vee \neq 0$ when $W = E$. A concrete continuous functional is $\phi \mapsto \phi(1)$. It is nonzero on

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V.$$

The function is even, Schwartz, and vanishes at zero. Integration by parts gives $\int x^4 e^{-\pi x^2} dx = \frac{3}{2\pi} \int x^2 e^{-\pi x^2} dx$, proving its integral is zero; its value at one is $2\pi(2\pi - 3)e^{-\pi} \neq 0$.

4. What the reversed bottom actually contains

In L^{op} the bottom is J_0 , the join operation is original intersection, and the top is the original empty mask. Thus the dual's bottom cohomology fibres are exactly $Q^\vee$ in degree -1 and $V^\vee$ in degree zero. In particular the latter is nonzero for $W = E$. The rule “every bottom fibre is zero” is therefore incompatible with this calculation.

The source's full reconstruction retains these fibres. For either cohomology diagram $H^i D$, it is

$$M^i = \coprod_{A \in L^{\text{op}}} H^i D_A,$$

with

$$(A, u) + (B, v) = (A \cap B, \rho_{A, A \cap B} u + \rho_{B, A \cap B} v), \quad r^\bullet(A, u) = (A, ru), \quad \tau(A, u) = (J_0, 0).$$

The global additive zero is $(J_0, 0)$, and $e(A, u) = (A, 0)$. Associativity transports three amplitudes to $A \cap B \cap C$; equality of the two iterated transports proves the two sums agree. The zero law uses the unique map from the bottom and its zero value. Linearity of transitions proves distributivity, and the fibrewise vector-space scalar laws prove the remaining semimodule laws. This is the exact reconstruction, including nonzero vectors in the fibre of its global zero under multiplication by e .

If an additional external-absence label is desired while retaining every original reversed label, it can also be constructed explicitly: adjoin a new bottom $\perp_\dagger < L^{\text{op}}$, put zero fibre there, and retain all old fibres and arrows. The new external zero is $(\perp_\dagger, 0)$. The label map sending $\perp_\dagger$ and J_0 both to J_0 , and fixing every other label, preserves joins and bottom. With identity maps on old fibres and the zero map on the new fibre it reconstructs a surjective split-linear comparison to M^i . Its only additional identification is between the new external zero and the zero vector at the old bottom J_0 . It does not remove any nonzero bottom vector. This proves the exact relation between the two declared conventions; it does not silently impose a zero bottom fibre on the original reversed diagram.

5. Strict natural splitting fails, but the derived splitting has a concrete roof

Let $C = V^\vee$, let Z_C be the diagram supported at the bottom J_0 with value C , and let H be the degree- -1 cohomology diagram just calculated. The maps $q^\vee : H[1] \rightarrow D$ and $\pi : D \rightarrow Z_C[0]$, where π is π_0 only at the joint mask in degree zero, are natural cochain maps.

There is no strict natural cochain section $s : Z_C[0] \rightarrow D$ of π when $C \neq 0$. At the joint mask write $s(c) = (a(c), b(c))$. Naturality to the plus leaf forces $a(c) = 0$, since the source at that leaf is zero. Naturality to the minus leaf forces $b(c) = 0$. Hence $\pi s = 0$, contradicting $\pi s = 1_C$. This is exactly a strict-section obstruction.

It is not a derived nonsplitting statement. Here is the actual surviving derived map. Write $b = J_0, x = +, y = -, t = \emptyset$ in the reversed diamond. For a vertex v , define the projective diagram

$$P_v(C)(w) = \begin{cases} C, & v \leq w, \\ 0, & v \not\leq w. \end{cases}$$

The augmented resolution of Z_C is

$$\mathcal{R}_C : P_t(C) \xrightarrow{c \mapsto (c, -c)} P_x(C) \oplus P_y(C) \xrightarrow{(a,b) \mapsto a+b} P_b(C) \xrightarrow{\epsilon} Z_C,$$

with degrees $-2, -1, 0$ before the augmentation. At b , the augmentation is identity. At either leaf the preceding map is identity and the augmented stalk is zero. At t , the stalk sequence is $0 \rightarrow C \rightarrow C^2 \rightarrow C \rightarrow 0$, with the displayed anti-diagonal and sum. This proves exactness stalkwise and on all arrows. Each $P_v(C)$ is projective because its Hom functor is $\text{Hom}_E(C, -_v)$, an exact functor over a field. Thus $\epsilon : \mathcal{R}_C \rightarrow Z_C[0]$ is a quasi-isomorphism.

Define $f : \mathcal{R}_C \rightarrow D$ as follows. In degree zero it is determined by

$$f_b^0(c) = (0, c), \quad f_x^0(c) = 0, \quad f_y^0(c) = c, \quad f_t^0(c) = 0.$$

In degree -1 , its map from $P_x(C)$ is zero, and its map from $P_y(C)$ is determined at y by

$$f_y^{-1}(c) = -c\mathcal{F}^{-1}\Lambda \in \mathcal{B}^\vee.$$

At t it is zero; in degree -2 it is zero. These maps are natural by their projective-source definitions and the zero target at t . At y , the chain equation is

$$\delta_y(-c\mathcal{F}^{-1}\Lambda) = c\mathcal{F}^{-1}\Lambda\Theta\mathcal{F} = c.$$

At x both sides are zero, and at t both sides are zero. These verify every chain equation. Finally $\pi f = \epsilon$, so

$$Z_C[0] \xleftarrow[\simeq]{\epsilon} \mathcal{R}_C \xrightarrow{f} D$$

is an explicit right inverse of π in the derived category. The combined cochain map

$$H[1] \oplus \mathcal{R}_C \longrightarrow D, \quad (h, r) \mapsto q^\vee h + f(r),$$

induces identity on degree -1 cohomology and the displayed isomorphism on degree zero cohomology. It is therefore a quasi-isomorphism. This proves the full derived relationship without replacing the original support arrows by equalities.

6. The original action and the actual defect of this roof

For $a > 0$, retain $U_a F(x) = F(x/a)$. On the source plus and minus legs the actions are U_a and $aU_{1/a}$. On $\mathcal{B}$ the action is U_a . Let $W = \mathcal{L}_w$ carry a^w , where the source supplies $w = 0, 1$. Then on the displayed dual complex the action is

$$\ell \mapsto a^w \ell U_{1/a}, \quad (\alpha, \beta) \mapsto (a^w \alpha U_{1/a}, a^{w-1} \beta U_a).$$

The source identities $\Theta U_a = U_a \Theta$ and $\mathcal{F} U_a = aU_{1/a} \mathcal{F}$ verify directly that these operators commute with every differential and support arrow. On cohomology they give

$$H^{-1} : \lambda \mapsto a^w \lambda \bar{U}_{1/a}, \quad H^0 : c \mapsto a^{w-1} c U_a.$$

For $w = 1$, these are the original weight-one dual action on $Q^\vee$ and the explicit action $c \mapsto c U_a$ on the retained bottom $V^\vee$.

The roof above need not be strictly equivariant because Λ is not asserted equivariant. Its degree-zero component is equivariant. At the minus generator of degree -1 , its exact failure is

$$U_a^D f_y^{-1}(c) - f_y^{-1}(U_a^C c) = a^w c \mathcal{F}^{-1}(U_{1/a} \Lambda - \Lambda U_{1/a}).$$

Here the first $U_{1/a}$ in parentheses acts on V , and the second on $\mathcal{B}$. The formula follows by substitution and $U_a \mathcal{F}^{-1} = a \mathcal{F}^{-1} U_{1/a}$. Its precomposition with Θ is zero, using $\Lambda \Theta = 1$ and the theta intertwining. It consequently factors through the exact injection $q^\vee : Q^\vee \rightarrow \mathcal{B}^\vee$, with functional

$$[F] \mapsto a^w c \mathcal{F}^{-1} (U_{1/a} \Lambda - \Lambda U_{1/a}) F.$$

This is the computed relation between the original action, the chosen derived roof, and the retained $Q^\vee$ fibre. No equivariant splitting is inferred by erasing this term. In this statement each projective $P_v(C)$ carries the displayed coefficient action $c \mapsto a^{w-1} c U_a$ at every nonzero stalk, and its restriction maps are identities; its resolution maps therefore intertwine the actions.

The derived class of this defect is also computable. Applying $\text{Hom}(-, H)$ to the displayed projective resolution gives the complex

$$\text{Hom}_E(C, Q^\vee) \xrightarrow{\ell \mapsto (\ell, \ell)} \text{Hom}_E(C, Q^\vee)^2 \rightarrow 0.$$

The last term is zero because $H_t = 0$. Thus

$$\text{Ext}^1(Z_C, H) \cong \text{Hom}_E(C, Q^\vee), \quad [(u, v)] \mapsto v - u, \quad d \mapsto [(0, d)].$$

The two maps are inverse: their first composite is identity, and subtracting $(0, v - u)$ from (u, v) gives the diagonal pair (u, u) . The action defect of the chosen roof has zero plus component and minus component $q^\vee d_a$, where

$$d_a(c)([F]) = a^w c \mathcal{F}^{-1} (U_{1/a} \Lambda - \Lambda U_{1/a}) F.$$

It therefore has the exact Ext class d_a . Equivalently, a nullhomotopy would be a degree-1 map from $P_b(C)$ to D , represented by the same functional on both leaves; its degree-zero equation forces that functional to annihilate ΘV . The leaf equations require its two copies to equal $(0, q^\vee d_a)$. Such a map exists exactly when $d_a = 0$. This proves that this particular derived section intertwines the action exactly when the displayed class vanishes; it does not assert that no other equivariant section can exist.

The class is the transpose of the original global theta cocycle by an exact formula. Put

$$s[F] = (1 - \Theta \Lambda) F, \quad k_b = \Lambda U_b s.$$

Every F has its original decomposition $F = \Theta \Lambda F + s q F$. Substitution, with $\Lambda \Theta = 1$ and $U_b \Theta = \Theta U_b$, gives

$$U_b \Lambda - \Lambda U_b = -k_b q.$$

Consequently the same class is

$$d_a(c) = -a^w c \mathcal{F}^{-1} k_{1/a} : Q \rightarrow W.$$

The minus sign, Fourier inverse, reciprocal parameter, and coefficient action are retained. The reversed-support defect and the source's scaling boundary are thus connected by a proved morphism.

7. Continuity and the finite residue inclusion remain attached

For $W = E$, all the formulas also hold for continuous duals with their strong topologies. The maps $\Theta, \mathcal{F}, q$ and the source's explicit Λ are continuous, and $\Lambda \Theta = 1$ splits the original topological sequence. Thus the plus and minus lifts used above are continuous; all quotient and inverse maps are continuous on the displayed finite products. Precomposition by a continuous linear map is strongly continuous because it takes bounded subsets to bounded subsets. This verifies the asserted topology of every displayed dual transport. The derived roof assertion in Section 5 is in the declared algebraic diagram category; no undeclared derived category of all locally convex spaces is used.

The explicit inverse pair on strong duals is

$$\mathcal{B}'_b \longrightarrow V'_b \oplus Q'_b, \quad \ell \longmapsto (\ell\Theta, \ell s), \quad (\alpha, \lambda) \longmapsto \alpha\Lambda + \lambda q.$$

Its composites are identity by $\Lambda\Theta = 1$, $qs = 1$, $\Lambda s = 0$, $q\Theta = 0$, and $\Theta\Lambda + sq = 1$. Strong continuity follows directly from $p_B(T'\ell) = p_{T(B)}(\ell)$ for a bounded subset B and continuous linear T . In particular q' identifies Q'_b with the annihilator of ΘV , with continuous inverse $\ell \mapsto \ell s$. The joint relation subspace is the closed kernel of π_0 , and $c \mapsto [(0, c)]$ is its continuous quotient inverse.

For the original reflection-stable finite arithmetic packet A_Z , keep $g = 2\xi$, every actual multiplicity, the complete jet map $J_Z : Q \rightarrow A_Z$, and the full residue form

$$R_Z(f, h) = \sum_{\rho \in \mathbb{Z}} \text{Res}_{s=\rho} \frac{f^\dagger(s)h(s)}{g(s)} ds, \quad f^\dagger(s) = \overline{f(1-\bar{s})}.$$

The source proves its perfectness, J_Z 's surjectivity and continuity, and its covariance with $\mathcal{L}_1$. Therefore the map

$$\overline{A_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto ([F] \mapsto R_Z(f, J_Z[F]))$$

is the same injective continuous residue map at each active vertex of $H^{-1}D$. The active support arrows are identities, so its three naturality squares commute exactly; at the empty vertex both maps are zero. Every nilpotent remains in the full A_Z and residue pairing. The new bottom $H^0 = V'_b$, the action defect of Section 6, and the original arithmetic residue inclusion now coexist in one completely specified reversed support calculation.

For the continuous-jet assertion one can retain an explicit estimate. Write $r = \Re \rho$ and choose an integer $n > |r|$. Differentiating the Mellin integral and splitting it at one gives, for every $j \geq 0$,

$$|(\mathcal{M}F)^{(j)}(\rho)| \leq j! \left(\frac{p_{n,0}(F)}{(n-r)^{j+1}} + \frac{p_{-n,0}(F)}{(n+r)^{j+1}} \right).$$

The two integrals used here are respectively $\int_1^\infty x^{r-n}(\log x)^j dx/x = j!/(n-r)^{j+1}$ and $\int_0^1 x^{r+n}|\log x|^j dx/x = j!/(n+r)^{j+1}$. They follow from substitution $y = |\log x|$ and j integrations by parts. Thus every derivative in the finite jet is continuous, with its actual order retained. The map descends by the original quotient topology since it annihilates ΘV .

The finished result is the explicit dual diagram, its nonzero bottom degree-zero fibre, the precise strict-section obstruction, the constructed derived roof that survives that obstruction, and its full scaling defect. None of these computations proves a theta weight bound, removes a finite-prime trace term, or replaces A_τ by restriction to σ .

Appendix A: complete retraction on the original theta source

A.1. Original spaces and the fixed auxiliary functions

Work over $\mathbb{C}$, with Fourier convention

$$\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

Retain the original spaces, with their indicated seminorm topologies:

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \hat{\phi}(0) = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x.$$

Use Schwartz seminorms $\sigma_{m,j}(u) = \sup_{x \in \mathbb{R}} (1+|x|)^m |u^{(j)}(x)|$. The Fourier transform preserves the even Schwartz space and satisfies $\mathcal{F}^2 \phi(x) = \phi(-x)$. Consequently it preserves V , since $\hat{\phi}(0) = \int \phi$ and $\int \hat{\phi} = \phi(0)$.

Fix the following explicit smooth functions for the entire construction:

$$h(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \chi(x) = \frac{h(1/16 - x^2)}{h(1/16 - x^2) + h(x^2 - 1/64)}, \quad \alpha = \beta = \chi.$$

All derivatives of h at zero vanish: each derivative for $t > 0$ is a polynomial in $1/t$ times $e^{-1/t}$, which tends to zero. The denominator defining χ never vanishes, since its two arguments cannot both be nonpositive. Thus χ is real, even, smooth, lies in $[0, 1]$, equals one for $|x| \leq 1/8$, and vanishes for $|x| \geq 1/4$. In particular

$$\|\chi\|_2^2 \leq \frac{1}{2}.$$

These are choices of reconstruction data on the original coordinates. The spaces, Fourier convention and arithmetic theta map remain exactly those displayed here.

A.2. Theta continuity and recovery on both exterior regions

The original maps are

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1}F(1/x).$$

Poisson summation gives $\Theta\mathcal{F} = J\Theta$ on V . For this identity, periodize the Schwartz function $t \mapsto \phi(xt)$; its Fourier coefficients are $x^{-1}\hat{\phi}(n/x)$, obtained by integrating over all translated unit intervals. Both the periodization and its Fourier series converge absolutely with derivatives. Evaluation at zero gives $\sum_n \phi(nx) = x^{-1} \sum_n \hat{\phi}(n/x)$. The two zero-index terms vanish by the original moments.

The theta map is continuous into $\mathcal{B}$. Indeed, for $x \geq 1$, an integer $N > 1$, and $s_{N,k}(\phi) = \sup_{t>0} t^N |D^k \phi(t)|$,

$$|D^k \Theta\phi(x)| \leq 2\zeta(N) s_{N,k}(\phi) x^{-N}.$$

Each $s_{N,k}$ is bounded by finitely many Schwartz seminorms. Choosing $N \geq b$ bounds the part of $p_{b,k}$ on $x \geq 1$. The identities

$$D^k J = J(1 - D)^k, \quad p_{b,k}(JF) \leq \sum_{j=0}^k \binom{k}{j} p_{1-b,j}(F)$$

follow by differentiating $x^{-1}F(1/x)$. For $x \leq 1$, apply the preceding theta estimate to $\hat{\phi}$ at $1/x$; choosing also $N \geq 1 - b$ bounds this part by $2\zeta(N) \sum_{j=0}^k \binom{k}{j} s_{N,j}(\hat{\phi})$. This proves every required bound and also continuity of J .

For $x \neq 0$, define the actual arithmetic inverse series

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|).$$

Here μ is the ordinary Möbius function; the factor $1/2$ inverts the sum over both nonzero integer signs. Put

$$P_j(T) = (-1)^j T(T+1) \cdots (T+j-1), \quad P_0 = 1, \quad C_{N,j}(F) = \sum_{k=0}^j |[T^k] P_j(T)| p_{N,k}(F).$$

The identity $x^j \partial_x^j = P_j(D)$ follows by induction. Therefore, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{\zeta(N)}{2} C_{N,j}(F) x^{-N-j}. \quad (1)$$

The same absolute bound holds for negative x . It proves locally uniform convergence of every differentiated series away from zero.

For $\phi \in V$, the divisor rearrangement is legitimate: for fixed $x > 0$, $\sum_{m,n \geq 1} |\phi(mnx)|$ is finite, by a Schwartz bound of degree $N > 1$ and $\sum_{m,n} (mn)^{-N} = \zeta(N)^2$. Thus

$$\mathcal{U}\Theta\phi(x) = \sum_{k \geq 1} \left(\sum_{d|k} \mu(d) \right) \phi(kx) = \phi(x).$$

The divisor sum is one for $k = 1$ and zero otherwise: prime factorization gives the product $\prod_{p|k} (1 - 1)$ when $k > 1$. Evenness gives recovery on negative x , and Poisson summation gives the second exterior recovery:

$$\boxed{\mathcal{U}\Theta\phi = \phi, \quad \mathcal{U}J\Theta\phi = \hat{\phi} \quad (x \neq 0).} \quad (2)$$

In particular Θ is injective, including at zero by continuity. No information about zeros of zeta in a critical strip enters this proof.

A.3. Fixed contraction and full Schwartz estimates

On $L^2(\mathbb{R}, dx)$, set

$$A = M_\chi, \quad B = \mathcal{F}^{-1}M_\chi\mathcal{F}, \quad T = AB.$$

Its integral kernel is $\chi(x)\tilde{\chi}(x-y)$. Plancherel and the change of variable $z = x - y$ give

$$\|T\|^2 \leq \|T\|_{\text{HS}}^2 = \|\chi\|_2^2 \|\tilde{\chi}\|_2^2 = \|\chi\|_2^4 \leq \frac{1}{4}.$$

Consequently the operator-norm convergent series supplies the actual inverse:

$$\boxed{(1 - T)^{-1} = \sum_{n=0}^{\infty} T^n, \quad \|(1 - T)^{-1}\| \leq 2.} \quad (3)$$

Multiplying its partial sums by $1 - T$ leaves $1 - T^{n+1}$, which tends to the identity, proving both inverse identities.

Define exterior functions, extended by zero near the origin, by

$$a_F = (1 - \chi)\mathcal{U}F, \quad b_F = (1 - \chi)\mathcal{U}JF, \quad YF = a_F + \chi\mathcal{F}^{-1}b_F.$$

Here are explicit bounds verifying all their Schwartz seminorms. Write $c_0 = 1$ and $c_r = \|\chi^{(r)}\|_\infty$ for $r \geq 1$. Leibniz's rule and (1), with $N = m + 2$, give

$$\sigma_{m,j}(a_F) \leq \frac{\zeta(m+2)}{2} g^m \sum_{r=0}^j \binom{j}{r} c_r 8^{2+j-r} C_{m+2,j-r}(F). \quad (4)$$

Indeed every nonzero differentiated term occurs at $|x| \geq 1/8$, where $1 + |x| \leq 9|x|$ and $|x|^{-2-j+r} \leq 8^{2+j-r}$. The same bound holds for b_F with JF in place of F ; the displayed bounds for J express it in finitely many original $p_{b,k}(F)$. Smoothness across the plateau endpoints follows from smoothness of the cutoffs and their identically zero interior factors.

Fourier inversion is continuous on these Schwartz seminorms. Explicitly, integration by parts gives

$$\sigma_{m,j}(\mathcal{F}^{-1}u) \leq \sum_{r=0}^m \binom{m}{r} (2\pi)^{j-r} \|\partial_\xi^r(\xi^j u)\|_1.$$

Each integral on the right is bounded by a finite sum of Schwartz seminorms, using Leibniz's rule and $\int_{\mathbb{R}} (1 + |\xi|)^{-2} d\xi = 2$. Together with (4), this proves that $Y : \mathcal{B} \rightarrow \mathcal{S}^{\text{even}}$ is continuous.

The operator T also maps L^2 continuously to Schwartz space. Put

$$b_j = \|(2\pi\xi)^j \chi(\xi)\|_2 \leq \frac{(\pi/2)^j}{\sqrt{2}}, \quad A_{m,j} = (5/4)^m \sum_{r=0}^j \binom{j}{r} \|\chi^{(r)}\|_\infty b_{j-r}.$$

Cauchy–Schwarz in the Fourier integral bounds $\|\partial^j Bf\|_\infty$ by $b_j\|f\|_2$. Leibniz’s rule and the support of χ therefore give

$$\sigma_{m,j}(Tf) \leq A_{m,j}\|f\|_2.$$

Set $\Phi F = (1 - T)^{-1}YF$, initially in L^2 . The equation $\Phi F = YF + T\Phi F$ proves that it is Schwartz, and supplies the explicit continuity estimate

$$\sigma_{m,j}(\Phi F) \leq \sigma_{m,j}(YF) + 2A_{m,j}\|YF\|_2. \quad (5)$$

The final L^2 norm is bounded by $\sqrt{2}\sigma_{1,0}(YF)$. Every operator preserves parity; hence ΦF is even. Equations (2) give

$$Y\Theta\phi = (1 - A)\phi + A(1 - B)\phi = (1 - T)\phi, \quad \Phi\Theta\phi = \phi. \quad (6)$$

A.4. The original moment correction and exact splitting

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int_{\mathbb{R}} \phi)$ are $(1, 1)$ and $(0, 1)$. For the integral constants, the square of the Gaussian integral is the planar polar integral $2\pi \int_0^\infty r e^{-\pi r^2} dr = 1$. Integrating $\partial_x(xe^{-\pi x^2})$ then gives $2\pi \int x^2 e^{-\pi x^2} dx = 1$. Thus the continuous projection

$$P_V u = u - u(0)\gamma_0 - \left(\int_{\mathbb{R}} u - u(0) \right) \gamma_2$$

has both original moments zero and is identity on V . Continuity follows from $|u(0)| \leq \sigma_{0,0}(u)$ and $|\int u| \leq 2\sigma_{2,0}(u)$. The required map is therefore

$$\boxed{\Lambda = P_V(1 - T)^{-1}Y : \mathcal{B} \longrightarrow V, \quad \Lambda\Theta = 1_V.} \quad (7)$$

All constants and seminorm dependence have been supplied in (1)–(5). For the original quotient $q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V$, put $K = 1 - \Theta\Lambda$. Equation (7) gives $K^2 = K$, $\ker K = \Theta V$, and $\text{im } K = \ker \Lambda$. The image ΘV is closed because it is the kernel of continuous K . The continuous map K descends by the quotient topology to

$$s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0, \quad \Theta\Lambda + sq = 1.$$

These equations prove the continuous inverse pair

$$V \oplus Q \longrightarrow \mathcal{B}, \quad (\phi, u) \longmapsto \Theta\phi + su, \quad F \longmapsto (\Lambda F, qF).$$

Finally, retain $U_a F(x) = F(x/a)$. It is continuous because $p_{b,k}(U_a F) = a^b p_{b,k}(F)$; its source action preserves V and obeys $U_a \Theta = \Theta U_a$. It therefore induces $\bar{U}_a$ on Q . Put $k_a = \Lambda U_a s$. The decomposition of $U_a s$ gives

$$U_a s = \Theta k_a + s\bar{U}_a, \quad k_{ab} = U_a k_b + k_a \bar{U}_b, \quad k_1 = 0.$$

For the middle identity, substitute the first identity with parameter b into $\Lambda U_a U_b s$, and use $\Lambda U_a \Theta = U_a$ and $\Lambda U_a s = k_a$. Writing $F = \Theta\Lambda F + sqF$ also proves

$$U_a \Lambda - \Lambda U_a = -k_a q.$$

Consequently the reversed-support dual roof’s action defect is exactly

$$\boxed{d_a(c) = -a^w c \mathcal{F}^{-1} k_{1/a}.$$

This connects the completed retraction to the original action on the full Q , retaining its Fourier inverse, weight factor, reciprocal parameter and sign. The calculation does not set this cocycle to zero.

Chapter 4

The original theta scaling cocycle

Finite dilation of the original arithmetic packet over the tau base

0. Scope and exact source correspondence

This calculation continues the original two-leg theta complex. The retained infinitesimal identity is (A11) of `arithmetic_input.tex`; the absolute-base pushforward, its two-leg action, and its tensor differential are (19), (21), (34), and (46) of *Arithmetic cohomology over the tau-base*. The full global retraction and its cocycle were already constructed in transcript turn **A1694**, node `1fa0ebbc-f569-40c1-b7ac-b523da19a5f0`, chain position 3576, and in *Tau_Global_Comparison*, equations (16), (19), (29)–(35), and (41). Those results are retained as prior work, not reported as newly discovered here.

The additional calculation is the explicit finite-packet dilation cocycle for every positive real dilation, its full Taylor-unit and nilpotent coefficients, convergent source-space integral, incomplete-gamma evaluation, exact finite/global change of representatives, two-leg homotopies, and tensor and boundary formulas. These formulas perform the group-level continuation of the existing infinitesimal extension. They do not assign positivity to a representative pairing.

The source bytes used are pinned at the end. Every integral on the positive half-line keeps the displayed measure. In particular the Hilbert pairing below uses dx , and its dilation mass is a , not 1. The notation $t = \log a$ is a real time parameter; the polynomial coordinate in $\mathbb{C}[X]/(h)$ is written X to avoid confusing it with that time. Thus $A = M_X$ here is exactly the source operator denoted M_t .

1. The original spaces, maps, and topology

Work over $\mathbb{C}$. Let

$$V = \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \quad \phi(0) = 0, \quad \int_{\mathbb{R}} \phi(x) dx = 0 \right\},$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \quad (b \in \mathbb{Z}, k \geq 0) \right\}, \quad D = -x\partial_x.$$

Use the Schwartz seminorms

$$q_{N,k}(\phi) = \sup_{x \in \mathbb{R}} (1 + |x|)^N |\partial_x^k \phi(x)|, \quad N, k \geq 0.$$

The space V is a closed subspace of the complete Schwartz space: parity, value at zero, and integration are continuous. For example $|\int \phi| \leq q_{2,0}(\phi) \int_{\mathbb{R}} (1 + |x|)^{-2} dx$. The space $\mathcal{B}$ is complete because a Cauchy sequence in all $p_{b,k}$ converges with all derivatives on compact subsets after $x = e^y$; the weighted

uniform bounds pass to its limit. These observations justify limits of Riemann sums in the actual spaces below.

Retain

$$\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1} F(1/x),$$

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

Schwartz estimates, followed by the original Poisson identity, give

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta\hat{\phi} = J\Theta\phi, \quad D\Theta = \Theta D.$$

Here is the continuity argument. At $x \geq 1$, each $D^k\phi(nx)$ is bounded by a finite sum of Schwartz seminorms times $(nx)^{-M}$, where M can be any sufficiently large positive integer. Summing n^{-M} gives every required bound at infinity. The Fourier transform preserves Schwartz seminorms continuously, preserves evenness, and interchanges the two vanishing moments. Poisson summation therefore gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; applying D^k and the preceding bounds gives every required estimate at zero. Each resulting estimate involves finitely many source seminorms. Termwise differentiation is justified uniformly on every $x \geq \delta > 0$ by the same summable majorants.

The theta map is injective. For fixed $x > 0$, absolute convergence permits

$$\frac{1}{2} \sum_{m \geq 1} \mu(m) \Theta\phi(mx) = \sum_{r \geq 1} \left(\sum_{m|r} \mu(m) \right) \phi(rx) = \phi(x).$$

Indeed the rearranged absolute sum is bounded by $\sum_{r \geq 1} d(r) |\phi(rx)| < \infty$, since $d(r) \leq r$ and ϕ is Schwartz. Thus no theta relation is lost when its source is recovered.

For $a > 0$ define $U_a F(x) = F(x/a)$, and use the same formula on V . Direct substitutions give

$$p_{b,k}(U_a F) = a^b p_{b,k}(F), \quad q_{N,k}(U_a \phi) \leq a^{-k} \max(1, a)^N q_{N,k}(\phi), \quad (1)$$

$$\mathcal{M}U_a F(s) = a^s \mathcal{M}F(s), \quad \widehat{U_a \phi} = a U_{1/a} \hat{\phi}, \quad \Theta U_a = U_a \Theta. \quad (2)$$

The two moments of $U_a \phi$ are $\phi(0)$ and $a \int \phi$, so V is invariant. Differentiating the defining formula gives

$$\frac{d}{dt} U_{e^t} \phi = D U_{e^t} \phi = U_{e^t} D \phi.$$

This is a derivative in every Schwartz seminorm. To verify it, apply the ordinary Taylor formula with integral remainder to $U_{e^{t+h}} \phi$; its second derivative is $U_{e^{t+h}} D^2 \phi$, uniformly bounded on each compact time interval by (1). The same argument, using the exact first identity in (1), proves smoothness in $\mathcal{B}$. Repeating with $D^j \phi$ proves derivatives of all orders. In particular this is a smooth group action on the retained Frechet spaces.

The four-point source poset is

$$+ < \eta < \sigma, \quad - < \eta < \sigma.$$

For the sheaf with stalks $V, V, \mathcal{B}, 0$ and restriction maps $\Theta, \Theta\mathcal{F}$, the source projective resolution computes

$$C = A_\tau(\mathcal{T}) = \left[V \oplus V \xrightarrow{d} \mathcal{B} \right], \quad d(v, w) = \Theta(v - \widehat{w}), \quad |V \oplus V| = 0, \quad |\mathcal{B}| = 1. \quad (3)$$

Its actual chain action is

$$\mathcal{U}_a^0(v, w) = (U_a v, aU_{1/a} w), \quad \mathcal{U}_a^1 F = U_a F. \quad (4)$$

Equation (2) proves $d\mathcal{U}_a^0 = \mathcal{U}_a^1 d$. On even Schwartz functions $\mathcal{F}^2 = 1$, so the exact degree-zero cycle inclusion is

$$i_0(w) = (\widehat{w}, w), \quad \mathcal{U}_a i_0(w) = i_0(aU_{1/a} w). \quad (5)$$

It follows from theta injectivity that $H^0(C) = i_0(V)$ and $H^1(C) = Q = \mathcal{B}/\Theta V$. These are the original global cohomology groups over the absolute tau base.

2. The finite arithmetic packet and its unchanged Taylor unit

Retain the original Gaussian source

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta \phi_*.$$

It has value zero at zero, and direct Gaussian integration gives zero integral. More precisely, with $M_+ \phi(s) = \int_0^\infty \phi(x) x^s dx/x$,

$$M_+ \phi_*(s) = \pi^{-s/2} (2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (6)$$

Consequently the entire function $g = \mathcal{M} f_0$ agrees, first by absolutely convergent integration in a right half-plane and then by continuation, with the original function

$$g(s) = 2\xi(s) = s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

Entireness of the Mellin transform of $f_0 \in \mathcal{B}$ follows directly from the bounds in $\mathcal{B}$. For $0 < \operatorname{Re} s < 1$, set

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} M_+ \phi(s).$$

Then

$$\mathcal{M} \Theta \phi = g H_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s). \quad (7)$$

For completeness, $M_+ \phi$ is holomorphic on $\operatorname{Re} s > -2$, since smooth evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$. In $\operatorname{Re} s > 1$, interchanging its theta series and Mellin integral gives $\mathcal{M} \Theta \phi = 2\zeta(s) M_+ \phi(s)$. Continue to the indicated strip to obtain the first equality in (7). Integrating $D\phi$ by parts proves $H_{D\phi} = s H_\phi$; the endpoint term $[x^s \phi(x)]_0^\infty$ vanishes on that strip. Equation (6) then proves the other two equalities.

Fix the original nonempty finite packet Z of actual nontrivial zeros, each with its complete order m_ρ , and put

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E = \mathbb{C}[X]/(h), \quad A = M_X,$$

$$v(s) = g(s)/h(s), \quad \varepsilon = j_h(h/g) \in E^\times.$$

The quotient h/g here is taken as a germ at each selected zero; these germs are holomorphic and nonzero because the complete multiplicity has been retained. The local unit is **not** replaced by 1. For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree less than d_h representing εu , and put

$$\ell(u) = [X^{d_h-1}]R_Z(u). \quad (8)$$

We record the inverse used to define the source section, rather than taking an unproved inverse as an extra hypothesis. On $F \in \mathcal{B}$ satisfying $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Differentiation gives $(D - \rho)S_\rho F = F$. The moment condition makes the upper integral the negative lower integral. If $\sigma = \operatorname{Re} \rho$, the upper estimate, for $N > \sigma$, is

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \sigma} x^{-N}.$$

The lower estimate, for $N + \sigma > 0$, is

$$|S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \sigma} x^N.$$

Using the upper estimate on $x \geq 1$ and the lower estimate on $x \leq 1$ gives every weighted bound. The differential identity expresses $D^k S_\rho F$ as $\rho^k S_\rho F$ plus a finite linear combination of $D^j F$, proving all derivative bounds and continuity. The homogeneous solution $cx^{-\rho}$ cannot belong to $\mathcal{B}$ unless $c = 0$, so the inverse is unique. Integration by parts gives

$$\mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho},$$

with the removable value filled in. Iterating this construction, removing one vanishing factor at a time, defines the continuous inverse S_h of $h(D)$ on $\ker(j_h \mathcal{M})$. The operation preserves the remaining zeros and lowers the order at the chosen zero by exactly one at each step. Conversely any $h(D)F$ has those vanishing jets. This proves

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D) \text{ is injective.} \quad (9)$$

The section in the question is therefore the actual well-defined map

$$R = R_{\text{ref}} : E \longrightarrow \mathcal{B}, \quad Ru = S_h \Theta(R_Z(u)(D)\phi_*). \quad (10)$$

Indeed its input has Mellin transform $gR_Z(u)$, which vanishes to all the required orders. Equations (7) and (9) give

$$\mathcal{M}Ru = vR_Z(u), \quad J_Z R = 1_E, \quad J_Z = j_h \mathcal{M}, \quad J_Z \Theta = 0. \quad (11)$$

The Mellin-jet map is continuous: split its integral at 1, dominate the logarithmic factors at zero by a sufficiently negative weight and those at infinity by a sufficiently positive weight. This yields a finite sum of seminorm bounds for each derivative at each ρ . In (11), $j_h(v)\varepsilon = 1$ proves the middle identity. Hence R and $\sigma = qR : E \rightarrow Q$ are injective, since J_Z descends to Q .

The infinitesimal boundary is precisely the retained one:

$$DR - RA = \Theta \phi_* \ell = f_0 \ell. \quad (12)$$

To prove it without losing its coefficient, the polynomial $XR_Z(u) - R_Z(Au)$ vanishes modulo h and has degree at most d_h . Since h is monic, its quotient by h is exactly $\ell(u)$. Apply $h(D)$ to (12), use (10), $D\Theta = \Theta D$, and that polynomial identity. Injectivity of $h(D)$ from (9) proves (12). For a single simple zero, $R_Z(u) = u/g'(\rho)$ and $\ell(u) = u/(2\xi'(\rho))$, with the original factor of two.

3. The finite dilation cocycle as an actual convergent source integral

For every real t , define $T_t = e^{tA}$ and

$$c_t(u) = \int_0^t U_{e^{t-r}} \phi_* \ell(T_r u) dr \in V. \quad (13)$$

If $t < 0$, $\int_0^t = -\int_t^0$; no orientation is changed. Write $c_a = c_{\log a}$ and $a^A = T_{\log a}$ when using the multiplicative parameter.

Convergence and continuity. Fix any norm on the finite-dimensional coefficient space, solely for estimates, and its induced operator norms. On the oriented interval between 0 and t , both $|r|$ and $|t - r|$ are at most $|t|$. Equation (1) and $\|T_r\| \leq e^{|r|\|A\|}$ yield

$$q_{N,k}(c_t u) \leq |t| e^{(N+k+\|A\|)|t|} q_{N,k}(\phi_*) \|\ell\| \|u\|. \quad (14)$$

The integrand is continuous in every seminorm. Its Riemann sums are Cauchy in every seminorm by uniform continuity on the compact interval. Completeness of V proves their common limit exists in V , and the inequality follows first for sums and then for their limit. Thus the integral is an actual source vector with the two original zero moments and even Schwartz regularity. It is not merely an integral in a weak dual or in L^2 .

Smoothness in t follows either by the differentiable compact-interval rule justified by the same estimates for all time derivatives, or by writing (13) as $t \int_0^1 U_{e^{t(1-r)}} \phi_* \ell(T_{tr} u) dr$. This latter formula works without a sign convention change for negative t . Every derivative has a finite sum of terms involving $D^j \phi_*$, powers of A , and bounded polynomial factors in r . Inequality (14) with those finitely many seminorms proves local uniform convergence of the differentiated Riemann sums. Therefore $t \mapsto c_t$ is smooth as a map to continuous linear maps $E \rightarrow V$, with their bounded-set topology. Finite dimensionality of E makes the preceding uniform estimates on its unit ball sufficient for this statement.

The exact dilation identity. For all $t \in \mathbb{R}$,

$$\boxed{U_{e^t} R - R T_t = \Theta c_t.} \quad (15)$$

For fixed u , differentiate in the Frechet space $\mathcal{B}$:

$$\frac{d}{dr} (U_{e^{t-r}} R T_r u) = -U_{e^{t-r}} (DR - RA) T_r u = -\Theta U_{e^{t-r}} \phi_* \ell(T_r u).$$

The fundamental theorem of calculus follows in every defining seminorm from the same Riemann-sum argument; integrating with the indicated orientation proves (15), including $t < 0$. Continuity of Θ permits moving it through the integral.

Cocycle and derivative. Directly splitting the defect in (15) gives

$$\boxed{c_{t+s} = U_{e^t} c_s + c_t T_s, \quad c_0 = 0, \quad c_{-t} = -U_{e^{-t}} c_t T_{-t}.} \quad (16)$$

Indeed applying Θ to the asserted first equality yields the identity obtained by inserting $U_{e^t}RT_s$ in $U_{e^{t+s}}R - RT_{t+s}$. Injectivity of Θ proves equality in the actual source V . The other two statements follow by setting $t = s = 0$ and $s = -t$. Differentiating (13), or the cocycle in either variable, gives the two equal expressions

$$\boxed{\partial_t c_t = Dc_t + \phi_* \ell T_t = U_{e^t} \phi_* \ell + c_t A, \quad \partial_t c_t|_{t=0} = \phi_* \ell.} \quad (17)$$

All derivatives are in V . In multiplicative notation (16) is

$$c_{ab} = U_a \circ c_b + c_a \circ b^A,$$

where the final product means the map $c_a \circ b^A$; to remove any typographical ambiguity it can also be written $c_{ab}(u) = U_a(c_b(u)) + c_a(b^A u)$.

The space $\mathcal{B}_Z = \Theta V + RE$ is a direct topological sum with coordinates $(\phi, u) \mapsto \Theta\phi + Ru$. Its inverse on that subspace is $(\Theta^{-1}(F - RJ_Z F), J_Z F)$; continuity of the inverse theta map is also proved in Section 6 below. Equations (15) and (16) integrate the source block operator to the exact group representation

$$U_a|_{\mathcal{B}_Z} = \begin{pmatrix} U_a|_V & c_a \\ 0 & a^A \end{pmatrix}. \quad (18)$$

This is the explicit group-level form of the already known extension (A12). In particular (15) proves the true quotient intertwining $\bar{U}_a \sigma = \sigma a^A$.

4. All nilpotents, the Mellin multiplier, and an incomplete-gamma evaluation

The Chinese remainder decomposition is

$$E = \bigoplus_{\rho \in Z} \mathbb{C}[z_\rho]/(z_\rho^{m_\rho}), \quad A|_\rho = \rho I + N_\rho,$$

where $N_\rho z_\rho^j = z_\rho^{j+1}$ and $N_\rho^{m_\rho} = 0$. Denote the summand projector by P_ρ , extending N_ρ by zero on other summands. These are the actual full local algebras. They can be constructed explicitly: if $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$, then P_ρ is multiplication by the remainder modulo h of

$$H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{(H_\rho^{-1})^{(j)}(\rho)}{j!} (X - \rho)^j.$$

At every other centre this has all required zero jets; at ρ it has jet 1. This verifies all its projector identities by the Chinese remainder map.

The action and the entire boundary integral therefore contain the full expressions

$$T_t = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho, \quad (19)$$

$$c_t u = \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} \ell(N_\rho^j P_\rho u) \int_0^t U_{e^{t-r}} \phi_* e^{r\rho} \frac{r^j}{j!} dr. \quad (20)$$

The unit entering every coefficient is still $\varepsilon = j_h(h/g)$ through (8). If the packet has one zero of order m , and $\varepsilon = \sum_{i < m} \varepsilon_i z^i$, $u = \sum_{i < m} u_i z^i$, then $\ell(u) = \sum_{i=0}^{m-1} \varepsilon_i u_{m-1-i}$. This follows because the top coefficient of a polynomial of degree less than m is the same in the X and $z = X - \rho$ bases. Thus even

the functional ℓ retains the lower coefficients of the Taylor unit, rather than only its nonzero constant value.

There is a closed expression for the integral kernels in (20). For $x > 0$ let

$$K_{\rho,t}(x) = \int_0^t e^{r\rho} \phi_*(xe^{r-t}) dr.$$

Define the oriented incomplete-gamma interval by

$$\Gamma_{[L,H]}(w) = \int_L^H y^{w-1} e^{-y} dy \quad (L, H > 0),$$

where $y^{w-1} = \exp((w-1)\log y)$ uses the real logarithm. Then substitution first of $y = xe^{r-t}$ and then of $z = \pi y^2$ gives

$$K_{\rho,t}(x) = e^{t\rho} x^{-\rho} \pi^{-\rho/2} \left[2\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 2) - 3\Gamma_{[\pi x^2 e^{-2t}, \pi x^2]}(\rho/2 + 1) \right]. \quad (21)$$

Both substitutions preserve orientation when $t < 0$. The coefficients 2 and -3 come respectively from $4\pi^2 x^4$ and $-6\pi x^2$, including $dy/y = dz/(2z)$. No Gaussian constant is removed.

Differentiation under the compact oriented r -integral now gives the finite formula

$$c_t u = \sum_{\rho \in \mathbb{Z}} \sum_{j=0}^{m_\rho-1} \frac{\ell(N_\rho^j P_\rho u)}{j!} \partial_\rho^j K_{\rho,t}. \quad (22)$$

The derivative here acts on the complex parameter in the entire kernel $K_{\rho,t}$ before evaluation at the indicated centre, not on the arithmetic zero or on the coefficient $\ell(N_\rho^j P_\rho u)$. The formula on $x > 0$ is extended evenly to $\mathbb{R}$, with its smooth value at 0 supplied by the integral defining K . That integral and Section 3 prove smoothness and both zero moments, so no regularity is inferred from cancelling the separate singular-looking factors in (21).

The full Mellin multiplier of the source boundary is also explicit:

$$H_{c_t u}(s) = \int_0^t e^{(t-r)s} \ell(T_r u) dr = \frac{e^{ts} R_Z(u)(s) - R_Z(T_t u)(s)}{h(s)}. \quad (23)$$

For the first equality in the original strip, use (7), $H_{U_a \phi_*} = a^s$, and the continuous source integral. For the second, differentiate $e^{(t-r)s} R_Z(T_r u)(s)$ in r . The polynomial identity proving (12) says its derivative is $-e^{(t-r)s} h(s) \ell(T_r u)$. Integration proves (23) wherever $h(s) \neq 0$. The numerator has every required vanishing jet: in the local algebra its jet is $e^{tA} \varepsilon u - \varepsilon T_t u = 0$. Hence the quotient in (23) has removable singularities at every chosen zero and is entire. This entire expression extends the original source multiplier; it is not a replacement of h or g .

For every integer $n \geq 0$ its actual Taylor coefficient at any complex centre s_0 is

$$\frac{H_{c_t u}^{(n)}(s_0)}{n!} = \int_0^t e^{(t-r)s_0} \frac{(t-r)^n}{n!} \ell(T_r u) dr. \quad (24)$$

Inserting (19) gives every mixed coefficient in $(t-r)^n r^j / (n! j!)$ and keeps all nilpotent terms. This is also a direct convergence proof for every jet, since the parameter interval is compact.

5. Actual two-leg homotopies and the support-face comparison

Regard $E[-1]$ as the complex with E in degree 1 and zero elsewhere. The finite section defines a degree-zero cochain map $r : E[-1] \rightarrow C$, with $r^1 = R$. The maps $\mathcal{U}_a r$ and ra^A have homotopies

$$\begin{aligned} H_a^+ : E[-1] &\longrightarrow C[-1], & (H_a^+)^1 u &= (c_a u, 0), \\ H_a^- : E[-1] &\longrightarrow C[-1], & (H_a^-)^1 u &= (0, -\widehat{c_a u}). \end{aligned} \quad (25)$$

Here the notation records maps of cohomological degree -1 ; the source only has degree 1. In ordinary component notation the exact identity is

$$\boxed{\mathcal{U}_a r - ra^A = dH_a^\pm + H_a^\pm d_{E[-1]} = dH_a^\pm.} \quad (26)$$

Indeed $d(c_a u, 0) = \Theta c_a u$, whereas $d(0, -\widehat{c_a u}) = \Theta \mathcal{F}^2 c_a u = \Theta c_a u$. This verifies the minus sign using the original differential. Equation (2) also gives the homotopy cocycle itself:

$$H_{ab}^\pm = \mathcal{U}_a^0 H_b^\pm + H_a^\pm b^A. \quad (27)$$

No extra Fourier scalar is inserted: $\widehat{U_a c} = aU_{1/a} \widehat{c}$ is exactly the second-leg action in (4).

The original support masks are $\lambda \in \{\{+\}, \{-\}, \{+, -\}\}$. The fibre C_λ^0 consists of the legs present in λ , $C_\lambda^1 = \mathcal{B}$, with the same differential. The $+$ homotopy in (25) is defined on $\{+\}$ and on the joint face. The $-$ homotopy is defined on $\{-\}$ and on the joint face. On the joint face their difference is the precisely retained cycle

$$H_a^+ - H_a^- = (c_a, \widehat{c_a}) = i_0(\widehat{c_a}). \quad (28)$$

It lies in $H^0(C)$ by (5) and theta injectivity. There is no term in degree -1 of C , so a nonzero such cycle is not a boundary in C .

The two singleton homotopies are forced. On $\{+\}$, injectivity of theta forces the first one; on $\{-\}$ it forces the second. Consequently a single homotopy that is strictly natural for both singleton-to-joint inclusions would require $(c_a, 0) = (0, -\widehat{c_a})$, which is equivalent to $c_a = 0$. This is the exact scope of the support-naturality obstruction; it does not remove the homotopies in (25), their difference (28), or the resulting quotient action.

For completeness, c_a as a linear map is nonzero for every $a \neq 1$. If $c_a = 0$, (15) would make the nonzero finite-dimensional subspace RE invariant under U_a , with invertible action a^A . It therefore has a nonzero eigenvector $F \in \mathcal{B}$ with $U_a F = \lambda F$, $\lambda \neq 0$. Replacing a by a^{-1} if needed, take $a > 1$. Choose $x_0 > 0$ with $F(x_0) \neq 0$. Iteration gives $F(a^n x_0) = \lambda^{-n} F(x_0)$ for $n \geq 0$. The bound $p_{N,0}(F) < \infty$ implies $a^{nN} |\lambda|^{-n} |F(x_0)| \leq x_0^{-N} p_{N,0}(F)$. Choose N with $a^N > |\lambda|$ and let $n \rightarrow \infty$ to obtain a contradiction. Thus no such finite invariant subspace exists. This extends the previously proved failure of a strict equivariant section by giving its exact nonzero finite-dilation defect; the failure itself was already present in (A12).

At a fixed admitted mask the split lift of every displayed map is

$$\tilde{f}(\lambda, u) = (\lambda, f(u)), \quad \tilde{f}(\tau) = \tau.$$

An amplitude $f(u) = 0$ has value $(\lambda, 0)$, the supported zero, including at $a = 1$. All homotopy identities are interpreted using addition and the supported scalar $(-1)^\bullet$ in that fibre. No absent leg is inserted by the face-specific formulas.

The known synchronization comparison can be retained explicitly when a common joint target is desired. Send every active degree-one mask to $(\{+, -\}, F)$ and send the degree-zero pair to its joint-supported pair with the same two amplitudes, putting a supported zero in a formerly missing coordinate. Denote this operation by $\mathfrak{r}$. It is the source map $(a, b) \mapsto (a + eb, b + ea)$, because multiplication by e gives a supported zero at a present coordinate. It preserves the differential, fixes joint masks, and is idempotent. The pair $(\text{supp } x, \mathfrak{r}x)$ recovers the original element by selecting the originally present coordinates, proving injectivity of that comparison. After this explicit synchronization, (25)–(28) hold in the common joint fibre. The original mask must stay attached to the comparison; forgetting it is not part of the homotopy calculation.

One can also see the extra degree-zero group in an exact equivariant roof. Let $C_Z = [V \oplus V \rightarrow \mathcal{B}_Z]$ with the same differential, and define

$$p_Z^0(v, w) = w, \quad p_Z^1 = J_Z.$$

This is a cochain map to $V[0] \oplus E[-1]$. It is equivariant for the action $aU_{1/a}$ on V and a^4 on E , and induces isomorphisms on both cohomology groups: the degree-zero assertion follows from (5), while the degree-one assertion follows from the direct sum $\mathcal{B}_Z = \Theta V \oplus RE$ and (11). Thus the exact two-leg version of the source roof is

$$V[0] \oplus E[-1] \xleftarrow{p_Z} C_Z \longrightarrow C.$$

Its ordinary cochain section is $i_Z^0(w) = (\widehat{w}, w)$ and $i_Z^1 = R$. If $F = \Theta\phi + Ru$, the map $h_Z^1(F) = (\phi, 0)$ obeys $1 - i_Z p_Z = dh_Z + h_Z d$: in degree zero it returns $(v - \widehat{w}, 0)$ and in degree one it returns $\Theta\phi$. Its failure of equivariance in degree one is exactly (25). This proves the replacement on the entire original two-leg complex, retaining its $V[0]$ summand.

The transported minus face has an explicitly nonzero extension component

The preceding roof also computes what happens to the original support arrows when the action is retained. Work over $\mathbb{C}[X]$, with $X = D$ on $\mathcal{B}_Z$, $X = A$ on E , and $X = 1 - D$ on the degree-zero second-leg module, denoted V_- . Let $r(X) = R_Z(1)$ and $\psi = r(D)\phi_*$. Then

$$h(D)R(1) = \Theta\psi.$$

The class of ψ in $V/h(D)V$ is nonzero: applying $j_h H$ gives $j_h r = \varepsilon \neq 0$, whereas $j_h H$ vanishes on $h(D)V$ by (7). The Fourier map carries this quotient isomorphically onto $V/h(1 - D)V$, since differentiating (2) gives $\mathcal{F}D = (1 - D)\mathcal{F}$.

For the minus-face extension the injection is $i_- = -\Theta\mathcal{F}$. Its extension representative, with the convention $h(D)R(1) = i_-(w)$, is therefore

$$w = -\widehat{\psi} \quad \text{in } V/h(1 - D)V.$$

The component of the minus support arrow in the $V_-[0]$ summand of the joint roof has the opposite sign to this extension representative. Here is the exact shifted-resolution calculation. The free cochain resolution of E in degrees $-1, 0$ has differential h . We use $(C[n])^j = C^{j+n}$ and $d_{C[n]} = (-1)^n d_C$; after shifting by -1 , that resolution is in degrees $0, 1$ with differential $-h$. Its degree-one lift to the minus-face complex sends 1 to $R(1)$, extended $\mathbb{C}[X]$ -linearly. Its degree-zero lift must send 1 to $-w$, because $i_-(-w) = -h(D)R(1)$. Projection to $V_-[0]$ is identity in degree zero. Thus the induced derived map $E[-1] \rightarrow V_-[0]$ represents exactly

$$[-w] = [\widehat{\psi}] \quad \text{in} \quad \text{Ext}_{\mathbb{C}[X]}^1(E, V_-) \cong V/h(1 - D)V. \quad (28a)$$

The isomorphism here follows by applying $\mathrm{Hom}_{\mathbb{C}[X]}(-, V_-)$ to that two-term free resolution; its cokernel is the stated quotient. The class (28a) is nonzero by the preceding source-jet calculation. The same arrow has identity as its $E[-1]$ component. By contrast, the plus support arrow has zero $V_-[0]$ component and identity $E[-1]$ component, because its degree-zero image has second coordinate zero. Consequently the two support arrows in the equivariant derived category do not both become inclusion of the $E[-1]$ summand: the exact difference carries (28a), with the original Taylor unit. The strict original cochain support diagram remains available before passage through the roof.

6. The exact bridge to the already constructed global section

We recall the global section in a form that makes its use here mathematically explicit. Choose the same fixed even real cutoffs $\alpha, \beta \in C_c^\infty(\mathbb{R})$, with $0 \leq \alpha, \beta \leq 1$, equal to 1 near zero. For $x \neq 0$ define

$$\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|).$$

If $P_j(D) = (-1)^j D(D+1) \cdots (D+j-1) = x^j \partial_x^j$, and $C_{N,j}(F)$ is the sum of $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then for $x > 0$, $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (29)$$

Each summand is bounded by $\frac{1}{2} C_{N,j}(F) n^{-N} x^{-N-j}$; summation proves the estimate and local uniform convergence of every derivative. Thus the cutoff exterior functions $(1-\alpha)\mathcal{U}F$ and $(1-\beta)\mathcal{U}JF$, extended by zero near zero, are continuous Schwartz-valued maps. The theta inversion in Section 1 and Poisson give $\mathcal{U}\Theta\phi = \phi$ and $\mathcal{U}J\Theta\phi = \hat{\phi}$ away from zero.

On $L^2(\mathbb{R}, dx)$ define

$$T = M_\alpha \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad YF = (1-\alpha)\mathcal{U}F + \alpha \mathcal{F}^{-1}((1-\beta)\mathcal{U}JF).$$

The kernel of T is $\alpha(x)\check{\beta}(x-y)$; its squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$, so T is compact. Its norm c is strictly less than 1. Otherwise compactness would give a unit vector attaining norm 1. Equality in the two contraction inequalities forces its Fourier transform to be supported where $\beta = 1$, and then the vector itself to be supported where $\alpha = 1$. A compactly supported Fourier transform in L^2 is in L^1 , and its inverse Fourier transform extends to an entire function by its finite-interval integral. That entire function would vanish on a real open interval because of the compact spatial support, hence vanish identically, a contradiction.

Thus $(1-T)^{-1} = \sum_{n \geq 0} T^n$ in operator norm, with bound $(1-c)^{-1}$. The operator $T : L^2 \rightarrow \mathcal{S}$ is continuous: Cauchy–Schwarz on the compact Fourier support bounds every derivative of $\mathcal{F}^{-1}(\beta\hat{f})$ by a constant times $\|f\|_2$, and multiplication by α gives compact spatial support and every Schwartz seminorm. It follows that $\Phi = (1-T)^{-1}Y$ is Schwartz-valued and continuous; the explicit identity $\Phi = Y + T\Phi$ gives

$$q_{N,k}(\Phi F) \leq q_{N,k}(YF) + \frac{C_{N,k}}{1-c} \|YF\|_2.$$

No uniform lower bound for $1-c$ as cutoffs vary is used. On a theta input, $Y\Theta\phi = (1-T)\phi$, so $\Phi\Theta\phi = \phi$. The range of Φ is the even Schwartz space $\mathcal{S}(\mathbb{R})^{\text{even}}$. To check this exact domain, $\mathcal{U}F$ is even because its defining series uses $|x|$, and the same is true of $\mathcal{U}JF$. The cutoffs α, β are even, and Fourier transformation and inverse Fourier transformation preserve evenness, so YF is even. Reflection $\phi(x) \mapsto \phi(-x)$ commutes with T ; hence it commutes with every partial sum of the norm-convergent inverse $(1-T)^{-1}$ and with its limit. Thus ΦF is even as an L^2 vector, and its Schwartz representative

is even pointwise by continuity. With the original Gaussian moment corrections, define the map on this precise domain by

$$P_V \phi = \phi - \phi(0)e^{-\pi x^2} - \left(\int_{\mathbb{R}} \phi - \phi(0) \right) 2\pi x^2 e^{-\pi x^2}, \quad \Lambda = P_V \Phi.$$

The moment columns of the two displayed Gaussians are $(1, 1)$ and $(0, 1)$ by direct Gaussian integration. The two subtractions therefore set the value at zero and the integral to zero, while preserving evenness. On V both correction coefficients vanish. Consequently $P_V : \mathcal{S}(\mathbb{R})^{\text{even}} \rightarrow V$ is a continuous projection and $P_V|_V = 1$. This is its stated domain; no projection of the entire Schwartz space onto the even subspace is asserted by that formula. It follows that

$$\Lambda\Theta = 1_V, \quad K = 1 - \Theta\Lambda, \quad s : Q \rightarrow \mathcal{B}, \quad s[F] = KF. \quad (30)$$

This proves all needed existence and continuity claims for the particular global construction used in A1694. Since $\ker K = \Theta V$, that image is closed. The expression for s is well-defined, $qs = 1$, $\Lambda s = 0$, and it is continuous by the definition of the quotient topology. In particular Θ^{-1} on its image is Λ , verifying the topological inverse asserted after (17).

The already existing global cocycle is $k_a = \Lambda U_a s$, with

$$U_a s - s\bar{U}_a = \Theta k_a.$$

This follows by applying $1 = K + \Theta\Lambda$ to $U_a s$. The finite/global comparison is now fully determined, with no new choice of arithmetic unit:

$$b_Z = \Lambda R : E \rightarrow V, \quad R^s = s\sigma = R - \Theta b_Z. \quad (31)$$

Define $c_a^s = k_a \sigma$. Substitution in (15) yields

$$\boxed{c_a^s = c_a - U_a b_Z + b_Z a^A.} \quad (32)$$

To verify the equality in V , use $\bar{U}_a \sigma = \sigma a^A$ from (18) to compute

$$\begin{aligned} U_a R^s - R^s a^A &= U_a R - R a^A - \Theta U_a b_Z + \Theta b_Z a^A \\ &= \Theta(c_a - U_a b_Z + b_Z a^A) = \Theta k_a \sigma. \end{aligned}$$

Theta injectivity proves (32). Its derivative is the exact infinitesimal bridge

$$\boxed{k_D \sigma = \phi_* \ell - D b_Z + b_Z A, \quad k_D = \Lambda D s.} \quad (33)$$

All terms are defined continuous maps; differentiation is allowed by Section 1. This proves that the explicitly evaluated Duhamel integral and the transcript's global continuous cocycle represent the same actual arithmetic extension, through the displayed coboundary.

The full extension class is unchanged. Applying $h(D)$ to (31) gives

$$h(D)R^s(u) = \Theta(R_Z(u)(D)\phi_* - h(D)b_Z u).$$

Modulo $h(D)V$, its source class equals that of $R_Z(u)(D)\phi_*$. Under $j_h H$ its value is exactly εu , by (7). There is no quotient operation that discards the nonconstant coefficients of ε .

At cochain level the two finite representative maps r^s and r have the difference

$$r^s - r = dL^\pm, \quad L^+(u) = (-b_Z u, 0), \quad L^-(u) = (0, \widehat{b_Z u}). \quad (34)$$

The minus-face sign follows by applying $d(0, \hat{b}) = -\Theta b$. Equations (25), (32), and (34) give the exact homotopy comparison

$$H_a^{s,\pm} = H_a^\pm + \mathcal{U}_a^0 L^\pm - L^\pm a^A.$$

These identities retain the joint degree-zero difference as in (28), rather than asserting strict agreement of the two face homotopies.

7. Tensor powers with every cohomological sign and jet coefficient

Let $k \geq 1$. On the original ordered tensor complex $C^{\otimes k}$ use

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_{j=1}^k (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k. \quad (35)$$

This is exactly the tau-base product resolution differential. It squares to zero because the two orders of applying differentials in different factors have opposite signs; a repeated differential in one factor is zero.

Write $f_a = U_a R : E \rightarrow C^1$, $g_a = R a^A : E \rightarrow C^1$, and $h_a = H_a^\pm : E \rightarrow C^0$ for any admitted choice of face in each factor. The explicit degree-1 homotopy on the top packet is

$$\mathcal{H}_{a,k}(u_1 \otimes \cdots \otimes u_k) = \sum_{j=1}^k (-1)^{j-1} f_a u_1 \otimes \cdots \otimes f_a u_{j-1} \otimes h_a u_j \otimes g_a u_{j+1} \otimes \cdots \otimes g_a u_k. \quad (36)$$

In this evaluated formula the factor $(-1)^{j-1}$ is displayed exactly once. If the same formula is instead written as a tensor of graded linear maps, it is the Koszul evaluation sign of the degree-1 map in slot j crossing the preceding $j-1$ inputs, each of degree 1; it must not be inserted a second time.

Only the slot containing $h_a u_j$ contributes to the differential, because the other slots are already in top degree. Its differential sign from (35) is $(-1)^{j-1}$, which cancels the prefactor. Since $dh_a = f_a - g_a$, the result is the telescoping identity

$$\boxed{d\mathcal{H}_{a,k} = (U_a R)^{\otimes k} - (R a^A)^{\otimes k}.} \quad (37)$$

For $k=2$ this reads $\mathcal{H}_{a,2} = h_a \otimes g_a - f_a \otimes h_a$ on evaluated inputs, and its differential is $(f_a - g_a) \otimes g_a + f_a \otimes (f_a - g_a)$. This displays the original minus sign without relying on an implicit convention.

Define

$$R_k = R^{\otimes k}, \quad A_k = \sum_{j=1}^k 1^{\otimes(j-1)} \otimes A \otimes 1^{\otimes(k-j)}, \quad T_{t,k} = T_t^{\otimes k} = e^{tA_k}.$$

For the ordered local block $(\rho_1, \dots, \rho_k)$ the action is exactly

$$T_{t,k} = e^{t \sum_i \rho_i} \sum_{0 \leq j_i < m_{\rho_i}} \frac{t^{j_1 + \cdots + j_k}}{j_1! \cdots j_k!} N_{\rho_1}^{j_1} \otimes \cdots \otimes N_{\rho_k}^{j_k}. \quad (38)$$

The formula follows by multiplying (19); the nilpotents in different factors commute, and no term has been discarded. The representative is still the tensor of the maps using ε in (10), hence the coefficient unit is $\varepsilon^{\otimes k}$ in the ordered local algebra.

For the multivariable Mellin transform $\mathcal{M}_k$ with measure $\prod_i dx_i/x_i$, the top boundary $B_{a,k} = d\mathcal{H}_{a,k}$ has, on a pure coefficient tensor, the exact expression

$$\begin{aligned} \mathcal{M}_k B_{a,k}(\mathbf{s}) &= \prod_{i=1}^k [a^{s_i} v(s_i) R_Z(u_i)(s_i)] \\ &\quad - \prod_{i=1}^k [v(s_i) R_Z(a^A u_i)(s_i)]. \end{aligned} \quad (39)$$

All integrals factor by absolute convergence of the Schwartz Mellin integrals. Applying the full packet jet in each factor gives zero, as also follows from (37). The homotopy (36) exhibits the actual tensor relation giving that zero. It is not inferred from a dimension count or an assumed vanishing tensor kernel.

Every map in (36) is continuous. More explicitly, for seminorms $p_1, \dots, p_k$ on the appropriate target factors, the projective tensor seminorm of one summand on a pure input is bounded by the product of the corresponding operator bounds from (1), (10), and (14), times $\prod_i \|u_i\|$. Sum over the k terms and use the defining infimum for the projective seminorm on arbitrary finite sums. Thus all maps extend to completed projective tensor products, and (37) persists by density and continuity. The coefficient domain $E^{\otimes k}$ is finite dimensional, but this continuity statement identifies its targets in the same completed topology used by the original global construction.

The support rule is applied to each admitted factor before the tensor balancing quotient: an external absent factor is absorbing, while a zero amplitude in an active factor remains its supported tensor zero. The chosen face in that factor determines which of (25) occurs. Changing that face replaces its homotopy by (28); substitution into (36) gives an explicit tensor of the retained H^0 cycle with the other degree-one factors. No identification of external tensor support with a Cartesian product of labels is needed or asserted.

This tensor homotopy difference can be detected without discarding a possible lower-degree boundary. Define cochain maps $\pi_0 : C \rightarrow V[0]$ by $\pi_0^0(v, w) = w$, $\pi_0^1 = 0$, and $\pi_1 : C \rightarrow Q[-1]$ by $\pi_1^0 = 0$, $\pi_1^1 = q$. In the difference between the all-plus and all-minus versions of (36), apply π_0 in slot j and π_1 in the others. Every summand but the j -th vanishes. The result is

$$(-1)^{j-1} (\sigma a^A)^{\otimes(j-1)} \otimes \hat{c}_a \otimes (\sigma a^A)^{\otimes(k-j)}.$$

It is a nonzero map whenever $a \neq 1$: σa^A is injective by (11), $\hat{c}_a$ is nonzero by Section 5, and a tensor of nonzero vectors over $\mathbb{C}$ is nonzero, as witnessed by a tensor of linear functionals nonzero on those vectors. The target complex has zero differential, so the detected class cannot be a boundary. This proves the exact survival of the homotopy difference in degree $k-1$, where one factor carries the original $H^0(C)$ and the others carry Q .

8. The complete dx boundary pairing, its derivative, and tensor mass

The inclusion $\mathcal{B} \hookrightarrow L^2(\mathbb{R}_{>0}, dx)$ is continuous. For example on $(0, 1)$ use $|F(x)| \leq p_{-1,0}(F)x$, and on $(1, \infty)$ use $|F(x)| \leq p_{1,0}(F)x^{-1}$. Both squared majorants are integrable. Use the inner product conjugate-linear in its first argument. Direct substitution, retaining dx , gives

$$\langle U_a F, U_a G \rangle = a \langle F, G \rangle. \quad (40)$$

Let $G = R^* R$ in the retained coefficient basis and let $B_a = \Theta c_a : E \rightarrow L^2(dx)$. Since $J_Z R = 1$, G is positive definite. Equation (15), $U_a R = R a^A + B_a$, gives the exact finite boundary identity

$$\boxed{(a^A)^* G a^A - a G = -(U_a R)^* B_a - B_a^* (U_a R) + B_a^* B_a.} \quad (41)$$

On vectors u, w , its right side is

$$-\langle U_a Ru, \Theta c_a w \rangle - \langle \Theta c_a u, U_a R w \rangle + \langle \Theta c_a u, \Theta c_a w \rangle.$$

To check the signs, solve $Ra^A = U_a R - B_a$ and expand its squared pairing. Equation (40) supplies aG . Equivalently the right side is $-(Ra^A)^* B_a - B_a^* Ra^A - B_a^* B_a$; these two forms agree by (15). Thus the positive sign in the version using $U_a R$ is forced by the expansion, not a sign choice.

Every term is finite and depends smoothly on a , by the Frechet bounds already proved and the continuous Hilbert inclusion. For a full generalized packet vector u , the left side is evaluated using (19), retaining every power of $\log a$. For a true eigenvector $Au = \rho u$ it is $(a^{2\text{Re } \rho} - a) u^* G u$.

The infinitesimal identity obtained by differentiating (41) at $t = \log a = 0$ is

$$W = A^* G + G A - G = -R^* f_0 \ell - \ell^* f_0^* R. \quad (42)$$

This agrees exactly with (12) and the original integration by parts. Indeed for $F, H \in \mathcal{B}$,

$$\langle DF, H \rangle + \langle F, DH \rangle = - \int_0^\infty x (\overline{F} H)' dx = \langle F, H \rangle,$$

since the endpoint term $[x \overline{F}(x) H(x)]_0^\infty$ vanishes by the displayed weighted bounds. This independently verifies the mass term $-G$ and the two negative boundary cross terms in (42).

The finite defect is exactly the integrated infinitesimal one:

$$\boxed{T_t^* G T_t - e^t G = \int_0^t e^{t-r} T_r^* W T_r dr.} \quad (43)$$

Set $F(t) = T_t^* G T_t - e^t G$. Direct differentiation gives $F'(t) = F(t) + T_t^* W T_t$ and $F(0) = 0$. Multiplying by e^{-t} and integrating proves (43) with the original orientation for $t < 0$. This is a finite-dimensional exact matrix integration, valid with all the nilpotents in (19).

For the global representative R^s from (31), replace R by R^s and c_a by the fully computed c_a^s in (32). The finite expansion (41) then holds with

$$G^s = (R - \Theta b_Z)^* (R - \Theta b_Z), \quad \Theta c_a^s = \Theta(c_a - U_a b_Z + b_Z a^A).$$

Its infinitesimal boundary is the changed map

$$B_D^s = \Theta k_D \sigma = \Theta(\phi_* \ell - D b_Z + b_Z A). \quad (43a)$$

Indeed differentiating the explicit $c_{e^t}^s$ at $t = 0$ gives $k_D \sigma$ by (33), and direct differentiation of $R^s = R - \Theta b_Z$ gives $DR^s - R^s A = B_D^s$ by (12). Apply the same dx integration-by-parts identity to $R^s u, R^s w$ for arbitrary coefficient vectors. Substituting $DR^s = R^s A + B_D^s$ and moving the two boundary terms to the other side proves exactly

$$W^s = A^* G^s + G^s A - G^s = -(R^s)^* B_D^s - (B_D^s)^* R^s. \quad (43b)$$

Thus the factor $f_0 \ell$ in (42) is replaced by B_D^s , not retained as the old infinitesimal source boundary. Finally differentiating $T_t^* G^s T_t - e^t G^s$ as in the proof of (43) gives $F_s'(t) = F_s(t) + T_t^* W^s T_t$, $F_s(0) = 0$, and therefore

$$T_t^* G^s T_t - e^t G^s = \int_0^t e^{t-r} T_r^* W^s T_r dr. \quad (43c)$$

This is exactly the restriction of the A1694 global pairing identity to the actual packet σE . It proves the finite/global relationship between both the cocycle and its boundary pairings, without assuming an unproved vanishing of either cross term.

For k factors use the Hilbert pairing with measure $dx_1 \cdots dx_k$. Fubini gives its value on pure tensors as the product of the original pairings, hence

$$\langle U_a^{\otimes k} F, U_a^{\otimes k} H \rangle = a^k \langle F, H \rangle.$$

Let $G_k = G^{\otimes k}$, $B_{a,k}$ be the actual boundary (37), and $\mathcal{R}_a = (U_a R)^{\otimes k}$. Then the exact tensor version of (41) is

$$(a^A)^{* \otimes k} G_k (a^A)^{\otimes k} - a^k G_k = -\mathcal{R}_a^* B_{a,k} - B_{a,k}^* \mathcal{R}_a + B_{a,k}^* B_{a,k}. \quad (44)$$

Indeed $R_k(a^A)^{\otimes k} = \mathcal{R}_a - B_{a,k}$ by (37), and expansion proves the formula. Its derivative has the exact ordered sum

$$W_k = A_k^* G_k + G_k A_k - k G_k = \sum_{j=1}^k G^{\otimes(j-1)} \otimes W \otimes G^{\otimes(k-j)}, \quad (45)$$

because each of the k copies of the term $-G$ in (42) contributes one copy of $-G_k$. The finite integral is

$$T_{t,k}^* G_k T_{t,k} - e^{kt} G_k = \int_0^t e^{k(t-r)} T_{r,k}^* W_k T_{r,k} dr. \quad (46)$$

The proof differentiates the left side exactly as in (43), with mass exponent k . Equations (38) and (46) retain all tensor nilpotents. They are actual formulas for the original representative boundary contribution on the original tau-base tensor complex.

9. Source pins and status of the calculation

The complete input files read for this calculation were:

- `arithmetic_input.tex` (original September 12 cumulative source), SHA256 a50b0fb587b644c4ea94dba45b2b423d038f2bfbe7fef188543939ba850e58d2.
- *Tau_Base_Cohomology*, NOTE.md, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3.
- Original transcript turn A1694.md, SHA256 a87bff4960c0a1742734e9e711780e0533ea76ba6f46ceab1fea115a2fd66f18; its original locator is recorded in Section 0.
- *Tau_Global_Comparison*, NOTE.tex, SHA256 50284fceb0dc859ac398a00839a29e15be60c008e02314ac1f35c31950ed01; its equations are cited at the beginning.

The completed new formulas are (13)–(24), their exact finite/global comparison (31)–(34), the evaluated tensor homotopy and jets (36)–(39), and the integrated finite and tensor boundary expressions (43)–(46). Sections 1, 2, 5, 6, and the unintegrated boundary pairing explicitly reprove the source interfaces needed to type and verify those formulas. The global retraction, global cocycle, and support comparison in A1694 remain credited to that earlier calculation.

The resulting boundary terms have been computed, including their orientations, topologies, full arithmetic coefficients, and product masses. None has been set to zero by calling it a theta boundary. No estimate forcing the sign or vanishing of the original global weight defect is asserted by these identities. That precise limit on the conclusion does not delete the displayed maps, the full quotient Q , or the original tau-base construction.

Chapter 5

Full-jet grading and its theta lift

This calculation completes the operator comparison left unspecified in shared transcript A0961, node a7bbd422-311c-4380-ba6c-a4bdd61fce2f, §7. It uses the original arithmetic objects of *Arithmetic cohomology over the tau-base*, equations (10)–(14), (19), (27), and (38)–(45). It preserves the actual function $g = 2\xi$, every local unit, multiplicity, coordinate, reflected-germ sign and support mask. The new grading is an explicitly constructed additional action. It is not identified with arithmetic Frobenius or with the original Mellin scaling.

1. Original arithmetic objects and analytic identities

Write $E = \mathbb{C}$ and retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The target is the original Fréchet space

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)|, \quad D = -x\partial_x.$$

Its Mellin transform is

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

For $F \in \mathcal{B}$ this is entire. Indeed, for any compact set of s and any derivative order, choose integers b_- below every $\Re s$ in that compact set and b_+ above every $\Re s$. On $(0, 1)$ and $(1, \infty)$ respectively, the bounds $|F(x)| \leq p_{b_-,0}(F)x^{-b_-}$ and $|F(x)| \leq p_{b_+,0}(F)x^{-b_+}$ give integrable dominating functions even after multiplication by any power of $|\log x|$. Differentiation under the integral is therefore justified.

Retain the two original maps

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{J}F(x) = x^{-1}F(1/x).$$

For every $\phi \in V$, $\Theta\phi \in \mathcal{B}$. At infinity this follows by termwise Euler differentiation and the Schwartz estimate, choosing a decay exponent greater than one so that its sum over n converges. At zero Poisson summation, with both original vanishing moments, gives

$$\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x).$$

The same infinity estimates for $\hat{\phi}$ apply after differentiating this identity; the resulting finite sums of Euler derivatives still decay faster than every power at zero. All sums and differentiated sums converge normally on compact positive intervals. Fourier transformation preserves V , and

$$\mathcal{J}\Theta\psi = \Theta\hat{\psi}.$$

Thus the original two-leg cochain complex is

$$C_\theta = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta\phi - \mathcal{J}\Theta\psi = \Theta(\phi - \widehat{\psi}),$$

in degrees 0, 1. Its degree-one quotient is $Q = \mathcal{B}/\Theta V$.

For clarity, the exact Mellin factor used below can be proved without replacing g by its leading local monomial. Put

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The entire continuation and reflection identity needed here follow directly from the Gaussian theta integral. With

$$\vartheta(x) = 2 \sum_{n \geq 1} e^{-\pi n^2 x^2},$$

Poisson summation gives $\vartheta(x) = x^{-1}\vartheta(1/x) + x^{-1} - 1$. Mellin integration for $\Re s > 1$, followed by splitting at 1 and substituting $x \mapsto 1/x$ in the lower integral, yields

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \vartheta(x)(x^s + x^{1-s}) \frac{dx}{x}.$$

The latter integral is entire by Gaussian decay. Consequently

$$g(s) = 1 + s(s-1) \int_1^\infty \vartheta(x)(x^s + x^{1-s}) \frac{dx}{x}, \quad g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s).$$

In particular g is not zero identically and satisfies $g^\dagger = g$ for the reflection below.

For $\phi \in V$, absolute convergence for $\Re s > 1$ gives the exact factorization

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)\mathcal{M}\phi(s) = g(s)H_\phi(s), \quad H_\phi(s) = \frac{2\pi^{s/2}\mathcal{M}\phi(s)}{s(s-1)\Gamma(s/2)}.$$

Here $\mathcal{M}\phi$ uses the positive half-line. To verify that H_ϕ is entire, split its integral at 1 and subtract finitely many terms of the even Taylor expansion of ϕ at zero. Since $\phi(0) = 0$, its only possible Mellin poles are simple poles at $s = -2, -4, \dots$, with the corresponding even Taylor coefficients as residues. Each is canceled by the simple zero of $1/\Gamma(s/2)$. At $s = 0$ the same reciprocal gamma factor cancels the displayed factor s . At $s = 1$, $\mathcal{M}\phi(1) = \frac{1}{2} \int_{\mathbb{R}} \phi = \frac{1}{2} \widehat{\phi}(0) = 0$ cancels $s - 1$. This proves the asserted entire continuation. The displayed identity then holds everywhere by continuation. No vanishing at an arithmetic zero was assumed in this argument.

2. Full actual jet packets, reflection, and residue pairing

Fix any finite set Z of actual zeros of this g which is stable under

$$\iota\rho = 1 - \bar{\rho}.$$

If Z is empty, set $A_Z = 0$, $J_Z = 0$, $S = 0$, and $L_q = 1$; all packet identities below are the unique identities on the zero space, and no minimum or maximum over Z is taken. For the remaining construction assume Z is nonempty. Retain the actual order $m_\rho \geq 1$ of each zero and the coordinate $z_\rho = s - \rho$. Write the actual germ, with its unit intact, as

$$g_\rho(z) = g(\rho + z) = u_\rho(z)z^{m_\rho}, \quad u_\rho(0) \neq 0.$$

The coefficient algebra is

$$A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho/(g),$$

where $\mathcal{O}_\rho$ is the ring of holomorphic germs at ρ . Since multiplication by the actual unit u_ρ is invertible, the ideal (g_ρ) equals (z^{m_ρ}) . This equality identifies classes with unique truncated Taylor coordinates; it does not replace the denominator g_ρ in a residue. Define

$$(J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j.$$

The preceding factorization proves $J_Z \Theta = 0$, so J_Z induces $\bar{J}_Z : Q \rightarrow A_Z$.

Reflection is the conjugate-linear involution

$$(f^\dagger)_\rho(z) = \overline{f_{\iota\rho}(-\bar{z})} = \sum_{j \geq 0} (-1)^j \overline{f_{\iota\rho,j}} z^j.$$

Because $g^\dagger = g$, reflection preserves the defining ideals, $m_{\iota\rho} = m_\rho$, and the actual units satisfy

$$u_\rho(z) = (-1)^{m_\rho} \overline{u_{\iota\rho}(-\bar{z})}.$$

Retain the original residue form, conjugate-linear in its first variable and linear in its second:

$$R_Z(f, h) = \sum_{\rho \in Z} \text{Res}_{z=0} \frac{(f^\dagger)_\rho(z) h_\rho(z)}{u_\rho(z) z^{m_\rho}} dz.$$

Changing either representative by a multiple of its actual g changes the integrand by a holomorphic germ, so this form is well-defined. It is perfect: if a nonzero component of h starts with $h_k z^k$, choose $(f^\dagger)_\rho = z^{m_\rho-1-k}$ and all other components zero. Its residue is $h_k/u_\rho(0) \neq 0$. Reflection is a bijection, so that choice of $f^\dagger$ comes from an f . Finite dimension then gives nondegeneracy in both variables.

The form is **skew-Hermitian**, with no phase change. In fact for a meromorphic germ $a(z) = \sum a_j z^j$, reflection changes its residue coefficient to $-\bar{a}_{-1}$, because $(-1)^{-1} = -1$. Applying that identity, relabeling ρ by $\iota\rho$, and using $g^\dagger = g$ gives

$$R_Z(h, f) = -\overline{R_Z(f, h)}.$$

This sign will not be replaced by a positivity assertion.

3. The exact additional grading and its inverse

For each real $q > 0$ define componentwise

$$(C_q f)_\rho(z) = f_\rho(z/q) \pmod{g_\rho(z)}.$$

This is an algebra automorphism of the original quotient. To check well-definedness with the original unit retained, compute

$$\frac{g_\rho(z/q)}{g_\rho(z)} = q^{-m_\rho} \frac{u_\rho(z/q)}{u_\rho(z)}.$$

The right side is a holomorphic unit at zero. Thus substitution sends the ideal (g_ρ) onto itself. Its inverse is

$$(C_q^{-1} f)_\rho(z) = f_\rho(qz),$$

as is seen directly by composing the substitutions. In particular

$$C_p C_q = C_{pq}, \quad C_1 = 1, \quad C_q(z_\rho^j) = q^{-j} z_\rho^j.$$

Since q is real and positive, substitution commutes with the reflected conjugation:

$$(C_q f)^\dagger = C_q(f^\dagger).$$

The separate packet labels ρ , local coordinates, multiplicities, and positive-degree jets have all been retained.

On a component let

$$N_\rho f = z_\rho f, \quad U_a|_\rho = a^\rho e^{(\log a)N_\rho}, \quad a > 0, \quad a^\rho = \exp(\rho \log a),$$

with the real logarithm of the original parameter a . These are exactly the jet operators induced by the original scaling $U_a F(x) = F(x/a)$. Multiplication by $a^{\rho+z}$ follows from a change of variables in the original Mellin integral. No centered or normalized scaling has been substituted.

Direct multiplication proves

$$U_a N_\rho U_a^{-1} = N_\rho, \quad C_q N_\rho C_q^{-1} = q^{-1} N_\rho.$$

Conjugating the full exponential gives the exact surviving comparison

$$C_q U_a|_\rho C_q^{-1} = a^\rho \exp\left(\frac{\log a}{q} N_\rho\right) = a^{\rho(1-1/q)} U_{a^{1/q}}|_\rho.$$

Every exponential is its full finite sum through degree $m_\rho - 1$. The factor $a^{\rho(1-1/q)}$ remains part of the formula. This is a relation between two distinct actions on the same actual arithmetic packet. The conjugated operator $C_q U_a C_q^{-1}$ still commutes with N_ρ , because its displayed formula is a polynomial in N_ρ times a scalar. It is C_q itself, not that conjugated arithmetic operator, which scales N_ρ by q^{-1} .

4. Exact obstruction to a literal simultaneous identification

Let $N = \bigoplus N_\rho$ and $U_a = \bigoplus U_a|_\rho$. Fix $q > 0$, $q \neq 1$, and retain any target vector space W with invertible F and an operator N_D satisfying the exact Deligne-type relation

$$F N_D F^{-1} = q^{-1} N_D.$$

Every linear map $T : A_Z \rightarrow W$ intertwining both proposed operators,

$$T U_a = F T, \quad T N = N_D T,$$

annihilates $N A_Z$. Indeed $F^{-1} T = T U_a^{-1}$, so

$$F N_D F^{-1} T = F T N U_a^{-1} = T U_a N U_a^{-1} = T N = N_D T.$$

The left side is also $q^{-1} N_D T$. Since $1 - q^{-1} \neq 0$, $N_D T = T N = 0$. Hence T factors uniquely through

$$A_Z \twoheadrightarrow A_Z / N A_Z = \bigoplus_{\rho \in Z} \mathbb{C},$$

the actual residue-value observation. An injective such T therefore exists only when every $N_\rho = 0$, which is equivalent to every $m_\rho = 1$.

This proves precisely the obstruction to the **literal simultaneous identification** of original Mellin scaling and the same full jet nilpotent with that pair. It does not forbid other functors or additional actions. There is a concrete surviving quotient map: evaluation at $z = 0$ intertwines U_a with $\text{diag}(a^\rho)$ and sends N to zero. On that quotient the target relation holds with $N_D = 0$, and it imposes no condition on $\Re \rho$. Alternatively, C_q retains all nilpotents and satisfies the target conjugacy relation, with its distinct semidirect interaction with U_a proved above.

5. Residue Jacobian, adjoint, and trace contraction

Define the actual unit in each component

$$\kappa_{q,\rho}(z) = q^{1-m_\rho} \frac{u_\rho(z)}{u_\rho(qz)}.$$

Changing the local coordinate of integration by $z = qw$ in the residue gives, with its Jacobian retained,

$$\begin{aligned} R_Z(C_q f, C_q h) &= \sum_\rho q \operatorname{Res}_{w=0} \frac{(f^\dagger)_\rho(w) h_\rho(w)}{g_\rho(qw)} dw \\ &= \sum_\rho \operatorname{Res}_{w=0} \frac{(f^\dagger)_\rho(w) \kappa_{q,\rho}(w) h_\rho(w)}{g_\rho(w)} dw. \end{aligned}$$

Thus

$$\boxed{R_Z(C_q f, C_q h) = R_Z(f, M_{\kappa_q} h).}$$

In particular the factor is q^{1-m_ρ} , not q^{-m_ρ} , and the quotient of the full local units remains. The unit relation from reflection proves, without dropping its sign,

$$(\kappa_q^\dagger)_\rho(z) = q^{1-m_\rho} \frac{(-1)^{m_\rho} u_\rho(z)}{(-1)^{m_\rho} u_\rho(qz)} = \kappa_{q,\rho}(z).$$

Put

$$D_q = M_{\kappa_q} C_q^{-1}.$$

Replacing the second $C_q h$ in the residue formula by an arbitrary h proves

$$R_Z(C_q f, h) = R_Z(f, D_q h).$$

Skew-Hermitian symmetry then gives the equally important other-side formula

$$R_Z(D_q f, h) = R_Z(f, C_q h).$$

For example, apply the first formula to (h, f) and take negative complex conjugates; no phase is needed. These are the exact residue adjoint identities.

Retain the original trace contraction

$$J_g = M_{g'} : A_Z \rightarrow A_Z.$$

The actual derivative is $g'_\rho(z) = m_\rho u_\rho(z) z^{m_\rho-1} + u'_\rho(z) z^{m_\rho}$. Modulo its original ideal $(g_\rho) = (z^{m_\rho})$ this gives

$$\begin{aligned} \kappa_{q,\rho}(z) g'_\rho(qz) &\equiv q^{1-m_\rho} \frac{u_\rho(z)}{u_\rho(qz)} m_\rho u_\rho(qz) q^{m_\rho-1} z^{m_\rho-1} \\ &= m_\rho u_\rho(z) z^{m_\rho-1} \equiv g'_\rho(z). \end{aligned}$$

Consequently

$$\boxed{D_q J_g C_q = J_g.}$$

The contracted form is therefore invariant:

$$R_Z(C_q f, J_g C_q h) = R_Z(f, J_g h).$$

Its exact value is

$$R_Z(f, J_g h) = \sum_{\rho \in \mathbb{Z}} m_\rho \overline{f_{\iota\rho}(0)} h_\rho(0) = \operatorname{Tr}(M_{f^\dagger h} | A_Z).$$

For the first equality, $g'_\rho/g_\rho = m_\rho/z + u'_\rho/u_\rho$ and the second term is holomorphic. For the second, multiplication by a germ on the basis $1, z, \dots, z^{m_\rho-1}$ is triangular with its constant coefficient on all m_ρ diagonal entries. The radical of this contracted form is exactly NA_Z ; the full residue form is perfect before contraction. C_q is invertible and preserves that radical. No nilpotent or multiplicity has been discarded.

6. A proved continuous section on the original theta target

We now construct a section of J_Z in the actual space $\mathcal{B}$, not in a substitute polynomial test family. Index the Taylor coordinates by

$$\Lambda = \{(\rho, j) : \rho \in Z, 0 \leq j < m_\rho\}, \quad M = |\Lambda|.$$

Choose fixed real numbers $\alpha < \beta$ and retain the bump

$$\chi(y) = \begin{cases} \exp\left(-\frac{1}{(y-\alpha)(\beta-y)}\right), & \alpha < y < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

It is smooth and all its endpoint derivatives vanish. For $i = (\rho, j)$ let

$$w_i(y) = e^{\rho y} \frac{y^j}{j!}, \quad H_{ik} = \int_{\mathbb{R}} \chi(y) w_i(y) \overline{w_k(y)} dy.$$

The matrix H is invertible. Here is the required independence proof. An identically zero linear combination of the w_i on a nonempty real interval has the form $\sum_{\rho} e^{\rho y} P_{\rho}(y) = 0$ with $\deg P_{\rho} < m_{\rho}$. By analyticity it vanishes everywhere. Fix ρ and apply

$$\prod_{\sigma \in Z, \sigma \neq \rho} (\partial_y - \sigma)^{m_{\sigma}}.$$

All other summands vanish. The remaining equation is

$$e^{\rho y} \prod_{\sigma \neq \rho} (\partial_y + \rho - \sigma)^{m_{\sigma}} P_{\rho}(y) = 0.$$

On polynomials of degree below m_{ρ} , each factor $\partial_y + c$ with $c \neq 0$ is invertible: its inverse is the finite sum

$$c^{-1} \sum_{r=0}^{m_{\rho}-1} (-c^{-1} \partial_y)^r.$$

Thus $P_{\rho} = 0$. Doing this for every ρ proves independence. For any nonzero column v ,

$$v^* H v = \int \chi(y) \left| \sum_i \overline{v_i} w_i(y) \right|^2 dy > 0,$$

because the combination cannot vanish on the whole interval where $\chi > 0$. Hence H is positive definite and invertible.

Define functions in the original positive variable by

$$F_k(x) = \chi(\log x) \sum_{\ell \in \Lambda} \overline{w_{\ell}(\log x)} (H^{-1})_{\ell k}.$$

Each is smooth and supported in $[e^{\alpha}, e^{\beta}]$, with all derivatives zero at its endpoints, so $F_k \in \mathcal{B}$. Direct substitution $y = \log x$ gives

$$\frac{(\mathcal{M} F_k)^{(j)}(\rho)}{j!} = \int \chi(y) w_{(\rho, j)}(y) \sum_{\ell} \overline{w_{\ell}(y)} (H^{-1})_{\ell k} dy = \delta_{(\rho, j), k}.$$

Thus the explicit finite-rank map

$$S : A_Z \rightarrow \mathcal{B}, \quad S\left(\sum_i c_i z_i^{j_i}\right) = \sum_i c_i F_i, \quad J_Z S = 1_{A_Z},$$

is a continuous linear section. The notation in the sum means the specified Taylor coordinates in their separate ρ -components.

For explicit continuity of the observation, choose integers

$$b_- < \min_{\rho \in \mathbb{Z}} \Re \rho, \quad b_+ > \max_{\rho \in \mathbb{Z}} \Re \rho.$$

For each $i = (\rho, j)$, direct integration of $|\log x|^j$ on the two half-lines proves

$$|(J_Z F)_{\rho, j}| \leq \frac{p_{b_-, 0}(F)}{(\Re \rho - b_-)^{j+1}} + \frac{p_{b_+, 0}(F)}{(b_+ - \Re \rho)^{j+1}}.$$

The $j!$ in the jet definition exactly cancels the $j!$ in those elementary integrals. All constants and original seminorms remain explicit. In particular J_Z and S are continuous, J_Z is surjective, and $\bar{J}_Z$ on Q is surjective. No assertion about closedness of ΘV or completeness of its quotient was needed.

7. An exact action on the original four-point theta sheaf

Define on $\mathcal{B}$

$$L_q = 1 + S(C_q - 1)J_Z.$$

Then

$$J_Z L_q = C_q J_Z, \quad L_q|_{\ker J_Z} = 1, \quad L_q \Theta = \Theta.$$

The identity $J_Z S = 1$ shows by expansion that

$$L_p L_q = 1 + S(C_p C_q - 1)J_Z = L_{pq}, \quad L_q^{-1} = 1 + S(C_q^{-1} - 1)J_Z.$$

These are exact inverses on the full original function space, not just on the observed packet.

Keep the original four-point upper-set space

$$+ < \eta < \sigma, \quad - < \eta < \sigma,$$

with coefficient sheaf $\mathcal{T}_+ = \mathcal{T}_- = V$, $\mathcal{T}_\eta = \mathcal{B}$, $\mathcal{T}_\sigma = 0$, and restriction maps Θ and $\mathcal{J}\Theta$. Define a sheaf automorphism to be the identity at $+$ and $-$, L_q at η , and the unique map on the zero vector space at σ . This is a sheaf morphism because L_q fixes both ΘV and $\mathcal{J}\Theta V = \Theta V$ pointwise. On derived global sections it is precisely

$$\begin{array}{ccc} V \oplus V & \xrightarrow{\Theta - \mathcal{J}\Theta} & \mathcal{B} \\ \downarrow 1 & & \downarrow L_q \\ V \oplus V & \xrightarrow{\Theta - \mathcal{J}\Theta} & \mathcal{B}. \end{array}$$

The signs and the original two source legs are unchanged.

Consequently L_q descends to an automorphism $\bar{L}_q$ of the exact quotient Q . It induces the identity on degree-zero cohomology and satisfies

$$\bar{J}_Z \bar{L}_q = C_q \bar{J}_Z.$$

Equip Q here with the quotient topology of the original $\mathcal{B}$. Both $\bar{L}_q$ and its inverse are continuous by the quotient universal property. This does not assert that Q is Hausdorff or that ΘV is closed. It is natural for the retained inclusions of nonempty support masks when the same S is used on their common $\mathcal{B}$ stalk. Indeed each $\mathcal{T}_A$ has its original active source legs, this same target, and the corresponding restrictions. The source identities and common L_q commute with every such inclusion. The split reconstruction acts by

$$\tau \mapsto \tau, \quad (A, F) \mapsto (A, L_q F).$$

It fixes $(A, 0)$ and keeps the label A . Thus neither synchronization of masks nor conversion of a supported zero into τ has occurred. At σ the reconstructed support skeleton is fixed.

The action has concrete analytic bounds. In the Taylor coordinate basis C_q is diagonal, so

$$p_{b,k}(L_q F) \leq p_{b,k}(F) + \sum_{\rho \in Z} \sum_{j=0}^{m_\rho-1} |q^{-j} - 1| p_{b,k}(F_{\rho,j}) |(J_Z F)_{\rho,j}|.$$

Combining this with the preceding observation estimate proves continuity in every original Fréchet seminorm. For $q \geq 1$, all $|q^{-j} - 1| \leq 1$, so the displayed family of bounds is uniform in $q \geq 1$. For its inverse replace these factors by $|q^j - 1|$; it grows at most with the explicitly retained maximum degree $\max_\rho(m_\rho - 1)$. Nothing here bounds the original U_a in terms of $a^{1/2}$: L_q is the new finite-packet grading, and its inverse need not remain bounded as q grows.

The construction is noncanonical in its chosen bump and section. If S' is a second section and $K = S' - S$, then $J_Z K = 0$ and

$$L'_q - L_q = K(C_q - 1)J_Z.$$

Its image is in $\ker J_Z$, it vanishes on $\ker J_Z$, and both choices induce the same C_q on A_Z . This formula records the exact dependence instead of asserting a canonical global arithmetic action.

8. Compatibility with the original right-adjoint dual map

For the declared diagram category on the four-point space, the original calculation is

$$A_\tau(F) = [F_+ \oplus F_- \xrightarrow{r_+ - r_-} F_\eta], \quad K_\tau(W) = [I_\eta(W) \xrightarrow{(+\text{res}, -\text{res})} I_+(W) \oplus I_-(W)]$$

in degrees 0, 1 and $-1, 0$ respectively. Here $I_x(W)(y) = W$ for $y \leq x$ and zero otherwise. Evaluation gives $\text{Hom}(F, I_x(W)) = \text{Hom}(F_x, W)$; applying this to the two displayed terms proves the right-adjoint identity with differential $\ell \mapsto (\ell r_+, -\ell r_-)$. Therefore

$$H^{-1} R \text{Hom}(\mathcal{T}, K_\tau(\mathcal{L}_1)) = \text{Hom}_E(Q, \mathcal{L}_1),$$

where the original moment line $\mathcal{L}_1$ has the original scaling action a . Its underlying vector space is E ; no action of the new grading on it is being inferred from arithmetic purity.

The original finite-packet morphism is the linear map from the conjugate vector space

$$\iota_Z : \overline{A_Z} \longrightarrow \text{Hom}_E(Q, \mathcal{L}_1), \quad \iota_Z(\bar{f})([F]) = R_Z(f, J_Z F).$$

It is well-defined because $J_Z \Theta = 0$, and injective by the just-proved surjectivity of J_Z and perfection of R_Z . Precomposition with $\bar{L}_q$ has the exact effect

$$\begin{aligned} (\bar{L}_q^* \iota_Z(\bar{f}))([F]) &= R_Z(f, J_Z L_q F) \\ &= R_Z(f, C_q J_Z F) \\ &= R_Z(D_q f, J_Z F). \end{aligned}$$

Thus the diagram into the **same** computed right-adjoint object satisfies

$$\boxed{\bar{L}_q^* \iota_Z(\bar{f}) = \iota_Z(\overline{D_q f}), \quad D_q = M_{\kappa_q} C_q^{-1}.}$$

Every arrow is now calculated on the original theta source and the full actual arithmetic jets. This is not merely a pairing on an unrelated finite algebra. All dual support arrows remain precomposition, and the zero functional at a present mask remains its supported zero functional.

9. Exact scope of the completed comparison

The new automorphism C_q retains the actual zero ρ , its complete local algebra, and its unit. It provides the conjugacy $N \mapsto q^{-1}N$. Its continuous lift L_q acts on the original theta sheaf and fixes the original relation space ΘV pointwise; its finite-packet effect and dual effect are the explicit diagrams above. These are completed maps with full proofs and analytic seminorm control.

The actual arithmetic scaling remains $U_a = a^\rho e^{(\log a)N}$. It commutes with N and has the exact semidirect relation with C_q . The grading eigenvalues are q^{-j} on the original Taylor coordinates. They do not contain a bound on $\Re \rho$. On simple packets C_q is the identity for every q , while U_a still has eigenvalue a^ρ .

More generally, multiplying C_q on a component by any nonzero scalar c leaves its conjugacy with N unchanged and changes all its eigenvalues to cq^{-j} . The residue formula and its full unit, not that conjugacy alone, state the actual additional information carried by our chosen C_q . Consequently this completed comparison supplies neither a canonical arithmetic Frobenius nor an absolute arithmetic weight estimate. The original tensor-growth calculation over A_τ remains the original problem; the new maps give a precise continuation instead of a false literal identification or an unsupported claim of irrelevance.

Provenance

Primary source: `output/split_zero_rh_tandem_2026-09-12/sources/Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA-256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`. Canonical route: `work/f1_geometry_current_source_route_20260913.md`, SHA-256 `a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74`. Exact transcript node and its scope are recorded at the beginning. The independent algebra review is retained in `GERM_PAIRING_CHECK.md`. No Lean run, remote publication or shared-master edit is claimed.

Chapter 6

Raw Mellin relations and surviving cycles

Completed Mellin-realization calculations on the original theta source

These calculations close two finite algebraic gaps identified by the fully read transcript section U0029–U0040. They use the original theta source, its Fourier convention, its differentiated Gaussian, its complete two-leg support, and the original generator $g = 2\xi$. They do not replace that source by an arbitrary polynomial family. They concern Mellin realization, not the separately assigned homotopy fibre of restriction to σ .

Retained objects and source equations

Let $E = \mathbb{C}$ and retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \hat{\phi}(0) = 0\}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx,$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Set

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad JF(x) = x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The primary Tau Base NOTE.md equations (10)–(14),(21)–(22),(38) and the actual comparison source retain the identities

$$J\Theta = \Theta\mathcal{F}, \quad \mathcal{F}D = (1 - D)\mathcal{F}, \quad \Theta D = D\Theta, \quad \mathcal{M}(DF) = s\mathcal{M}F.$$

The fixed source function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \widehat{\phi}_* = \phi_*,$$

and

$$b_* = \Theta\phi_*, \quad \mathcal{M}b_* = g, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The entire function g is not zero. For example, the retained theta-boundary calculation gives $g(0)=1$. The arithmetic quotient is $Q = \mathcal{B}/\Theta V$. The original two-leg complex is

$$C^\bullet = [V \oplus V \xrightarrow{\Theta\phi - J\Theta\psi} \mathcal{B}], \quad C^0 = V \oplus V, \quad C^1 = \mathcal{B}.$$

Its H^0 is the actual Fourier graph V and its H^1 is Q . A single-leg complex has $H^0 = 0$; it is not the same complex.

All sheaf arguments below are stalkwise over $R = \mathcal{O}_{\mathbb{C}, s_0}$, with its original coordinate s , and hence glue. This local analytic ring is an integral domain. A nonzero entire function has a nonzero germ at every point by the identity theorem, even if its value at that point is zero.

Exact Fourier calculation for the retained theta source

We prove the Fourier self-duality used in the preceding definition, with the original convention and every coefficient. Put $u(x) = e^{-\pi x^2}$. Its integral is one. Indeed, its positive integral I satisfies, by Tonelli's theorem and the polar-coordinate Jacobian,

$$I^2 = \int_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^\infty e^{-\pi r^2} r dr d\vartheta = 2\pi \frac{1}{2\pi} = 1,$$

so $I = 1$.

Differentiation under the Fourier integral is justified by the integrable majorants $(2\pi|x|)^k u(x)$, for each fixed derivative order k . Since $u' = -2\pi x u$, integration by parts gives

$$\begin{aligned} \frac{d}{d\xi} \hat{u}(\xi) &= \int_{\mathbb{R}} (-2\pi i x) u(x) e^{-2\pi i x \xi} dx \\ &= i \int_{\mathbb{R}} u'(x) e^{-2\pi i x \xi} dx \\ &= i(2\pi i \xi) \hat{u}(\xi) = -2\pi \xi \hat{u}(\xi). \end{aligned}$$

The boundary term is zero because $u(x)$ tends to zero at both ends of the real line. The differential equation, together with $\hat{u}(0) = 1$, gives

$$\hat{u}(\xi) = e^{-\pi \xi^2}.$$

For every Schwartz function f , the same differentiation and integration by parts prove the typed operator identities

$$\mathcal{F}(xf) = \frac{i}{2\pi} \partial_\xi \hat{f}, \quad \mathcal{F}(f') = 2\pi i \xi \hat{f}.$$

The Gaussian derivatives are

$$\begin{aligned} u''(\xi) &= (4\pi^2 \xi^2 - 2\pi) u(\xi), \\ u^{(4)}(\xi) &= (16\pi^4 \xi^4 - 48\pi^3 \xi^2 + 12\pi^2) u(\xi). \end{aligned}$$

Consequently,

$$\begin{aligned} \mathcal{F}(x^2 u)(\xi) &= -\frac{1}{4\pi^2} u''(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) u(\xi), \\ \mathcal{F}(x^4 u)(\xi) &= \frac{1}{16\pi^4} u^{(4)}(\xi) = \left(\xi^4 - \frac{3}{\pi} \xi^2 + \frac{3}{4\pi^2} \right) u(\xi). \end{aligned}$$

Apply these formulas to the original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)u$:

$$\begin{aligned} \widehat{\phi_*}(\xi) &= \left[4\pi^2 \left(\xi^4 - \frac{3}{\pi} \xi^2 + \frac{3}{4\pi^2} \right) - 6\pi \left(\frac{1}{2\pi} - \xi^2 \right) \right] u(\xi) \\ &= (4\pi^2 \xi^4 - 12\pi \xi^2 + 3 - 3 + 6\pi \xi^2) u(\xi) \\ &= (4\pi^2 \xi^4 - 6\pi \xi^2) e^{-\pi \xi^2} = \phi_*(\xi). \end{aligned}$$

This establishes the literal equality $\widehat{\phi_*} = \phi_*$. The function is even and Schwartz, $\phi_*(0) = 0$, and $\widehat{\phi_*}(0) = \phi_*(0) = 0$; hence it belongs to the exact source space V .

The same calculation also verifies the source operator expression and the sign of its Fourier intertwiner. Since

$$Du = 2\pi x^2 u, \quad D^2 u = (-4\pi x^2 + 4\pi^2 x^4) u,$$

we have

$$D(D-1)u = (4\pi^2 x^4 - 6\pi x^2)u = \phi_*.$$

For every Schwartz f ,

$$\begin{aligned} \widehat{Df}(\xi) &= -\mathcal{F}(xf')(\xi) = -\frac{i}{2\pi} \partial_\xi (2\pi i \xi \hat{f}(\xi)) \\ &= \hat{f}(\xi) + \xi \partial_\xi \hat{f}(\xi) = (1 - D_\xi) \hat{f}(\xi). \end{aligned}$$

Here $D_\xi = -\xi\partial_\xi$, with the same sign as the original operator. This proves $\mathcal{F}D = (1 - D)\mathcal{F}$ on its stated Schwartz domain. Moreover, D preserves V : it preserves evenness and the Schwartz class, $(D\phi)(0) = 0$, and

$$\int_{\mathbb{R}} D\phi(x) dx = -[x\phi(x)]_{-\infty}^{\infty} + \int_{\mathbb{R}} \phi(x) dx = 0 \quad (\phi \in V).$$

The boundary term vanishes by Schwartz decay. Thus every $D^m\phi_*$ and every $(1 - D)^m\phi_*$ occurring in the two-leg calculation remains in the original domain.

The Mellin constant can also be obtained directly from the same function. For $\Re s > 0$, substitution $t = \pi x^2$ and $\Gamma(z + 1) = z\Gamma(z)$ give

$$\begin{aligned} M_+\phi_*(s) &= \int_0^\infty (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2} x^s \frac{dx}{x} \\ &= \pi^{-s/2} \left(2\Gamma\left(\frac{s}{2} + 2\right) - 3\Gamma\left(\frac{s}{2} + 1\right) \right) \\ &= \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \left[2\frac{s}{2} \left(\frac{s}{2} + 1\right) - 3\frac{s}{2} \right] \\ &= \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right). \end{aligned}$$

The recurrence follows by integration by parts in the defining Gamma integral on this half-plane; its boundary terms vanish at zero and infinity. For $\Re s > 1$, the full nonzero-integer theta sum is absolutely integrable term by term, because

$$\sum_{n \neq 0} \int_0^\infty |\phi_*(nx)| x^{\Re s} \frac{dx}{x} = 2 \sum_{n \geq 1} n^{-\Re s} \int_0^\infty |\phi_*(y)| y^{\Re s} \frac{dy}{y} < \infty.$$

Therefore

$$\begin{aligned} \mathcal{M}\Theta\phi_*(s) &= 2\zeta(s)M_+\phi_*(s) \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\zeta(s) = g(s). \end{aligned}$$

This retains both the factor two from the full integer sum and the factor one half in the source Mellin integral. The source's entire continuation identifies these same functions away from this initial half-plane; no division by g at an arithmetic zero has occurred.

The raw two-leg syzygy at every source order

For each integer $m \geq 1$, retain the actual functions

$$b_{+,m} = \Theta D^m \phi_* = D^m b_*, \quad b_{-,m} = J\Theta D^m \phi_* = (1 - D)^m b_*.$$

Their Mellin transforms are exactly

$$\mathcal{M}b_{+,m} = s^m g, \quad \mathcal{M}b_{-,m} = (1 - s)^m g.$$

Thus the same source input $D^m\phi_*$ on the two original charts gives these two images. No independent spectral function has been chosen.

Put $a_m(s) = s^m$ and $v_m(s) = (1 - s)^m$. Define

$$\begin{aligned} \delta_m : R^2 &\longrightarrow R \otimes_E \mathcal{B}, & (f, h) &\longmapsto f \otimes b_{+,m} - h \otimes b_{-,m}, \\ \beta : R \otimes_E \mathcal{B} &\longrightarrow R, & f \otimes F &\longmapsto f \mathcal{M}F. \end{aligned}$$

These give the literal commutative comparison

$$\begin{array}{ccc} R^2 & \xrightarrow{\delta_m} & R \otimes_E \mathcal{B} \\ 1 \downarrow & & \downarrow \beta \\ R^2 & \xrightarrow{(f,h) \mapsto g(a_m f - v_m h)} & R. \end{array}$$

First, $b_{+,m}$ and $b_{-,m}$ are E-linearly independent. A relation $cb_{+,m} + db_{-,m} = 0$ gives $g(s)(cs^m + d(1-s)^m) = 0$. On the nonempty open set on which g does not vanish the polynomial factor vanishes, so that polynomial is identically zero. Evaluation at $s=0$ gives $d=0$; evaluation at $s=1$ gives $c=0$. Tensoring this two-dimensional E-subspace with R is injective because vector spaces split over E . Therefore δ_m is injective.

Here are the exact polynomial coefficients used for its composite:

$$A_m(s) = \sum_{k=0}^{m-1} \binom{m+k-1}{k} s^k, \quad B_m(s) = \frac{1 - A_m(s)(1-s)^m}{s^m}.$$

The power series $(1-s)^{-m}$ has coefficients $\binom{m+k-1}{k}$, so its truncation A_m gives a numerator divisible by s^m . Thus B_m is a polynomial and

$$B_m a_m + A_m v_m = 1.$$

The germ g is a non-zero-divisor. Consequently $\beta\delta_m(f, h) = 0$ is equivalent to $a_m f = v_m h$. Set $r = B_m h + A_m f$. Then

$$v_m r = B_m v_m h + A_m v_m f = B_m a_m f + A_m v_m f = f,$$

$$a_m r = B_m a_m h + A_m a_m f = B_m a_m h + A_m v_m h = h.$$

Conversely $(f, h) = (v_m r, a_m r)$ has $a_m f - v_m h = 0$. The parametrization is injective because its two coefficients generate the unit ideal. Hence

$$\ker \delta_m = 0, \quad \ker(\beta\delta_m) = R(v_m, a_m).$$

In particular

$$z_m = v_m \otimes b_{+,m} - a_m \otimes b_{-,m} \neq 0, \quad \beta z_m = 0.$$

This proves the exact all-order extension of A1361's $m=1$ witness.

Its support is not optional. Let $L = \mathcal{P}(\{+, -\})$, with empty mask corresponding to external absence. On the $+$ face δ_m is $f \mapsto f \otimes b_{+,m}$; on the $-$ face it is $h \mapsto -h \otimes b_{-,m}$; on the joint face it is the map above. All maps retain their masks. The raw maps on either singleton face, and their Mellin composites, are injective because $b_{\pm,m}$ and ga_m, gv_m respectively are nonzero. The new syzygy occurs only in the joint face. Its lifted image is

$$(\{+, -\}, z_m) \mapsto (\{+, -\}, 0),$$

not τ . At $m=0$ the two source images coincide, and the raw map already has kernel $R(1,1)$; thus $m=0$ is not falsely included in the new-relation assertion.

This result excludes a quasi-isomorphism of these particular raw complexes: the first H^0 is zero while the second H^0 is R . It says nothing by itself about nonvanishing of the full realized H^1 kernel; that different calculation is given in Section 6.

Operator balancing removes the raw tensor relation, and retains honest joint cycles

Let $A = E[t]$. On $\mathcal{B}$ let t act by D , and on R let t act by multiplication by s . The quotient map

$$\pi_{\text{bal}} : R \otimes_E \mathcal{B} \longrightarrow R \otimes_A \mathcal{B}$$

imposes the R -linear span of the actual relations

$$sh \otimes F - h \otimes DF.$$

These are Mellin-null because $\mathcal{M}(DF) = s\mathcal{M}F$.

For every $j \geq 1$, the exact telescoping identity is

$$h \otimes D^j b_* - h s^j \otimes b_* = \sum_{k=0}^{j-1} (h s^k \otimes D^{j-k} b_* - h s^{k+1} \otimes D^{j-k-1} b_*).$$

Each summand is the negative of a displayed elementary balancing relation. Applying this identity to all coefficients of the original polynomials t^m and $(1-t)^m$ proves

$$\pi_{\text{bal}}(h \otimes P(D)b_*) = hP(s) \otimes b_* \quad (P \in E[t]).$$

It follows without removing any sign that

$$\pi_{\text{bal}}(z_m) = v_m a_m \otimes b_* - a_m v_m \otimes b_* = 0.$$

More explicitly, with $\Delta_P = 1 \otimes P(D)b_* - P(s) \otimes b_*$,

$$z_m = v_m \Delta_{t^m} - a_m \Delta_{(1-t)^m}.$$

The coefficients of $(1-t)^m$ remain $(-1)^j \binom{m}{j}$.

Balancing is not a reason to erase genuine degree-zero cycles in the original two-leg source. The exact saturated source calculation makes this precise. On the plus leg let t act by D ; on the minus leg let t act by $1-D$. These are the infinitesimal actions induced by primary (34), since the minus scaling is $aU_{1/a}$. Define

$$\begin{aligned} i_{+,m} : A &\longrightarrow V, & P &\longmapsto P(D)D^m \phi_*, \\ i_{-,m} : A &\longrightarrow V, & Q &\longmapsto Q(1-D)D^m \phi_*. \end{aligned}$$

They are injections. Indeed, Θ applied to the first has Mellin transform $P(s)s^m g$; $J\Theta$ applied to the second has transform $Q(s)(1-s)^m g$, and polynomial multiplication by either nonzero entire function is injective. Let Ab_* be the cyclic A -submodule of $\mathcal{B}$ generated by b_* . It too is free rank one because $P(D)b_* = 0$ implies $P(s)g(s)=0$ and therefore $P=0$.

The original two-leg differential restricted to these saturated subspaces is therefore exactly

$$C_m^{\text{cyc}} = \left[A^2 \xrightarrow{(P,Q) \mapsto (t^m P - (1-t)^m Q)b_*} Ab_* \right].$$

Fourier supplies the minus term:

$$J\Theta(Q(1-D)D^m \phi_*) = Q(D)(1-D)^m b_*.$$

The same Bezout polynomials show the differential is onto and that its kernel is $A((1-t)^m, t^m)$. Hence

$$\boxed{H^0(C_m^{\text{cyc}}) = A((1-t)^m, t^m), \quad H^1(C_m^{\text{cyc}}) = 0.}$$

This cycle is an actual Fourier-compatible two-leg cycle before Mellin realization. It is not a spurious raw tensor relation.

The A -module R is torsion-free: multiplication by a nonzero polynomial $P(s)$ is injective in the analytic domain R . A torsion-free module over a PID is flat. One direct proof writes it as the directed union of its finitely generated submodules; those submodules are finitely generated torsion-free modules over a PID and hence free, and tensoring commutes with directed colimits. A filtered colimit of exact sequences of modules is exact. Thus R is flat, and tensoring the displayed exact cyclic sequence gives

$$H^0(R \otimes_A C_m^{\text{cyc}}) = R((1-s)^m, s^m), \quad H^1 = 0.$$

No joint H^0 has been lost.

For completeness, its analytic comparison has $\beta_{\text{cyc}}(h \otimes b_*) = hg$ and identity in degree zero:

$$[R^2 \xrightarrow{a_m f - v_m h} Rb_*] \longrightarrow [R^2 \xrightarrow{g(a_m f - v_m h)} R].$$

The degree-one map $\times g$ is injective with cokernel $R/(g)$, and the degree-zero map is identity. Hence there is an exact sequence of these cochain complexes whose quotient is $R/(g)$ concentrated in degree one. The induced map on H^0 is identity on the displayed R-syzygy; the analytic target has $H^1 = R/(g)$. This calculation locates precisely what changes for this saturated cyclic subcomplex.

The cyclic subcomplex is not the full theta complex. Its inclusion is the literal pair $(i_{+,m}, i_{-,m})$ in degree zero and $Ab_* \hookrightarrow \mathcal{B}$ in degree one, and its differential is the restriction just proved. Extending these maps to the corresponding supported faces retains all masks. No equality with the full balanced kernel has been inferred.

An explicit free submodule in the raw realization kernel

This makes A1427's nonvanishing statement constructive with the actual finite-jet maps.

Let Z be any finite set of actual zeros of g , with their actual positive multiplicities m_ρ . Retain the complete jet space

$$A_Z = \bigoplus_{\rho \in Z} E[z_\rho]/(z_\rho^{m_\rho}), \quad z_\rho = s - \rho,$$

and the primary map

$$J_Z : Q \longrightarrow A_Z, \quad [F] \mapsto \left(\sum_{j < m_\rho} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z_\rho^j \right)_\rho.$$

The source's smooth compactly supported right inverse can be written in its original measure. Fix $0 < R' < R_0$, let

$$\eta(y) = \begin{cases} \exp(-1/(R'^2 - y^2)), & |y| < R', \\ 0, & |y| \geq R', \end{cases}$$

and for $\alpha = (\rho, j)$ put

$$r_\alpha(x) = x^{\bar{\rho}-1} \frac{(\log x)^j}{j!},$$

$$B_{\alpha\beta} = \frac{1}{j!k!} \int_{-R'}^{R'} \eta(y) e^{(\rho+\bar{\sigma}-1)y} y^{j+k} dy \quad (\beta = (\sigma, k)).$$

For a coefficient vector c ,

$$c^* B c = \int_{-R'}^{R'} \eta(y) e^{-y} \left| \sum_\alpha \bar{c}_\alpha e^{\rho y} \frac{y^j}{j!} \right|^2 dy.$$

The exponential polynomials with distinct exponents and all orders below the retained multiplicities are linearly independent on every open interval. To verify this, a vanishing sum is an entire function of y and vanishes identically. For a selected exponent ρ , applying $\prod_{\sigma \neq \rho} (\partial_y - \sigma)^{m_\sigma}$ kills the other blocks. On $e^{\rho y} P_\rho(y)$, each factor becomes $(\partial_y + \rho - \sigma)^{m_\sigma}$ on the polynomial P_ρ ; its constant term is nonzero, and in the degree-ordered polynomial basis its matrix is triangular with nonzero diagonal. Its product is therefore invertible. Thus $P_\rho = 0$ for each ρ . Since η is positive on the interval, $c^* B c = 0$ forces $c=0$. Hence B is positive definite and invertible.

For $v \in A_Z$ define

$$F_v(x) = \eta(\log x) \sum_\beta (B^{-1}v)_\beta r_\beta(x).$$

It lies in $C_c^\infty((e^{-R_0}, e^{R_0})) \subset \mathcal{B}$. Substituting $x = e^y$ in each Mellin derivative gives $J_Z[F_v] = v$, with exactly the displayed factorials and exponent -1 . Thus

$$s_Z : A_Z \longrightarrow Q, \quad v \mapsto [F_v]$$

is an explicit E-linear section of J_Z . It is not asserted to be equivariant under scaling; that additional property is neither needed nor inferred.

At each analytic stalk retain

$$\bar{\beta} : R \otimes_E Q \longrightarrow R/(g), \quad h \otimes [F] \longmapsto [h \mathcal{M}F], \quad K_{\text{raw}} = \ker \bar{\beta}.$$

Define

$$\boxed{\iota_Z : R \otimes_E A_Z \longrightarrow K_{\text{raw}}, \quad h \otimes v \longmapsto gh \otimes s_Z(v).}$$

The image is in the kernel because its Mellin evaluation is a multiple of g . If $\iota_Z(w) = 0$, applying $1 \otimes J_Z$ gives $gw=0$ in $R \otimes_E A_Z$. This latter module is free of rank $\sum_{\rho \in Z} m_\rho$, and g is a non-zero-divisor. Thus $w=0$. Therefore ι_Z is an injective R -linear map.

This gives an explicit free rank- $\sum_{\rho \in Z} m_\rho$ submodule of the raw kernel for every nonempty retained packet Z . The map is natural under stalk restriction for the fixed smooth section. Every derivative direction below each actual multiplicity appears in its domain. In particular the proof does not merely cite existence of a Hamel basis for Q or reduce the packet to its values.

At each original nonempty support mask L , lift the map as

$$(L, w) \longmapsto (L, \iota_Z w),$$

with $\tau \mapsto \tau$. It remains injective, retains its labelled zero, and commutes with the identity transports among the primary nonempty H^1 fibres. The target is the all-support realization-kernel fibre, not an identification with external absence.

Neither this free raw submodule nor the balancing calculation proves that the global balanced kernel is nonzero or zero. They prove the exact raw obstruction and the exact repair of its exhibited operator relation. The surviving full map is

$$\mathcal{O} \otimes_{E[t]} Q \longrightarrow \mathcal{O}/(g),$$

together with its actual kernel and the original strong-Schwartz/quotient topology. It must be connected through the original A_τ and K_τ ; the present calculations have made no replacement of those objects.

Source pins and transcript locators

The complete primary source was read: `Tau_Base_Cohomology_2026-09-12/NOTE.md`, SHA-256 `d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3`. The canonical route was read completely, SHA-256 `a8cdd18817a2c3596778c60a04ef9d76125a03ae5404759835a2bc2b18fa4c74`.

The all-order calculation answers A1361 (node `c9e551ee-298f-438b-9bdc-2e23c2c0613a`), source Section 1, and A1427 (node `d7e752ce-cb39-4130-b089-b20333037990`), source Section 4, while retaining A1304's actual two-leg source. The smooth section retains A1167 (node `157eecd-7f96-4ec5-96bd-6dd376276ea6`), source Section 5, with its original measure, factorials, and full jet orders. These are exact transcript UUID locators; line endings of the readable extraction were repaired during the audit without changing message text.

These are written proofs. No Lean session, remote write, numerical zero computation, or independently rerun historical checker is claimed.

Chapter 7

Original-source moment constants

Exact weighted theta-source constants

This calculation finishes the finite moment reduction left explicit in U0065/A2497 and U0066/A2559 of the retained transcript. It calculates the weighted-source Grams controlling the circle length, and the derivative Grams controlling its Fourier cutoff, on the original theta source. It supplies no asymptotic bound on their growing-degree inverses.

Original objects and domains

Retain the source's actual finite packet polynomial $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, with the full selected orders, $g = 2\xi$, and its specified inverse Mellin representative $F_h \in \mathcal{B}$, where

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}.$$

Write $v_h(s) = \mathcal{M}F_h(s) = g(s)/h(s)$, $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$, and $D = -x\partial_x$. For $k \geq 1$, set

$$c = k/2, \quad D^{(k)} = \sum_{j=1}^k D_j, \quad \mathcal{V}P = P(D^{(k)})F_h^{\otimes k}, \quad S = \sum_{j=1}^k s_j. \quad (1)$$

The polynomial source of degree at most N is $\mathcal{P}_N$, with the original basis $1, S, \dots, S^N$. Its arithmetic quotient map is the unchanged remainder map $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi_{h,k})$. Its full observation is

$$J^{(k)}\mathcal{V} = \eta_{h,k}J_N, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad v_h = j_h(g/h). \quad (2)$$

No coefficient of this unit is changed by the calculations below.

The original half-density coordinates are

$$r = \log x_k, \quad z_i = \log x_i - \log x_k \quad (i < k), \quad (3)$$

and

$$(\mathcal{U}_k F)(r, \mathbf{z}) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r). \quad (4)$$

The coordinate inverse in (3) has determinant one in logarithmic coordinates. Consequently (4) is an isometry from $L^2((0, \infty)^k, d^k x)$ to $L^2(\mathbb{R}, dr; L^2(\mathbb{R}^{k-1}, d\mathbf{z}))$, and $\partial_r \mathcal{U}_k = -\mathcal{U}_k(D^{(k)} - c)$. These statements follow by substitution in the measure and by the chain rule, respectively; for $k = 1$ the relative space is $\mathbb{C}$.

Weighted lines and exact finite moments

For every real a , define

$$w_a(t) = \frac{|v_h(1/2 + a + it)|^2}{2\pi}, \quad \mu_j(a) = \int_{\mathbb{R}} t^j w_a(t) dt, \quad \mathcal{M}_a(z) = \sum_{j \geq 0} \mu_j(a) \frac{z^j}{j!}. \quad (5)$$

The last expression is a formal series: each coefficient below uses only finitely many of its entries. No analytic convergence assertion about that series is needed.

All the integrals in (5) are finite. Indeed, for any real σ and nonnegative integers b, j , $y^b \partial_y^j (e^{\sigma y} F_h(e^y))$ decays exponentially at both ends. To check this, expand the derivative into finitely many terms $e^{\sigma y} D^l F_h(e^y)$, use a positive integer exponent larger than σ at $+\infty$, and an integer exponent less than σ at $-\infty$ in the definition of $\mathcal{B}$. Any polynomial in y is absorbed by the resulting strict exponential margins. Thus $e^{\sigma y} F_h(e^y)$ is Schwartz. Its Fourier transform with kernel e^{ity} is exactly $v_h(\sigma + it)$, which is therefore Schwartz on that line.

Let

$$m_{k,a} = w_0^{*(k-1)} * w_a, \quad \mu_j^{(k,a)} = j! [z^j] \mathcal{M}_0(z)^{k-1} \mathcal{M}_a(z), \quad (6)$$

where the exponent zero in the case $k = 1$ means the identity for convolution. Expansion of $(t_1 + \dots + t_k)^j$, with Fubini justified by absolute polynomial moments, proves $\mu_j^{(k,a)} = \int t^j m_{k,a}(t) dt$.

Theorem 7.0.1 (The actual weighted source Gram). *For $P, Q \in \mathcal{P}_N$, the original weighted observation is*

$$\begin{aligned} M_{k,N}(a)(P, Q) &:= \int_{(0,\infty)^k} x_k^{2a} \overline{\mathcal{V}P(\mathbf{x})} \mathcal{V}Q(\mathbf{x}) d^k x \\ &= \int_{\mathbb{R}} \overline{P(c + a + iu)} Q(c + a + iu) m_{k,a}(u) du. \end{aligned} \quad (7)$$

In the unchanged basis $1, S, \dots, S^N$, its entries are exactly

$$[M_{k,N}(a)]_{b,d} = \sum_{r=0}^b \sum_{s=0}^d \binom{b}{r} \binom{d}{s} (c+a)^{b+d-r-s} (-i)^r i^s (r+s)! [z^{r+s}] \mathcal{M}_0^{k-1} \mathcal{M}_a. \quad (8)$$

In particular these matrices require only $\mu_j(0), \mu_j(a)$ for $0 \leq j \leq 2N$. Their masses are

$$[M_{k,N}(a)]_{0,0} = \mu_0(0)^{k-1} \mu_0(a). \quad (9)$$

Proof. Put $y_j = \log x_j$, use $\sigma_j = 1/2$ for $j < k$ and $\sigma_k = 1/2 + a$, and apply ordinary Fourier Plancherel to $e^{\sum_j \sigma_j y_j} \mathcal{V}P(e^{\mathbf{y}})$. Integration by parts has no boundary term by the estimates just proved. It sends each D_j to multiplication by $s_j = \sigma_j + it_j$. Hence its Fourier transform is

$$P(c + a + i \sum_j t_j) \prod_{j < k} v_h(1/2 + it_j) v_h(1/2 + a + it_k).$$

The factor $(2\pi)^{-k}$ in Plancherel is exactly the product of the $(2\pi)^{-1}$ factors in (5). Expanding the complex inner product and integrating over the sum coordinate proves (7). The binomial formula for each original monomial gives (8), including both phases. Formula (9) is its constant entry. In particular the shift $c + a$ has not been replaced by c , and the sum has not been replaced by an average. $\square$

Each matrix $M_{k,N}(a)$ is positive definite. The entire nonzero Mellin function v_h has isolated zeros on each vertical line, so $w_a > 0$ almost everywhere. For $k \geq 2$, a convolution in (6) is positive at every real sum: for almost every choice of the first $k - 1$ variables all factors of its integrand are positive. For $k = 1$, the almost-everywhere positivity suffices. A nonzero polynomial has only finitely many zeros on the integration line in (7), so its squared norm is strictly positive.

Let $M = M_{k,N}(0)$. The constants required by A2497 and A2559 are therefore the explicitly defined finite-matrix values

$$\kappa_{a,N} = \lambda_{\max}(M^{-1}(M_{k,N}(a) + M_{k,N}(-a))), \quad a > 0, \quad (10)$$

where eigenvalues are understood in the positive inner product M . In particular there is a fully specified upper value avoiding an additional extreme-eigenvalue calculation:

$$0 < \kappa_{a,N} \leq \mathfrak{k}_{a,N} := \text{Tr}(M^{-1}(M_{k,N}(a) + M_{k,N}(-a))). \quad (11)$$

To prove (11), conjugate by $M^{1/2}$; the resulting matrix is positive, and each eigenvalue is at most the sum of its eigenvalues. The congruence is used to prove the inequality, not to replace the source basis or its metric. With $\kappa_{\pm} = \lambda_{\max}(M^{-1}M_{k,N}(\pm a))$, the same argument gives the individual upper values $\text{Tr}(M^{-1}M_{k,N}(\pm a))$.

Derivative and Fourier-cutoff Grams, including the cross terms

Write

$$a_h(t) = v_h(1/2 + it)/\sqrt{2\pi}, \quad n(t) = |a'_h(t)|^2, \quad b(t) = \overline{a'_h(t)}a_h(t). \quad (12)$$

All three measures $w_0(t)dt, n(t)dt, b(t)dt$ have finite absolute polynomial moments by the same Schwartz argument. The last is a complex measure; it is not replaced by a positive one. Define

$$\begin{aligned} m_k &= w_0^{*k}, & n_k &= w_0^{*(k-1)} * n, & b_k &= w_0^{*(k-1)} * b, \\ \mathcal{N}(z) &= \sum_{j \geq 0} \int t^j n(t) dt \, z^j / j!, & \mathcal{C}(z) &= \sum_{j \geq 0} \int t^j b(t) dt \, z^j / j!. \end{aligned} \quad (13)$$

Let $p \geq 1$, $R_P(S) = (S - c)^p P(S)$, and let primes on R_P mean polynomial derivatives in the unchanged coordinate S . The derivative Gram in A2497/A2559 is

$$H_{p,N}(P, Q) = \int_{\mathbb{R} \times \mathbb{R}^{k-1}} (1 + r^2) \overline{\partial_r^p \mathcal{U}_k \mathcal{V} P} \partial_r^p \mathcal{U}_k \mathcal{V} Q \, dr \, dz. \quad (14)$$

Theorem 7.0.2 (Complete derivative Gram). *With all polynomials in the following formula evaluated at $c + iu$,*

$$\begin{aligned} H_{p,N}(P, Q) &= \int_{\mathbb{R}} (\overline{R_P} R_Q (m_k + n_k) + \overline{R'_P} R'_Q m_k \\ &\quad + i \overline{R_P} R'_Q b_k - i \overline{R'_P} R_Q \overline{b_k})(u) \, du. \end{aligned} \quad (15)$$

Every entry is a finite expression in the moments encoded by $\mathcal{M}_0^k$, $\mathcal{M}_0^{k-1} \mathcal{N}$, and $\mathcal{M}_0^{k-1} \mathcal{C}$, through degree $2(N + p)$.

Proof. The chain rule in (4) gives $\partial_r^p \mathcal{U}_k \mathcal{V} P = (-1)^p \mathcal{U}_k \mathcal{V} R_P$. The two occurrences of $(-1)^p$ cancel in (14). The term without r^2 is (7) at $a = 0$, with polynomial R_P . For the second term return to the independent logarithmic coordinates $y_1, \dots, y_k$; its multiplier is exactly $r = y_k$. For the Fourier kernel $e^{i \sum t_j y_j}$, multiplication by y_k transforms to $-i \partial_{t_k}$. The factor $-i$ cancels against its conjugate in the inner product. After retaining the Plancherel constants in the product of the a_h 's, the differentiated last factor is

$$a'_h(t_k) R_P(c + i \sum_j t_j) + i a_h(t_k) R'_P(c + i \sum_j t_j).$$

Multiplication by its conjugate for P and by the displayed expression for Q gives, respectively, the n_k term, the derivative m_k term, and the cross terms $+i \overline{R_P} R'_Q b_k$ and $-i \overline{R'_P} R_Q \overline{b_k}$. Integrating the other variables gives (15). Expansion in powers of the sum and Fubini prove the stated coefficient products. The largest degree occurs in $\overline{R_P} R_Q$ and is at most $2(N + p)$. $\square$

A finite choice of comparison parameters and its exact scope

Set

$$\mathfrak{h}_{p,N} = \text{Tr}(M^{-1}H_{p,N}) \geq 0. \quad (16)$$

Both sides in (16) are defined on the original coefficient space. The same positive-matrix argument used for (11) gives $H_{p,N} \preceq \mathfrak{h}_{p,N}M$. Here are the explicit parameters obtained from these completed calculations. For prescribed positive error contributions η_1, η_2 , choose

$$L \geq a^{-1} \log(1 + \mathfrak{k}_{a,N}/\eta_1), \quad (17)$$

and, for the twisted lattice used by all scalar holonomy phases, take an integer

$$J \geq \max \left\{ 2, 1 + \frac{L}{2\pi} \left(\frac{\mathfrak{h}_{p,N}}{(2p-1)\eta_2} \right)^{1/(2p-1)} \right\}. \quad (18)$$

The ordinary lattice can omit the extra one; it has denominator J instead of $J-1$.

For completeness, the source estimates underlying this parameter choice have direct proofs. Write $\Psi = \mathcal{U}_k \mathcal{V}$ and $\mathcal{C}(s) = \int \Psi(r)^* \Psi(r+s) dr$. Weighted Cauchy–Schwarz gives, for $s > 0$,

$$|v^* \mathcal{C}(s) v| \leq e^{-as} \sqrt{v^* M_{k,N}(-a) v \ v^* M_{k,N}(a) v}.$$

The negative shifts exchange the two weights. Unfolding one period gives the Gram difference as the sum of the nonzero correlations at shifts mL ; holonomy only multiplies each by a unit complex number. Summing the geometric series and using $2\sqrt{xy} \leq x + y$ proves its absolute quadratic-form bound $(M_{k,N}(a) + M_{k,N}(-a))/(e^{aL} - 1)$. This is at most $\eta_1 M$ by (11) and (17).

Integration by parts p times in the retained Fourier transform, then Cauchy–Schwarz with $\int (1+r^2)^{-1} dr = \pi$, gives $\|\widehat{\Psi}v(u)\|^2 \leq \pi |u|^{-2p} v^* H_{p,N} v$. For the ordinary lattice, summing $|n| > J$ and bounding $\sum_{n>J} n^{-2p} \leq J^{1-2p}/(2p-1)$ gives

$$0 \leq M(L) - M(L, J) \leq \frac{1}{2p-1} \left(\frac{L}{2\pi J} \right)^{2p-1} H_{p,N}. \quad (19)$$

The constants arise from π/L in each Fourier coefficient and the two lattice directions; thus the factor in (19) is literal. For phases $0 \leq \theta < 2\pi$, $|n + \theta/(2\pi)| \geq |n| - 1$ for the omitted indices. Repeating the integral bound replaces J by $J-1$, and (18) makes this at most $\eta_2 M$. With $\eta_1 + \eta_2 < 1$, the resulting finite source Gram lies between $(1 - \eta_1 - \eta_2)M$ and $(1 + \eta_1 + \eta_2)M$.

For this quotient conclusion put $q = \deg \chi_{h,k}$ and take $N \geq q-1$, so the unchanged remainder map is onto E . Taking the infimum over the unchanged affine sets $\{P : J_N P = v\}$ gives the same inequalities for the canonical quotient Grams. Their sections differ by the original $\ker J_N = \chi_{h,k} \mathcal{P}_{N-q}$, with $\mathcal{P}_{-1} = 0$, hence by the original theta relations under (1)–(2). At a retained support label that correction maps to its supported zero; it never becomes external absence.

These estimates compute the finite error parameters from the actual one-factor shifted Mellin moments and their full derivative cross terms. They do not bound $\mathfrak{k}_{a,N}$ or $\mathfrak{h}_{p,N}$ uniformly as k, N grow. For the original four endpoints, one may use $N = 2q_k + 2$ in (17)–(18), exactly as in A2497; all smaller polynomial spaces are restrictions of that single source. The inverse M^{-1} and every original mass remain visible, so this result cannot be mistaken for the missing sub-threshold determinant estimate.

Finally, the weighted observations are norms of the already specified representatives, not new maps identifying the arithmetic cohomology with the Boolean stalk. In the primary Tau Base source the cohomology is $\mathbf{A}_\tau = R\Gamma(P, -)$, its right adjoint is $K_\tau(W) \simeq S_\eta W[1]$, and its actual top tensor quotient is $Q^{\otimes k}$. Equations (1)–(2) remain the same finite source and full arithmetic observation into that construction. Multiplying the source norm by x_k^{2a} has not been asserted to descend as an endomorphism of $Q^{\otimes k}$.

Chapter 8

Circle resolvent with full resonant packets

Circle resolvent with all arithmetic jets

U0065/A2470 promised a resolvent derived from theta boundaries. The subsequent A2497 and its retained note calculate the first- and second-order source defects. This note completes the resolvent calculation from those exact defects, including the resonant case. It makes no purity inference from the circle spectrum.

Original operator and its domain

Fix $k \geq 1$, $L > 0$, $c = k/2$, and the relative Hilbert space $\mathcal{K} = L^2(\mathbb{R}^{k-1})$; for $k = 1$ use $\mathcal{K} = \mathbb{C}$. The scalar holonomy phase is $0 \leq \theta < 2\pi$. Let

$$\mathcal{H} = L^2([0, L]; \mathcal{K}), \quad \mathcal{D} = -\partial_r + c, \quad \text{Dom } \mathcal{D} = \{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}. \quad (1)$$

The original orthonormal scalar functions are $e_n(r) = L^{-1/2} e^{-i\omega_n r}$, $\omega_n = (2\pi n + \theta)/L$. Fourier series, including $n = 0$, identify (1) with the diagonal

$$\mathcal{D}(e_n \otimes v) = (c + i\omega_n) e_n \otimes v, \quad \sum_n (1 + \omega_n^2) \|f_n\|_{\mathcal{K}}^2 < \infty. \quad (2)$$

In particular (1) is closed and normal: the adjoint of the diagonal has conjugate entries and the same domain. Its spectrum is exactly $\{c + i\omega_n : n \in \mathbb{Z}\}$. To check the last assertion, each listed value has its nonzero eigenvectors, while at any other complex number z the inverse diagonal is bounded and maps onto the stated domain. Its exact operator norm is

$$\|T(z)\| = \delta(z)^{-1}, \quad T(z) = (\mathcal{D} - z)^{-1}, \quad \delta(z) = \min_{n \in \mathbb{Z}} |c + i\omega_n - z| > 0. \quad (3)$$

The minimum exists because the distances tend to infinity. Equality in the norm follows by taking any unit vector in a minimizing mode.

Retain the actual cyclic packet $E = \mathbb{C}[S]/(\chi_{h,k})$, $A = M_S$, and the phase-periodized canonical representative $R : E \rightarrow \mathcal{H}$. Its source is still $P(D_1 + \dots + D_k) F_h^{\otimes k}$, with $\mathcal{M}F_h = 2\xi/h$, and full arithmetic observation $v_h^{\otimes k} P(A_h^{(k)}) 1$. Use an admitted finite degree $N \geq \deg \chi_{h,k} - 1$. For this source comparison choose the circle length by the proved formula (17) of `shifted_mellin_source_constants.tex`, with $a = 1$ and $\eta_1 = 1/2$: thus $L \geq \log(1 + 2\mathfrak{f}_{1,N})$. Its phase source Gram is at least one half of the positive original Gram. Periodization is therefore injective on the admitted polynomial source; composing with the canonical section proves that R is injective. It has values in every twisted Sobolev domain, and the actual theta boundary is

$$B = \mathcal{D}R - RA. \quad (4)$$

It is the scalar-holonomy periodization of the retained source vector $D^{(k)}r_N - r_N A = dK_N^{\text{prim}}$, not the zero vector obtained after applying its arithmetic quotient.

The resolvent comparison and the complete nonresonant packet

For z outside both spectra, the exact comparison is

$$T(z)R - R(A - z)^{-1} = -T(z)B(A - z)^{-1}. \quad (5)$$

Indeed, (4) says $(\mathcal{D} - z)R = R(A - z) + B$. Multiplying on the left by $T(z)$ and on the right by $(A - z)^{-1}$ gives (5), including its minus sign. All products are defined because E is finite dimensional and $R(E) \subset \text{Dom } \mathcal{D}$.

For a complete primary block $E_\lambda \subset E$, put $N_\lambda = (A - \lambda)|_{E_\lambda}$ and retain its full length r , so $N_\lambda^r = 0$. Restrict (4) to that block. If $\lambda \notin \text{Spec } \mathcal{D}$, it gives $R_\lambda = T(\lambda)R_\lambda N_\lambda + T(\lambda)B_\lambda$. Substitute this expression successively r times. Its last term contains N_λ^r and vanishes, giving the finite formula

$$R_\lambda = \sum_{j=0}^{r-1} T(\lambda)^{j+1} B_\lambda N_\lambda^j. \quad (6)$$

No jet was discarded: all r powers occur. Conversely the right side satisfies (4) on E_λ , by telescoping after multiplication by $\mathcal{D} - \lambda$; the last nilpotent term is zero. It is the unique such solution, because the same iteration applied to a homogeneous solution makes that solution zero. For the original finite coefficient norm, (3) also proves

$$\|R_\lambda\| \leq \sum_{j=0}^{r-1} \delta(\lambda)^{-j-1} \|B_\lambda N_\lambda^j\|. \quad (7)$$

Thus the displayed map is a calculated reconstruction from the retained boundary, not a vanishing assertion.

In particular, for an actual eigenvector $0 \neq v \in E_\lambda$ with $N_\lambda v = 0$,

$$Rv = T(\lambda)Bv, \quad \frac{\|Bv\|}{\|Rv\|} \geq \delta(\lambda) \geq |\Re \lambda - k/2|. \quad (8)$$

The denominator is positive by the injectivity of the canonical source. The first inequality follows either from (3) or directly from (2). Every off-weight primary block is nonresonant, regardless of its imaginary part. Hence making its quotient boundary zero does not make its source boundary zero.

The resonant block with its full kernel and nilpotents

Suppose $\lambda = c + i\omega_{n_0}$. Let P be the orthogonal projection onto $e_{n_0} \otimes \mathcal{K}$, and write $Q = I - P$. On $Q\mathcal{H}$, the inverse

$$T_Q = (\mathcal{D} - \lambda)^{-1}|_{Q\mathcal{H}} \quad \text{has norm } L/(2\pi). \quad (9)$$

This follows from the nonzero distances $|\omega_n - \omega_{n_0}| = 2\pi|n - n_0|/L$. Projection of (4) and the same nilpotent iteration now yield

$$QR_\lambda = \sum_{j=0}^{r-1} T_Q^{j+1} QB_\lambda N_\lambda^j, \quad PB_\lambda = -PR_\lambda N_\lambda. \quad (10)$$

These are all components of the comparison; (9) has not been used on the resonant mode.

For clarity, write $\chi_{h,k} = (S - \lambda)^r h_\lambda$. The two displayed factors are relatively prime. Choose polynomials a_λ, b_λ with $a_\lambda(S - \lambda)^r + b_\lambda h_\lambda = 1$, and set $\varepsilon_\lambda = [b_\lambda h_\lambda]_{\chi_{h,k}}$. Its residues are one in the λ -local factor and zero in every other primary factor. The Chinese remainder map and its inverse therefore identify

$E_\lambda = \varepsilon_\lambda E$ with $\mathbb{C}[S - \lambda]/((S - \lambda)^r)$. The actual embedded increasing power basis is $e_j = \varepsilon_\lambda (S - \lambda)^j$. Thus $N_\lambda e_j = e_{j+1}$ for $j < r - 1$, and $N_\lambda e_{r-1} = 0$. Write $X = PR_\lambda$, $Y = PB_\lambda$. Equation (10) says exactly

$$Ye_{r-1} = 0, \quad Xe_{j+1} = -Ye_j \quad (0 \leq j < r - 1), \quad Xe_0 \text{ is free in } P\mathcal{H}. \quad (11)$$

Necessity follows by applying $Y = -XN_\lambda$ to every basis vector. Conversely the values in (11) define a linear map satisfying that equation. Hence (11) proves both sufficiency and the complete freedom, with no assumed semisimplicity. The actual source supplies its particular Xe_0 . A squarefree sampled-value quotient would retain only a further observation of this data; it is not the map in (10)–(11).

The original Laplacian, including its two boundary terms

Set

$$\mathcal{L} = \mathcal{D}^2 - k\mathcal{D} = \partial_r^2 - k^2/4, \quad A^{\text{lap}} = A^2 - kA, \quad C = (\mathcal{D} - k)B + BA. \quad (12)$$

The domain of $\mathcal{L}$ is exactly

$$\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), \quad f'(L) = e^{-i\theta} f'(0)\},$$

equivalently the Fourier Sobolev domain of order two. Its eigenvalues are exactly $-\omega_n^2 - k^2/4$; the zeroth Fourier index remains included. Applying $\mathcal{D}$ to (4) and subtracting k times (4) proves

$$\mathcal{L}R - RA^{\text{lap}} = C. \quad (13)$$

In particular both terms of C in (12) are necessary. For z outside both relevant spectra, exactly the proof of (5) gives

$$(\mathcal{L} - z)^{-1}R - R(A^{\text{lap}} - z)^{-1} = -(\mathcal{L} - z)^{-1}C(A^{\text{lap}} - z)^{-1}. \quad (14)$$

On E_λ , retain

$$\mu = \lambda^2 - k\lambda, \quad M_\lambda = (2\lambda - k)N_\lambda + N_\lambda^2, \quad M_\lambda^* = 0. \quad (15)$$

If $\mu \notin \text{Spec } \mathcal{L}$, (13) gives the complete finite reconstruction

$$R_\lambda = \sum_{j=0}^{r-1} (\mathcal{L} - \mu)^{-j-1} C_\lambda M_\lambda^j. \quad (16)$$

The proof is the same finite substitution as (6). Formula (15) retains the exceptional coefficient $2\lambda - k$, including the case where it is zero, and retains the N_λ^2 term; it does not replace the original primary block by a simple root.

The linear source maps in (4)–(16) lift at each original support label as $(\ell, v) \mapsto (\ell, fv)$, with $\tau \mapsto \tau$. A source theta boundary has a supported-zero image under the original arithmetic quotient and retains its Hilbert vector in (6) or (10). Its necessary nonvanishing on each nonresonant eigenvector is proved by (8); the resonant boundary can vanish, with its full freedom given by (10)–(11). This resolves the promised resolvent calculation while proving its precise scope: a negative circle Laplacian controls its own resolvent and leaves the displayed arithmetic boundary terms. No bound for the four arithmetic quotient volumes follows by deleting those terms.

Chapter 9

Source cutoff and theta scaling

Exact fluid-cutoff transport into the retained theta complex

This is a completed calculation prompted by A0618, node ba85139d-2899-4a3b-b05e-9f30a7341c82, chain 1146, transcript-segment lines 2698–2818. That answer constructed an admissible cutoff test but left its equivariance and quotient class uncalculated. We calculate the exact scaling morphism and its image in the original theta complex; we do not assert a sign for its arithmetic pairing or replace the original fluid by a self-similar ansatz.

The arithmetic target is the original Tau_Base_Cohomology_2026-09-12/NOTE.md, SHA256 d03e71ad18f187bd89880cb8ca6fc9c5d6f260b3de8e8e5702c27db97e7b63c3: equations (10)–(14), (19)–(22), (32)–(45). All spaces and actions below retain those definitions. The factor connecting the early centered test with this target is an explicit invertible multiplication map, not an identification of their coordinates.

1. Original spaces and the cutoff map

Fix viscosity $\nu > 0$, a time interval I , and a smooth real vector field u on $\mathbb{R}^3 \times I$. Define $e_\theta = (-\sin \theta, \cos \theta, 0)$ and retain the **unaveraged** angular-momentum integral

$$a_u(r, t) = \int_0^{2\pi} r e_\theta \cdot u(r \cos \theta, r \sin \theta, 0, t) d\theta.$$

Let $\chi(r, t)$ be a smooth cutoff whose radial support is compact in $(0, \infty)$, locally uniformly in t . Set

$$D = r\partial_r, \quad b(r, t) = \chi(r, t)a_u(r, t), \quad g(r, t) = (D^2 - \tfrac{1}{4})b(r, t).$$

For every fixed t , $g_t \in C_c^\infty(\mathbb{R}_{>0})$. Define the original centered admissible space

$$\mathcal{T} = \{g \in C_c^\infty(\mathbb{R}_{>0}) : \int_0^\infty g(r)r^{1/2} dr/r = \int_0^\infty g(r)r^{-1/2} dr/r = 0\}.$$

The Mellin transform used in the tau source is $\mathcal{M}h(s) = \int_0^\infty h(r)r^s dr/r$. The support of b_t lies in some $[\alpha, \beta] \subset (0, \infty)$; it and every derivative vanish near both integration endpoints. Therefore, for every complex s ,

$$\mathcal{M}(Db)(s) = [b(r)r^s]_0^\infty - s\mathcal{M}b(s) = -s\mathcal{M}b(s),$$

and a second integration by parts gives

$$\mathcal{M}g(s) = (s^2 - \tfrac{1}{4})\mathcal{M}b(s). \tag{1}$$

At $s = \pm \frac{1}{2}$, this proves both admissibility equations, with all boundary terms zero for the stated support reason. Smooth parameter dependence follows by differentiation under the integral on each common compact support interval.

2. The explicit map to the original uncentered theta target

Retain the tau source's space

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{r>0} r^j |(-r\partial_r)^k F(r)| < \infty \text{ for all } j \in \mathbb{Z}, k \geq 0\}.$$

Multiplication defines a linear injection

$$\mathcal{C} : \mathcal{T} \longrightarrow \mathcal{B}, \quad (\mathcal{C}g)(r) = r^{-1/2}g(r). \quad (2)$$

Its image lies in $C_c^\infty \subset \mathcal{B}$, and its inverse on that image is multiplication by $r^{1/2}$. No endpoint extension is used. For $F = \mathcal{C}g$, equation (1) gives the exact entire identity

$$\boxed{\mathcal{M}F(s) = s(s-1)\mathcal{M}b(s - \tfrac{1}{2})}. \quad (3)$$

Thus the early two centered moment cancellations become the uncentered zeros at $s = 0, 1$, through the stated multiplication map.

The test space retains its usual inductive-limit topology over compact support intervals. On each such interval the multiplier $r^{-1/2}$ and all its derivatives are bounded; the product rule therefore bounds every displayed $\mathcal{B}$ seminorm by finitely many test-function seminorms. Thus (2) is continuous into $\mathcal{B}$. The quotient and adjoint calculations below retain the source's algebraic quotient and algebraic dual; they make no assertion that ΘV is closed or that this dual is the continuous dual of a completed quotient.

Retain the actual source spaces V , its Fourier transform with exponent $-2\pi i x \xi$, $\Theta\phi(r) = \sum_{n \neq 0} \phi(nr)$, $JF(r) = r^{-1}F(1/r)$, and

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta\phi - J\Theta\psi, \quad Q = \mathcal{B}/\Theta V.$$

Formula (2) followed by the quotient map gives a concrete linear map $\mathcal{T} \rightarrow H^1 \mathbf{A}_\tau(\mathcal{T}) = Q$. It does not say that every input class is nonzero. Equation (3), and the full jet formula below, compute its actual finite arithmetic observations without making that assumption.

The early theta background and its exact factor of two

A0598 used $\mathcal{E}\phi(r) = r^{1/2} \sum_{n \geq 1} \phi(nr)$, whereas the retained tau source uses $\Theta\phi(r) = \sum_{n \neq 0} \phi(nr)$. For an even $\phi \in V$, pairing the terms $n, -n$ gives the exact identity

$$\mathcal{C}(\mathcal{E}\phi) = \tfrac{1}{2}\Theta\phi. \quad (3a)$$

Here multiplication $\mathcal{C}$ extends to the displayed smooth functions by its same formula; this sentence does not assert that the uncut $\mathcal{E}\phi$ belongs to the compactly supported domain $\mathcal{T}$. On every compact positive r -interval the Schwartz bounds permit termwise differentiation of the theta sum, giving $D\Theta\phi = \Theta(D\phi)$. The operator D preserves even Schwartz functions and $\phi(0) = 0$; integration by parts gives $\int_{\mathbb{R}} D\phi = -\int_{\mathbb{R}} \phi = 0$. It therefore preserves V . Direct differentiation of $r^{1/2}h$ gives

$$r^{-1/2}(D^2 - \tfrac{1}{4})(r^{1/2}h) = (D^2 + D)h.$$

Consequently the uncut background maps to the original theta relation

$$\mathcal{C}(D^2 - \tfrac{1}{4})\mathcal{E}\phi = \tfrac{1}{2}\Theta((D^2 + D)\phi) \in \Theta V. \quad (3b)$$

For a compact cutoff χ , the exact image instead is

$$\begin{aligned} \mathcal{C}(D^2 - \tfrac{1}{4})(\chi\mathcal{E}\phi) &= \tfrac{1}{2}\chi\Theta((D^2 + D)\phi) + (D\chi)\Theta(D\phi) \\ &\quad + \tfrac{1}{2}(D^2\chi + D\chi)\Theta\phi. \end{aligned} \quad (3c)$$

The first term still contains its cutoff; membership of the uncut theta image does not erase that factor. Thus (3c) retains all three pieces of the actual early cutoff background in the original Q , without assuming that any one belongs to the relation space.

3. Constant parabolic scaling, with the same viscosity

For a constant $\lambda > 0$, put

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t), \quad f_\lambda(x, t) = \lambda^3 f(\lambda x, \lambda^2 t), \quad I_\lambda = \{t : \lambda^2 t \in I\}.$$

Differentiating each component gives

$$\partial_t u_\lambda = \lambda^3 \partial_t u, \quad (u_\lambda \cdot \nabla) u_\lambda = \lambda^3 (u \cdot \nabla) u, \quad \nabla p_\lambda = \lambda^3 \nabla p, \quad \Delta u_\lambda = \lambda^3 \Delta u, \quad \nabla \cdot u_\lambda = \lambda^2 \nabla \cdot u,$$

where the right sides are evaluated at $(\lambda x, \lambda^2 t)$. Consequently the forced equation $\partial_t u + (u \cdot \nabla) u = -\nabla p + \nu \Delta u + f$ is carried to that same equation with exactly the same ν . This is a map between fields at different coordinates, not an assertion that one fixed solution obeys a self-similarity identity.

Set $\chi_\lambda(r, t) = \chi(\lambda r, \lambda^2 t)$. Substitution in the unaveraged angular integral gives

$$a_{u_\lambda}(r, t) = \int_0^{2\pi} (\lambda r) e_\theta \cdot u(\lambda r \cos \theta, \lambda r \sin \theta, 0, \lambda^2 t) d\theta = a_u(\lambda r, \lambda^2 t).$$

The operator D commutes with $R_\lambda h(r) = h(\lambda r)$. Thus

$$b_\lambda(r, t) = b(\lambda r, \lambda^2 t), \quad g_\lambda(r, t) = g(\lambda r, \lambda^2 t). \quad (4)$$

At corresponding times the theta target satisfies

$$\boxed{F_\lambda = \lambda^{1/2} U_{1/\lambda} F, \quad \mathcal{M} F_\lambda(s) = \lambda^{1/2-s} \mathcal{M} F(s),} \quad (5)$$

where the source's action remains $U_a F(r) = F(r/a)$. Indeed $r^{-1/2} g(\lambda r) = \lambda^{1/2} (\lambda r)^{-1/2} g(\lambda r)$.

Equation (5) lifts to the **original two-leg chain map**

$$(\phi, \psi) \mapsto (\lambda^{1/2} U_{1/\lambda} \phi, \lambda^{-1/2} U_\lambda \psi), \quad F \mapsto \lambda^{1/2} U_{1/\lambda} F. \quad (6)$$

The first leg follows from $\Theta U_a = U_a \Theta$. On the second leg the source identity $U_a J = a J U_{1/a}$ gives $\lambda^{1/2} U_{1/\lambda} J \Theta = \lambda^{-1/2} J \Theta U_\lambda$. Both are therefore precisely the two terms of d , with the original minus sign retained. Each scaling preserves V , since the zero value and zero integral remain zero, and fixes every support mask. It consequently descends to Q . At σ the coefficient map remains the zero map on the zero coefficient space, with the original labelled support skeleton fixed; no class is deleted by this comparison.

4. Full actual-zero jets and the right-adjoint dual

Let $G(s) = 2\xi(s)$, and retain a finite reflection-stable set Z of actual zeros, with actual orders m_ρ . Write $z_\rho = s - \rho$, $A_Z = \bigoplus_{\rho \in Z} \mathcal{O}_\rho / (G)$, and let N_ρ denote multiplication by z_ρ . The finite observation is the source's map

$$(J_Z F)_\rho = \sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M} F)^{(j)}(\rho)}{j!} z_\rho^j.$$

Equation (3) supplies every coefficient of this polynomial from the original cutoff b . In particular no unproved nonvanishing or simplicity assertion enters its definition. Expanding the exponential multiplier in (5) gives

$$\boxed{(J_Z F_\lambda)_\rho = \lambda^{1/2-\rho} \sum_{k=0}^{m_\rho-1} \frac{(-\log \lambda)^k}{k!} N_\rho^k (J_Z F)_\rho.} \quad (7)$$

Equivalently, coefficient j is

$$\lambda^{1/2-\rho} \sum_{k=0}^j \frac{(-\log \lambda)^k}{k!} \frac{(\mathcal{M} F)^{(j-k)}(\rho)}{(j-k)!}.$$

This retains all nilpotent jets, not only the residue-value eigenline.

The source computes $K_\tau(\mathcal{L}_1) = [I_\eta(\mathcal{L}_1) \rightarrow I_+(\mathcal{L}_1) \oplus I_-(\mathcal{L}_1)]$ in degrees $-1, 0$, with differential $(+\text{res}, -\text{res})$, and $H^{-1}R\text{Hom}(\mathcal{T}, K_\tau(\mathcal{L}_1)) = \text{Hom}(Q, \mathcal{L}_1)$. Thus the retained residue injection evaluates the computed class by

$$\bar{h} \mapsto \ell_h, \quad \ell_h([F]) = \sum_{\rho \in \mathbb{Z}} \text{Res}_\rho \frac{h^\dagger(s) \mathcal{M}F(s)}{G(s)} ds, \quad h^\dagger(s) = \overline{h(1 - \bar{s})}. \quad (8)$$

For a scaling packet $T_\lambda h(s) = \lambda^{1/2-s} h(s)$ in each local algebra,

$$(T_\lambda h)^\dagger(s) = \lambda^{s-1/2} h^\dagger(s).$$

Multiplying by the multiplier in (5) cancels these two displayed factors exactly. Hence the original residue pairing and its trace contraction are invariant:

$$R_Z(T_\lambda h, T_\lambda k) = R_Z(h, k), \quad R_Z(T_\lambda h, G' T_\lambda k) = R_Z(h, G' k). \quad (9)$$

The derivative multiplier G' commutes with T_λ , since both act by multiplication in the same local algebra. No phase changes the source's raw skew-Hermitian residue orientation; $\mathcal{L}_1$ has not been replaced by a different coefficient line. In particular (9) concerns the explicitly displayed transport $T_\lambda = \lambda^{1/2} U_{1/\lambda}$. The original uncentered action still has $R_Z(U_a h, U_a k) = a R_Z(h, k)$; here its factor $a = 1/\lambda$ cancels the two real scalar factors $\lambda^{1/2}$ carried by the two inputs.

5. The unchanged source Weil scalar is invariant under this transport

Retain exactly

$$(g_1 * g_2)(x) = \int_0^\infty g_1(y) g_2(x/y) dy/y, \quad g^*(x) = \overline{g(x^{-1})}.$$

Then

$$(g * g^*)(x) = \int_0^\infty g(y) \overline{g(y/x)} dy/y.$$

For $g_\lambda = R_\lambda g$, substitute $v = \lambda y$ to obtain

$$\boxed{(g_\lambda * g_\lambda^*)(x) = \int_0^\infty g(v) \overline{g(v/x)} dv/v = (g * g^*)(x).} \quad (10)$$

This is equality of the entire test function, before any arithmetic functional is applied. In particular it retains every prime term and the complete archimedean subtraction in

$$W_p(f) = (\log p) \sum_{m \geq 1} p^{-m/2} (f(p^m) + f(p^{-m})),$$

$$W_{\mathbb{R}}(f) = (\log 4\pi + \gamma) f(1) + \int_1^\infty (f(x) + f(x^{-1}) - 2x^{-1/2} f(1)) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}.$$

Therefore the source scalar $\sum_v W_v(g * g^*)$, whose required sign in A0618 is nonpositive under RH, is unchanged by (4) at corresponding times. No estimate, omitted boundary term, discarded prime or replacement norm proves this: equation (10) does.

6. Time-dependent concentration and its exact extra force

Let $\lambda(t) > 0$ and $s(t) \in I$ be smooth. Define

$$u^\#(x, t) = \lambda(t) u(\lambda(t)x, s(t)), \quad p^\#(x, t) = \lambda(t)^2 p(\lambda(t)x, s(t)),$$

and transport χ by the same arguments. Equations (4)–(10) still hold instantaneously with source time $s(t)$. They do not by themselves claim a fluid-equation intertwiner for this varying parameter.

Write $X = \lambda(t)x$ and $\mathcal{N}(u, p) = (u \cdot \nabla_X)u + \nabla_X p - \nu \Delta_X u$. Direct differentiation, using the original equation $\partial_s u + \mathcal{N}(u, p) = f$, yields the complete transformed force

$$\begin{aligned} \partial_t u^\# + (u^\# \cdot \nabla_x)u^\# + \nabla_x p^\# - \nu \Delta_x u^\# \\ = \lambda s' f(X, s) + (\lambda^3 - \lambda s') \mathcal{N}(u, p)(X, s) \\ + \frac{\lambda'}{\lambda} (u^\# + x \cdot \nabla_x u^\#). \end{aligned} \quad (11)$$

Indeed $\partial_t u^\# = (\lambda'/\lambda)(u^\# + x \cdot \nabla_x u^\#) + \lambda s' \partial_s u$, whereas each nonlinear, pressure and viscosity term carries λ^3 . This proves (11) term by term. Divergence is $\lambda^2 \nabla_X \cdot u$, so incompressibility is retained. If $s' = \lambda^2$, the middle term cancels and the remaining force is

$$\lambda^3 f(X, s) + \frac{\lambda'}{\lambda} (u^\# + x \cdot \nabla_x u^\#). \quad (12)$$

The second summand is a calculated force contribution; its endpoint regularity cannot be inferred from the original force's regularity.

For the test and theta amplitude, direct differentiation gives

$$\begin{aligned} \partial_t g^\# &= \frac{\lambda'}{\lambda} Dg^\# + s' R_\lambda \partial_s g, \\ \partial_t F^\# &= \frac{\lambda'}{\lambda} (D + \tfrac{1}{2}) F^\# + s' \lambda^{1/2} U_{1/\lambda} \partial_s F. \end{aligned} \quad (13)$$

Writing w for the Mellin variable in this sentence, to distinguish it from the source-time function $s(t)$, the first transformed generator is exactly $(\lambda'/\lambda)(\frac{1}{2} - w)$; on the ρ -jet it is $(\lambda'/\lambda)((\frac{1}{2} - \rho)I - N_\rho)$. Equation (10) shows at each time that the transported Weil scalar equals the source scalar at $s(t)$. Thus parameter variation in λ alone contributes zero to that scalar, whereas changes in the actual source profile or a cutoff not transported by this formula need their own calculation.

Proven scope

We have completed the exact transport of A0618's cutoff construction into the original A_τ complex, Q , the full actual-zero jets and the retained right-adjoint dual. Pure transported parabolic concentration preserves the entire autocorrelation and the source Weil scalar; this does not rule out effects from the actual NS solution's changing anisotropic profile, interactions or cutoff. The explicit extra force in (11)–(12) shows why a time-dependent coordinate transport cannot be called a new smooth-forced solution without calculating that term. Neither a vanished supported quotient kernel nor a global weight estimate has been assumed.

Chapter 10

Retained physical formulas: exact completed comparisons

Finished calculations on the retained physical and arithmetic formulas

These calculations use the objects and constants displayed in A0656 and A0706. They close bounded algebraic omissions. They do not identify the modular surface, the physical wedge, or the fluid/gravity expansion with the later split-base cohomology by assumption. The source coordinates below are LF coordinates in the pinned transcript segment.

1. The modular scattering coefficient retains every zero multiplicity

A0656, lines 2964–3096, displays

$$S(k) = \pi^{-2ik} \frac{\Gamma(\frac{1}{2} + ik)}{\Gamma(\frac{1}{2} - ik)} \frac{\zeta(1 + 2ik)}{\zeta(1 - 2ik)}, \quad E_* = \frac{\hbar^2}{2mR^2} > 0, \quad \mathcal{E}(k) = E_*(k^2 + \tfrac{1}{4}).$$

Take any nontrivial zeta zero in the specified lower half-plane, with its actual multiplicity:

$$\rho = a - i\gamma, \quad 0 < a < 1, \quad \gamma > 0, \quad q = \text{ord}_{s=\rho} \zeta(s) \geq 1.$$

Solving the denominator equation exactly gives

$$k_\rho = \frac{\gamma}{2} - \frac{i(1-a)}{2}, \quad 1 - 2ik_\rho = \rho, \quad 1 + 2ik_\rho = 2 - \rho, \quad \tfrac{1}{2} - ik_\rho = \tfrac{\rho}{2}, \quad \tfrac{1}{2} + ik_\rho = 1 - \tfrac{\rho}{2}.$$

Both Gamma arguments have positive real part. Gamma is finite and nonzero there. Also $\Re(2 - \rho) = 2 - a > 1$, where the absolutely convergent Euler product is nonzero. Consequently the analytic numerator

$$N(k) = \pi^{-2ik} \frac{\Gamma(\frac{1}{2} + ik)}{\Gamma(\frac{1}{2} - ik)} \zeta(1 + 2ik)$$

is nonzero at k_ρ . Taylor expansion at the actual zero gives

$$\zeta(1 - 2ik) = \frac{\zeta^{(q)}(\rho)}{q!} (-2i)^q (k - k_\rho)^q + O((k - k_\rho)^{q+1}).$$

Thus the pole order is exactly q , and its leading Laurent coefficient is exactly

$$a_{-q} = \frac{q!N(k_\rho)}{(-2i)^q \zeta^{(q)}(\rho)} \neq 0.$$

The map to energy has derivative $2E_*k_\rho \neq 0$, so it is locally invertible at this pole. The energy-plane pole has the same order and leading coefficient

$$b_{-q} = a_{-q}(2E_*k_\rho)^q.$$

Direct squaring, without removing either real correction, gives

$$\mathcal{E}_\rho = E_* \left[\frac{\gamma^2}{4} - \frac{(1-a)^2}{4} + \frac{1}{4} \right] - i \frac{E_*\gamma(1-a)}{2}.$$

Let $\sum_{j=1}^q b_{-j}(\mathcal{E} - \mathcal{E}_\rho)^{-j}$ be the complete principal part. The positively oriented local contour contribution with the explicitly specified convention $(2\pi i)^{-1} \oint e^{-i\mathcal{E}t/\hbar} \mathbf{S}(k(\mathcal{E})) d\mathcal{E}$ equals

$$e^{-i\mathcal{E}_\rho t/\hbar} \sum_{j=1}^q \frac{b_{-j}}{(j-1)!} \left(-\frac{it}{\hbar} \right)^{j-1}.$$

Indeed, expand the exponential in powers of $\mathcal{E} - \mathcal{E}_\rho$; its coefficient of power $j-1$ is the displayed factor, which is exactly the residue paired with the order- j pole. A lower-half-plane contour used in a global inverse transform may carry a different overall orientation; the convention above fixes this sign and makes no unproved contour deformation. Every multiple zero contributes the complete polynomial of degree $q-1$, with nonzero top coefficient, multiplying the decaying exponential. Its exponential amplitude factor is

$$\exp \left[-\frac{E_*\gamma(1-a)t}{2\hbar} \right].$$

The source's simple-pole width $\Gamma_\rho = E_*\gamma(1-a)$ is therefore retained, and the previously undisplayed multiplicities are now explicit. This calculation concerns the principal part of the continued scattering coefficient. It does not assert a complete time-domain expansion for every wavepacket, or convert poles of a continued coefficient into nonreal eigenvalues of a self-adjoint Hamiltonian. Those claims require a specified inversion contour, packet class, and bounds on the remaining integral.

2. Channel redistribution is exact and need not be positive

A0656, lines 3104–3137, uses a Hilbert space $\mathcal{H}$, a unitary $U : \mathcal{H} \rightarrow \mathcal{H}$, and an orthogonal projection $P = P^* = P^2$. For every $\psi \in \mathcal{H}$,

$$\langle \psi, (U^*PU - P)\psi \rangle = \langle U\psi, PU\psi \rangle - \langle \psi, P\psi \rangle = \|PU\psi\|^2 - \|P\psi\|^2.$$

Moreover,

$$\frac{1}{2}U^*[2P - I, U] = \frac{1}{2}U^*((2P - I)U - U(2P - I)) = U^*PU - P.$$

No domain issue occurs: both operators are bounded. Unitarity does not imply positivity of this operator. In $\mathbb{C}^2$ with its usual inner product, take

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$U^*PU - P = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The two coordinate unit vectors have expectations -1 and $+1$. This retains the precise useful interpretation of a change in the selected observable. Any use of this identity to prove an arithmetic positivity statement still requires the actual arithmetic subspace and pairing; the identity by itself supplies no such sign.

3. Equal marginal histories leave a phase family even after cross-energy measurement

A0706, lines 3835–3901, displays a thermofield vector and the partial-trace map. Work on the exact countable energy eigenbasis appearing there. Put

$$p_n = \frac{e^{-\beta E_n}}{Z(\beta)}, \quad \rho_\beta = \sum_n p_n |n\rangle\langle n|, \quad \sum_n p_n = 1.$$

For every real phase sequence $\theta = (\theta_n)$, the norm-convergent vector

$$|\Psi_\theta\rangle = \sum_n \sqrt{p_n} e^{i\theta_n} |n\rangle_L |n\rangle_R$$

has norm one because its summands are mutually orthogonal. In the partial trace of $|\Psi_\theta\rangle\langle\Psi_\theta|$, the right basis inner product is $\langle m|n\rangle = \delta_{mn}$; hence both marginal density operators are exactly ρ_β . The product density $\rho_\beta \otimes \rho_\beta$ has the same marginals. This proves the source's many-to-one statement.

When $\sum_n p_n E_n^2 < \infty$, diagonal summation gives

$$\langle H_L H_R \rangle_{\Psi_\theta} = \sum_n p_n E_n^2, \quad \langle \delta H_L \delta H_R \rangle_{\Psi_\theta} = \sum_n p_n E_n^2 - \left(\sum_n p_n E_n \right)^2.$$

This value is independent of all phases. It is zero in the product state after centering because the expectation factors. Thus the particular cross-energy observable in A0706 distinguishes the displayed product state from a nonzero-variance thermofield vector, but it does not recover the phase pattern inside the entire pure-state family.

An explicit bounded cross observable that does retain this missing phase is

$$O_{mn} = (|m\rangle\langle n|)_L \otimes (|m\rangle\langle n|)_R, \quad m \neq n.$$

Direct action on the series gives

$$\langle \Psi_\theta | O_{mn} | \Psi_\theta \rangle = \sqrt{p_m p_n} e^{i(\theta_n - \theta_m)}.$$

Its Hermitian real and imaginary parts measure respectively the cosine and sine of this relative phase, with the exact factor $\sqrt{p_m p_n}$. This is a concrete continuation beyond the source's single cross-energy observable.

For completeness, partial trace after arbitrary local unitary driving obeys

$$\text{Tr}_R[(U_L \otimes U_R) \rho (U_L^* \otimes U_R^*)] = U_L (\text{Tr}_R \rho) U_L^*.$$

To prove it, pair both sides with any bounded A_L ; cyclicity of the trace replaces $(A_L \otimes I)$ by $(U_L^* A_L U_L \otimes I)$, and the defining property of partial trace gives equality of all such expectations. The

same argument applies on the right. Therefore equal initial marginals give equal histories for the same specified local unitary drives. This is an obstruction to reconstructing the joint state from those histories alone. It does not obstruct a reconstruction that includes the displayed cross observables, a joint evolution law, or an independently specified joint state.

4. The arithmetic occupation map, modular clock, and trace-class boundary

A0706, lines 3903–3980, gives the BC Hamiltonian

$$H|n\rangle = E_* \log n |n\rangle, \quad E_* > 0.$$

Let $\mathfrak{P}$ denote the primes. The bosonic Fock occupation basis over $\ell^2(\mathfrak{P})$ is indexed by sequences $m = (m_p)_{p \in \mathfrak{P}}$ of nonnegative integers of finite total sum. Such a sequence has finite support. Unique factorization is a bijection

$$n \longleftrightarrow (v_p(n))_{p \in \mathfrak{P}}, \quad (m_p)_p \longmapsto \prod_p p^{m_p}.$$

Sending one orthonormal basis to the other gives a surjective isometry $\mathcal{U}$, hence a unitary. On its full diagonal-operator domain,

$$\mathcal{U}H\mathcal{U}^* = \sum_p E_* \log p N_p.$$

The equality of domains is explicit: a Fock coefficient sequence c_m belongs to this domain exactly when

$$\sum_m \left(\sum_p E_* m_p \log p \right)^2 |c_m|^2 < \infty,$$

which is the image of $\sum_n (E_* \log n)^2 |c_n|^2 < \infty$. Thus this is an operator equality, including its domain, rather than a formal equality of symbols.

Write $\sigma = \beta E_*$. The positive trace is

$$\mathrm{Tr}(e^{-\beta H}) = \sum_{n \geq 1} n^{-\sigma}.$$

It is finite exactly when $\sigma > 1$, by comparison with $\int_1^\infty x^{-\sigma} dx$; for $\sigma \leq 0$ the terms do not even tend to zero. For any finite prime set F , summing the independent geometric occupation series gives

$$\sum_{\mathrm{supp}(m) \subset F} \prod_{p \in F} p^{-\sigma m_p} = \prod_{p \in F} (1 - p^{-\sigma})^{-1} \quad (\sigma > 0).$$

Increasing F through all primes and using nonnegative monotone sums recovers $\sum_n n^{-\sigma}$. Hence the full Gibbs density and doubled vector exist in the displayed Hilbert-space representation exactly for $\beta E_* > 1$. In that range,

$$|\mathrm{TFD}_\beta\rangle = \zeta(\beta E_*)^{-1/2} \sum_{n \geq 1} n^{-\beta E_*/2} |n\rangle_L |n\rangle_R,$$

and its prime factors are exactly those displayed in A0706. Every energy moment used in the preceding calculation is finite because $\sum_n n^{-\sigma} (\log n)^j < \infty$ for every fixed nonnegative integer j and $\sigma > 1$, as follows from the same integral comparison after $x = e^y$.

The frequency dictionary is $\omega_p = E_* \log p / \hbar$. Substitution into the source's thermal-coefficient match gives

$$\beta E_* \log p = \frac{2\pi c \omega_p}{a} \iff \beta = \frac{2\pi c}{\hbar a}.$$

No prime-dependent choice remains. Combining this equality with the actual trace-class domain yields the precise domain of this global vector correspondence:

$$0 < a < \frac{2\pi c E_*}{\hbar}.$$

At the upper endpoint, $\beta E_* = 1$, the vector normalization diverges through the harmonic series, although every individual prime's geometric factor is finite. This is an omitted global boundary in the retained response. It is a limitation of this trace-class arithmetic representation, not a theorem that an arbitrary Rindler observer lacks a KMS state.

The modular sign and proper clock also agree exactly with the source convention. On the thermofield representation the modular operator for the left algebra is

$$\Delta = \rho_\beta \otimes \rho_\beta^{-1}.$$

For a bounded left observable, the convention used in A0706 gives

$$\Delta^{-is}(A \otimes I)\Delta^{is} = (e^{i\beta s H} A e^{-i\beta s H}) \otimes I.$$

The scalar partition factors cancel. Physical Heisenberg time is $\alpha_t(A) = e^{itH/\hbar} A e^{-itH/\hbar}$, so $t = \beta \hbar s = 2\pi c s / a$, exactly the source relation $s = a\tau / (2\pi c)$. This proves a thermal representation and clock correspondence. It does not construct a spatial observable net, a relativistic stress tensor, or a cooling generator.

5. The force pullback has a real terminal-control issue

A0706, lines 3660–3697, constructs a one-form through the actual preterminal fluid flow:

$$A_t = -(\Phi_t^{-1})^* \int_0^t \Phi_s^* f_s ds, \quad h_{00} = -2v^j A_j.$$

Pulling back by Φ_t , differentiating, and using the chain rule for a transported one-form gives

$$\Phi_t^*[(\partial_t + \mathcal{L}_v)A_t] = -\Phi_t^* f_t, \quad \partial_t A_i + v^j \partial_j A_i + A_j \partial_i v^j = -f_i.$$

Therefore the boundary-force expression is exactly

$$\frac{1}{2} \partial_i h_{00} - \partial_t A_i + v^j (\partial_i A_j - \partial_j A_i) = -\partial_t A_i - v^j \partial_j A_i - A_j \partial_i v^j = f_i.$$

In Lagrangian coordinates the full derivative factor is

$$A_t(\Phi_t(x)) = -D\Phi_t(x)^{-T} \int_0^t D\Phi_s(x)^T f_s(\Phi_s(x)) ds.$$

This identifies exactly what must be computed on the source's actual flow, rather than silently inferring terminal regularity from f .

The general implication from terminal-flat forcing to smooth coefficients is false even among smooth preterminal forced incompressible NS solutions. The following explicit comparison tests that implication without replacing the source's finite-energy field. Retain arbitrary viscosity $\nu > 0$, source time $0 \leq t < 1$, and put

$$u = 1 - t, \quad g(t) = e^{-1/u^2}, \quad v(t, x) = \left(-\frac{x_1}{u}, \frac{x_2}{u}, 0 \right), \quad p(t, x) = g(t)x_1 - \frac{x_2^2}{u^2}, \quad f = g(t) dx^1.$$

The force extends smoothly by zero to $t = 1$, with all terminal time derivatives zero: every derivative is a polynomial in u^{-1} times e^{-1/u^2} , which tends to zero. The velocity has divergence $-1/u + 1/u = 0$ and zero Laplacian. Its first material-acceleration component is zero and its second is $2x_2/u^2$; adding $\nabla p = (g(t), -2x_2/u^2, 0)$ gives exactly f . Thus

$$\partial_t v + v \cdot \nabla v - \nu \Delta v + \nabla p = (g(t), 0, 0)$$

for every retained $\nu > 0$.

The flow and pulled-back force are

$$\Phi_t(x) = (ux_1, u^{-1}x_2, x_3), \quad \Phi_s^* f_s = (1 - s)g(s) dx^1.$$

Writing $I(t) = \int_0^t (1 - s)g(s) ds$, the prescribed construction yields

$$A_t = -\frac{I(t)}{u} dx^1, \quad h_{00} = -\frac{2I(t)x_1}{u^2}.$$

Here $I(t)$ tends to the strictly positive finite number $\int_0^1 (1 - s)e^{-1/(1-s)^2} ds$, so these coefficients do not extend smoothly to the endpoint. This example is not compactly supported and does not have finite total kinetic energy; it therefore does not settle the actual NS manuscript's terminal behavior. It proves exactly that the displayed preterminal force-map identity alone cannot make that inference. The strongest continuation remains evaluating its full flow derivatives and the gravitational corrections on the actual retained NS profile.

Chapter 11

Source viscosity and cutoff clock

Exact source-viscosity and cutoff-clock morphism

The retained A0656 states a viscosity-one NS stage (LF2866). A0706 displays cutoff viscosity r_c (LF3571–3577) and later the coordinate embedding $\tau = t$ with unchanged spatial coordinates (LF3611–3627). The retained response does not display the equation-preserving conversion between those parameters. The linked full research package was not acquired by this bounded worker; this finding is about the retained response.

Keep the original source viscosity $\nu_s > 0$, source coordinates (t, x) , and the complete forced source equations

$$\partial_t u_i + u^j \partial_j u_i - \nu_s \Delta u_i + \partial_i p = f_i, \quad \partial_i u^i = 0.$$

Keep the target cutoff parameter $r_c > 0$ and its coordinates (τ, x) . Set

$$b = \frac{r_c}{\nu_s}, \quad t = b\tau, \quad v_i(\tau, x) = b u_i(b\tau, x),$$

$$P(\tau, x) = b^2 p(b\tau, x), \quad f_i^{\text{cut}}(\tau, x) = b^2 f_i(b\tau, x).$$

Here b has the time-coordinate conversion units required by the displayed equations; no equality of source and cutoff clocks is presumed. The map on points is

$$j_c^{\nu_s, r_c} : (t, x) \mapsto \left(\tau = \frac{\nu_s}{r_c} t, r = r_c, x \right).$$

Direct differentiation gives

$$\partial_\tau v_i = b^2 \partial_t u_i, \quad v^j \partial_j v_i = b^2 u^j \partial_j u_i, \quad r_c \Delta v_i = r_c b \Delta u_i = b^2 \nu_s \Delta u_i, \quad \partial_i P = b^2 \partial_i p.$$

All right-hand source fields are evaluated at $(b\tau, x)$. Consequently

$$\partial_\tau v_i + v^j \partial_j v_i - r_c \Delta v_i + \partial_i P = f_i^{\text{cut}}, \quad \partial_i v^i = b \partial_i u^i = 0.$$

The map is invertible: $\tau = t/b$, $u_i(t, x) = b^{-1} v_i(t/b, x)$, $p(t, x) = b^{-2} P(t/b, x)$, and $f_i(t, x) = b^{-2} f_i^{\text{cut}}(t/b, x)$. This is an exact bijection between the displayed forced-equation data, with the original viscosity retained. It is not a modification of the source solution.

The source's cutoff proper clock is $dt_c = \sqrt{r_c} d\tau/c$. With origins chosen at source $t = 0$, the same morphism gives

$$t_c = \frac{\nu_s}{c\sqrt{r_c}} t.$$

The original source endpoint ($t = 1, x = 0$) is therefore sent, under this particular same-spatial-coordinate map, to

$$\left(\tau = \frac{\nu_s}{r_c}, r = r_c, x = 0 \right), \quad t_c = \frac{\nu_s}{c\sqrt{r_c}}.$$

The simple identification $\tau = t$, $v = u$, $P = p$, and unchanged spatial coordinates intertwines these equations only in the parameter-matching case $r_c = \nu_s$: otherwise the target residual differs by $(\nu_s - r_c)\Delta u_i$. One may instead use a different explicitly typed spatial-coordinate conversion, but it must be written and its action on the source support, wavelengths, force, and norms must be carried through.

For this exact map, the source spatial support and wave numbers at corresponding times are unchanged, whereas

$$\begin{aligned} \int |v(\tau, x)|^2 dx &= b^2 \int |u(b\tau, x)|^2 dx, \\ \int_{\tau_0}^{\tau_1} \int |\nabla v|^2 dx d\tau &= b \int_{b\tau_0}^{b\tau_1} \int |\nabla u|^2 dx dt. \end{aligned}$$

These follow from the pointwise factor b in v and the exact change $dt = b d\tau$. The displayed Brown–York coefficient therefore becomes

$$[\mathfrak{B}_2(v, P)]_{\tau\tau} = \frac{c^4 b^2}{16\pi G_5 \sqrt{r_c}} |u(b\tau, x)|^2.$$

Finally, this change of clock does not remove the actual higher-derivative problem. At the corresponding source Fourier wave number k , the magnitude ratio of the displayed correction $-\frac{3}{2}r_c^2 \Delta^2 v_i$ to $-r_c \Delta v_i$ remains exactly $\frac{3}{2}r_c |k|^2$. Its growth must still be calculated on the source’s actual decreasing wavelengths.

Chapter 12

Coordinated arithmetic determinant transport

This calculation retains the arithmetic coefficient complex over the absolute pointed base $\mathbf{F}_{1,\tau} = \{\tau, 1\}$. It computes the signed arithmetic/reference determinant term on the original finite theta source. The new formulas are a complete first and second variation, an exact four-window projection trace, and a quantitative bound in terms of the actual common source Gram. No arithmetic source multiplier is assumed bounded above or bounded away from zero.

Original objects and the fixed comparison

The retained tau-base construction has the four-point chart $+, - < \eta < \sigma$ and the coefficient sheaf $\mathcal{T}$ with $(\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_\eta, \mathcal{T}_\sigma) = (V, V, \mathcal{B}, 0)$. Its two nonzero restrictions are Θ and $J\Theta = \Theta\mathcal{F}$. The actual projective resolution gives

$$\mathbf{A}_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}], \quad H^1 \mathbf{A}_\tau(\mathcal{T}) = Q = \mathcal{B}/\Theta V. \quad (\text{AT1})$$

These are the fixed source objects of the original Tau Base note, equations (18)–(25), supplied in the source closure. The tensor resolution in its equations (46)–(48) retains the original differential signs and identifies its top cohomology with $Q^{\otimes k}$. The operation here uses that pushforward, with its arithmetic classes. Its separately computed restriction to σ remains an available comparison; it is not substituted for (AT1).

Fix an original actual packet h of full zero orders of $g = 2\xi$, and a positive integer k . Retain the full tensor-sum operator, its original cyclic annihilator $\chi = \chi_{h,k}$ of degree $q \geq 1$, and $E = \mathbb{C}[S]/(\chi)$ in the increasing-power remainder frame. The original construction supplies

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V}P = \eta_{h,k}J_N P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_h^{(k)})1, \quad (\text{AT2})$$

where $v_h = j_h(g/h)$ is the full invertible Taylor unit, J_N is literal remainder modulo χ , and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the original injective full-jet observation, $E_h = \mathbb{C}[s]/(h)$. The original section $\sigma_h : E_h \hookrightarrow Q$ gives the separate arithmetic cohomology injection and equality

$$q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}J_N P, \quad \sigma_h^{\otimes k}\eta_{h,k} : E \hookrightarrow Q^{\otimes k}. \quad (\text{AT2a})$$

The supplied arithmetic-input and cyclic-sum proofs construct these maps; all operations below keep them fixed. In particular the sum coordinate is $S = k/2 + iu$, with no division by k .

For $N \geq q - 1$, set $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and let $B_N : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$ be multiplication by the literal monic χ . Write $s_N : E \rightarrow \mathcal{P}_N$ for the degree-less-than- q remainder section. Polynomial division proves

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad J_N s_N = I_E. \quad (\text{AT3})$$

The arithmetic norm has density $m_1(u) = m_{h,k}(u) = w_h^*(u)$, $w_h(u) = |(g/h)(1/2 + iu)|^2/(2\pi)$. The retained Gamma reference has density

$$m_0(u) = r_{1/4,k}(u) = \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a). \quad (\text{AT4})$$

Its mass is $(2\pi)^{k/2}$; the arithmetic mass is the original μ_h^k . The source proofs included with this calculation establish finite moments of every degree and positivity almost everywhere for both densities. The calculation below only integrates polynomials against these two actual measures and their sum. It needs no estimate on the pointwise ratio m_1/m_0 .

For $0 \leq t \leq 1$ define, in the unchanged coefficient frame,

$$\begin{aligned} m_t &= (1-t)m_0 + tm_1, \quad \Delta m = m_1 - m_0, \quad v_N(u) = (1, S(u), \dots, S(u)^N), \\ M_N(t) &= \int v_N(u)^* v_N(u) m_t(u) du, \quad \Delta M_N = M_N(1) - M_N(0), \\ H_N(t) &= B_N^* M_N(t) B_N. \end{aligned} \quad (\text{AT5})$$

Every $M_N(t)$ is positive definite: a nonzero polynomial has finitely many real zeros, and m_t is positive almost everywhere. B_N is injective, so $H_N(t)$ is positive definite on its actual relation domain. When $N = q - 1$, that domain is zero; its determinant is one, its unique endomorphism is its inverse, and all compositions through that domain are zero. The real parameter t interpolates norms on (AT3); it changes neither the absolute base nor a cochain map.

Canonical sections, determinant lines and full variations

Theorem 12.1. *On the fixed exact sequence (AT3), its least-norm section and quotient Gram are*

$$\begin{aligned} R_N(t) &= s_N - B_N H_N(t)^{-1} B_N^* M_N(t) s_N, \\ J_N R_N(t) &= I_E, \quad B_N^* M_N(t) R_N(t) = 0, \\ G_N(t) &= R_N(t)^* M_N(t) R_N(t) = (J_N M_N(t)^{-1} J_N^*)^{-1}, \\ V_N(t) &:= \det G_N(t) = \frac{\det M_N(t)}{\det H_N(t)}. \end{aligned} \quad (\text{AT6})$$

In the original polynomial frame the exact first and second variations are

$$\begin{aligned} \dot{R}_N &= -B_N H_N^{-1} B_N^* \Delta M_N R_N, \quad \dot{G}_N = R_N^* \Delta M_N R_N, \\ \ddot{G}_N &= -2R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N \preceq 0. \end{aligned} \quad (\text{AT7})$$

The exact logarithmic variation, with both trace terms retained, is

$$\begin{aligned} \frac{d}{dt} \log V_N &= \text{Tr}(M_N^{-1} \Delta M_N) - \text{Tr}(H_N^{-1} B_N^* \Delta M_N B_N) \\ &= \text{Tr}(G_N^{-1} R_N^* \Delta M_N R_N). \end{aligned} \quad (\text{AT8})$$

Its second derivative is

$$\begin{aligned} \frac{d^2}{dt^2} \log V_N &= -2 \text{Tr}(G_N^{-1} R_N^* \Delta M_N B_N H_N^{-1} B_N^* \Delta M_N R_N) \\ &\quad - \text{Tr}(G_N^{-1} \dot{G}_N G_N^{-1} \dot{G}_N) \leq 0. \end{aligned} \quad (\text{AT9})$$

Proof. Every lift of $x \in E$ is $s_N x + B_N y$. Expanding its squared norm and completing its finite quadratic expression gives the unique minimizer $y = -H_N^{-1} B_N^* M_N s_N$. This proves the first two lines of (AT6). For the inverse expression, every polynomial vector p decomposes as $B_N y + R_N J_N p$.

Orthogonality therefore gives $R_N^* M_N p = G_N J_N p$ for every p . Taking adjoints yields $M_N R_N = J_N^* G_N$. Substituting this into $J_N R_N = I_E$ gives $(J_N M_N^{-1} J_N^*) G_N = I_E$, which proves the inverse expression.

The determinant calculation preserves orientation. Put $r = N - q + 1$ and $L_N = [B_N \ s_N]$. In the ordered monomial bases, $[s_N \ B_N]$ has determinant one, since χ is monic and the lower diagonal block is triangular with diagonal one. Thus $\det L_N = (-1)^{qr}$. The upper-left block of $L_N^* M_N L_N$ is H_N , and its Schur complement is G_N . Consequently $\det H_N \det G_N = |(-1)^{qr}|^2 \det M_N = \det M_N$. Equivalently the map

$$\det \mathcal{P}_{N-q} \otimes \det E \longrightarrow \det \mathcal{P}_N, \quad (y_1 \wedge \cdots \wedge y_r) \otimes (x_1 \wedge \cdots \wedge x_q) \longmapsto B_N y_1 \wedge \cdots \wedge B_N y_r \wedge s_N x_1 \wedge \cdots \wedge s_N x_q \quad (\text{AT10})$$

is the fixed determinant-line isomorphism. Replacing any lift by a relation does not change its wedge. Orthogonal lifts give the product metric just computed. This proves the determinant identity including the empty relation space, without changing any scalar mass.

Differentiate $J_N R_N = I$; then $\dot{R}_N$ lies in $\text{im } B_N$. Differentiating $B_N^* M_N R_N = 0$ determines its unique coefficient and proves the first identity of (AT7). On differentiating $R_N^* M_N R_N$, the two terms involving $\dot{R}_N$ vanish by that orthogonality. A second differentiation gives the two equal negative terms in (AT7), since ΔM_N is Hermitian. This proves the claimed second derivative and its sign. Applying the finite determinant derivative to (AT6) gives the first equality of (AT8); applying it to G_N gives the second. One more derivative gives (AT9). The first trace there is the squared Hilbert–Schmidt norm of $H_N^{-1/2} B_N^* \Delta M_N R_N G_N^{-1/2}$; the second is the squared Hilbert–Schmidt norm of $G_N^{-1/2} \dot{G}_N G_N^{-1/2}$. These identities establish their signs and retain the original forms. All inverse matrices are smooth on a neighborhood of the compact interval $[0, 1]$, by positivity and continuity. No infinite-dimensional inverse has been used. $\square$

The derivative $\dot{R}_N$ in (AT7) is a literal polynomial relation. Applying the original $\mathcal{V}$ in (AT2) therefore retains its original theta primitive and sends its arithmetic class to zero; $J^{(k)} \mathcal{V} R_N(t) = \eta_{h,k}$ for every t . At each nonempty original support label A , the cochain identities lift by $(A, P) \mapsto (A, fP)$, while $\tau \mapsto \tau$. Thus a relation-valued derivative has a supported zero class at label A . Its source vector and its full nonnegative energy in (AT7)–(AT9) remain present. These formulas compute their contribution rather than deleting it after taking cohomology.

The signed four-endpoint term on one retained source

Let $N_* = 2q$. Every space in (AT3) for the four endpoints sits in $\mathcal{P}_{N_*}$ by its actual coefficient inclusion. Give this common space the form $M_*(t) = M_{N_*}(t)$. Write $P_N(t)$ for its orthogonal projection onto $\mathcal{P}_N$, and $Q_N(t)$ for its orthogonal projection onto the relation subspace $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$. The latter is zero at $N = q - 1$. For the coefficient inclusion $I_N : \mathcal{P}_N \hookrightarrow \mathcal{P}_{N_*}$, the formulas, with their domains retained, are

$$P_N = I_N M_N^{-1} I_N^* M_*, \quad Q_N = I_N B_N H_N^{-1} B_N^* I_N^* M_*, \quad C(t) = M_*(t)^{-1} \Delta M_*. \quad (\text{AT11})$$

Both P_N and Q_N are $M_*(t)$ -self-adjoint idempotents with their displayed images, verified by substitution. Since $\mathcal{R}_N \subset \mathcal{P}_N$, $P_N Q_N = Q_N P_N = Q_N$. Their difference is the orthogonal projection onto the finite canonical quotient section and has trace q . Cyclicity of finite matrix trace in (AT11) proves the exact common-source version of (AT8):

$$\frac{d}{dt} \log V_N(t) = \text{Tr}((P_N(t) - Q_N(t))C(t)). \quad (\text{AT12})$$

Put $i_0 = q - 1$, $i_1 = q$, $j_0 = 2q - 1$, and $j_1 = 2q$. Define the two positive operators

$$\begin{aligned} U(t) &= (Q_{j_0} - Q_{i_0}) + (Q_{j_1} - Q_{i_1}), \\ W(t) &= (P_{j_0} - P_{i_0}) + (P_{j_1} - P_{i_1}). \end{aligned} \quad (\text{AT13})$$

Each difference in this display is an orthogonal projection, because the two spaces in its pair are nested in the same Hilbert space. Each has rank q . The sums need not be projections and are not asserted to be orthogonal sums; they are positive operators with $\text{Tr } U = \text{Tr } W = 2q$.

Write

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{i_0}(t)V_{i_1}(t)}{V_{j_0}(t)V_{j_1}(t)}, \quad T_N = V_N(1)/V_N(0), \\ \Delta \mathcal{B} &= \mathcal{B}(1) - \mathcal{B}(0) = \log \frac{T_{q-1}T_q}{T_{2q-1}T_{2q}}. \end{aligned} \tag{AT14}$$

Subtracting the four exact formulas (AT12), with their original signs, and integrating gives

$$\Delta \mathcal{B} = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt. \tag{AT15}$$

This is the actual arithmetic correction in the audit's formula (54). It compares the two added relation windows with the two added polynomial windows. Concavity of each individual $\log V_N(t)$ in (AT9) does not assign a sign to their signed combination in (AT15); that combination has now been computed.

There is also an exact pointwise integral, using no division by the arithmetic density. Define

$$\begin{aligned} k_N(t, u) &= v_N(u)M_N(t)^{-1}v_N(u)^*, \\ r_N(t, u) &= |\chi(S(u))|^2 v_{N-q}(u)H_N(t)^{-1}v_{N-q}(u)^*, \\ d_N(t, u) &= k_N(t, u) - r_N(t, u), \quad r_{q-1}(t, u) = 0, \\ d_N(t, u) &= v_N(u)R_N(t)G_N(t)^{-1}R_N(t)^*v_N(u)^* \geq 0. \end{aligned} \tag{AT16}$$

The last equality follows by writing the orthogonal splitting of M_N^{-1} as $B_N H_N^{-1} B_N^* + R_N G_N^{-1} R_N^*$; multiplying by M_N and evaluating on $\text{im } B_N \oplus \text{im } R_N$ proves that inverse identity. In particular $\int d_N(t, u)m_t(u) du = q$. Consequently (AT8) also reads

$$\frac{d}{dt} \log V_N(t) = \int d_N(t, u) \Delta m(u) du, \quad \Delta \mathcal{B} = \int_0^1 \int (d_{i_0} + d_{i_1} - d_{j_0} - d_{j_1})(t, u) \Delta m(u) du dt. \tag{AT17}$$

All coefficients of these fixed-degree polynomials in u and $\bar{S}(u)$ are bounded for $t \in [0, 1]$. Since $|\Delta m| \leq m_0 + m_1$ and these measures have the required finite moments, absolute integrability proves the exchange of integrals. This remains valid at every zero of m_1 .

An exact finite-source bound for the correction

Let $b_{\min}$ and $b_{\max}$ be the extreme eigenvalues of the actual positive $M_*(0)$ -self-adjoint operator $D_* = M_*(0)^{-1}M_*(1)$, and set $\kappa_* = b_{\max}/b_{\min}$. These are finite original source data at the common cutoff $N_* = 2q$. They are not presumed uniform in either the packet or k .

Theorem 12.2. *The exact correction on the original finite source satisfies*

$$|\Delta \mathcal{B}| \leq 2q \log \kappa_*. \tag{AT18}$$

Equivalently, retaining the original Gamma and arithmetic endpoint quantities,

$$\mathcal{B}_{h,k}^\Gamma - 2q \log \kappa_* \leq \mathcal{B}_{h,k}^{\text{ar}} \leq \mathcal{B}_{h,k}^\Gamma + 2q \log \kappa_*. \tag{AT19}$$

Proof. The operator $C(t)$ is $M_*(t)$ -self-adjoint. If its extreme eigenvalues are $c_{\min}(t), c_{\max}(t)$, positivity of U, W and their exact traces give

$$|\text{Tr}((U - W)C)| \leq 2q(c_{\max}(t) - c_{\min}(t)). \tag{AT20}$$

For example $\text{Tr}(UC) \leq c_{\max} \text{Tr} U$ follows by taking the trace of the product of U and the positive operator $c_{\max}I - C$ in the original Hilbert space. The lower bound is identical, and subtraction proves (AT20), without assuming that U, W, C commute.

In the fixed coefficient space,

$$C(t) = ((1-t)I + tD_*)^{-1}(D_* - I).$$

Thus each eigenvalue is $c_b(t) = (b-1)/(1-t+tb)$ for an eigenvalue $b > 0$ of D_* . Its derivative with respect to b is $(1-t+tb)^{-2} > 0$. The extreme values therefore correspond to the same $b_{\min}, b_{\max}$ throughout the interval, including repeated eigenvalues. Direct integration gives

$$\int_0^1 (c_{\max}(t) - c_{\min}(t)) dt = \log b_{\max} - \log b_{\min} = \log \kappa_*. \quad (\text{AT21})$$

Integrating (AT20) in the exact identity (AT15) proves (AT18). No pointwise multiplier bound or asymptotic estimate has entered. $\square$

When $M_*(1) = cM_*(0)$ with its literal $c > 0$, the exact source form at time t is $(1-t+tc)M_*(0)$. Each quotient determinant is $(1-t+tc)^q V_N(0)$, so all four factors in (AT14) cancel and $\Delta\mathcal{B} = 0$. This is also forced by $\kappa_* = 1$ in (AT18). The individual masses and volume factors remain as written; cancellation occurs only in this specified ratio. An arbitrary fixed change of the coefficient frame conjugates D_* , transports every subspace and its projection, and multiplies each quotient determinant by the same squared frame determinant. Hence (AT15)–(AT21) preserve their original values under that explicitly stated map.

For the reflection-stable quartet calculation of the supplied Gamma audit, let its actual displacement be $\delta > 0$ and retain $q = [1 + k(m-1)](k+1)^2$. Its fully written Gamma theorem gives $\mathcal{B}_{h,k}^\Gamma \geq 4q \log(\delta k / (2\sqrt{5}))$. Substitution in (AT19) gives the new quantitative comparison on the same source:

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log\left(\frac{\delta k}{2\sqrt{5}}\right) - 2q \log \kappa_*, \quad \log \kappa_* \geq 2 \log\left(\frac{\delta k}{2\sqrt{5}}\right) - \frac{\mathcal{B}_{h,k}^{\text{ar}}}{2q}. \quad (\text{AT22})$$

The quartet and actual packet restriction are retained in this application; a reference test polynomial has not been declared to divide 2ξ . Equations (AT15) and (AT17) give the full signed quantity behind this bound. In particular they provide the exact integrand that an arithmetic cancellation calculation must evaluate, with both relation windows and both polynomial windows present. No subthreshold arithmetic bound or weight purity is asserted by this source comparison alone.

Source dependencies and precise status

The complete retained Tau Base note supplies (AT1), its tensor pushforward, right adjoint and supported maps. The original arithmetic-input and cyclic-sum proofs supply (AT2), full packet units and the theta primitives. The Gamma/source-density notes supply the actual measures and finite moments in (AT4)–(AT5). The new Deligne-bound note supplies the full quartet Gamma lower bound used only in (AT22). Each is preserved in the accompanying source closure, with its actual edition and hash.

The new proved calculations here are (AT6)–(AT21) on those unchanged input objects, and their substitution (AT22). They give an analytic calculation on the canonical finite complex mapping to $\mathcal{A}_{\tau,k}(\mathcal{J}^{\boxtimes k})$. They neither replace that pushforward by a generic stalk nor deduce a verdict about the entire tau-based program from an auxiliary model.

Chapter 13

Coordinated relation-window spectral transport

This calculation continues arithmetic determinant transport on its original finite source over the absolute tau base. It computes the full spectra of the two retained window operators, constructs an isometry intertwining them in the actual source metric, and strengthens the signed determinant bound. The source measures, their masses, the cyclic polynomial, the Taylor unit, and the cohomology observation remain fixed.

The original source and its four windows

Use a full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and the original cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Let $E = \mathbb{C}[S]/(\chi)$, $E_h = \mathbb{C}[s]/(h)$, and $v_h = j_h(g/h) \in E_h^\times$. The original tensor-sum observation is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1. \quad (\text{AW1})$$

The original section $\sigma_h : E_h \hookrightarrow Q = \mathcal{B}/\Theta V$ gives the separate injection $\sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}$. Here $Q^{\otimes k}$ is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau} = \{\tau, 1\}$; the one-factor complex is $[V \oplus V \xrightarrow{\Theta(\phi-\psi)} \mathcal{B}]$. The complete Tau Base and cyclic source proofs accompany the preceding arithmetic determinant transport calculation (AT).

For $N \geq q - 1$, retain the increasing-power spaces and maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{B_N = \times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0, \quad \mathcal{P}_{-1} = 0. \quad (\text{AW2})$$

The common source is $H = \mathcal{P}_{2q}$, of dimension $n = 2q + 1$. Its original coordinate on the integration line is $S = k/2 + iu$. Set $v_N(u) = (1, S(u), \dots, S(u)^N)$ and retain the two densities

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k |\Gamma(k/4 + iu/2)|^2}{c_{k/4} 2\pi}, & c_a &= 2^{1-2a} \Gamma(2a). \end{aligned} \quad (\text{AW3})$$

Their actual masses are μ_h^k and $(2\pi)^{k/2}$. For the real parameter $0 \leq t \leq 1$, put $m_t = (1-t)m_0 + tm_1$, $M_N(t) = \int v_N^* v_N m_t$, and $M(t) = M_{2q}(t)$. The source density proofs give all finite moments and positivity almost everywhere, so all these finite Grams are positive definite. We use the original form $\langle x, y \rangle_t = x^* M(t) y$ on H .

Let $P_N(t)$ be the orthogonal projection onto $\mathcal{P}_N \subset H$, and $Q_N(t)$ the projection onto $\chi \mathcal{P}_{N-q} \subset H$. Set $Q_{q-1} = 0$. For the coefficient inclusion $I_N : \mathcal{P}_N \rightarrow H$, their fully typed coefficient formulas are

$$\begin{aligned} P_N &= I_N M_N^{-1} I_N^* M, \\ Q_N &= I_N B_N (B_N^* M_N B_N)^{-1} B_N^* I_N^* M. \end{aligned} \quad (\text{AW4})$$

The zero relation domain at $N = q - 1$ contributes the zero operator. Let $G_N(t)$ be the minimum-lift Gram on E from (AW2) and $V_N(t) = \det G_N(t)$. Retain

$$\begin{aligned}\mathcal{B}(t) &= \log \frac{V_{q-1}(t)V_q(t)}{V_{2q-1}(t)V_{2q}(t)}, \\ C(t) &= M(t)^{-1}(M(1) - M(0)), \\ U(t) &= (Q_{2q-1} - Q_{q-1}) + (Q_{2q} - Q_q), \\ W(t) &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q).\end{aligned}\tag{AW5}$$

The AT proof, included in full, establishes $V_N = \det M_N / \det(B_N^* M_N B_N)$ with the original orientation $(-1)^{q(N-q+1)}$, and

$$\Delta\mathcal{B} = \mathcal{B}(1) - \mathcal{B}(0) = \int_0^1 \text{Tr}((U(t) - W(t))C(t)) dt.\tag{AW6}$$

Thus the following computation evaluates the operators in the already proved signed arithmetic correction.

Full spectra and an explicit source isometry

Theorem 13.1. *In the original $M(t)$ metric, both $U(t)$ and $W(t)$ have spectrum*

$$0 \text{ with multiplicity } q, \quad 1 \text{ with multiplicity } 2, \quad 2 \text{ with multiplicity } q - 1.\tag{AW7}$$

The last multiplicity is zero when $q = 1$. There is an explicitly constructed smooth $M(t)$ -unitary $T(t) : H \rightarrow H$ such that $U(t) = T(t)W(t)T(t)^{-1}$.

Proof. Every subspace in either of the following flags is the literal displayed polynomial subspace, with its original coefficient inclusion. For the relation flag use

$$F_0 = \chi\mathcal{P}_0 \subset F_1 = \chi\mathcal{P}_{q-1} \subset F_2 = \chi\mathcal{P}_q \subset H.\tag{AW8}$$

Their dimensions are $1, q, q + 1$, since multiplication by χ is injective. When $q = 1$, $F_0 = F_1$ is retained as that equality. Projections of nested subspaces commute and their products equal the projection of the smaller subspace: each larger space is the orthogonal direct sum of the smaller space and its orthogonal complement within the larger space. On the original orthogonal decomposition $F_0 \oplus (F_1 \cap F_0^\perp) \oplus (F_2 \cap F_1^\perp) \oplus F_2^\perp$, the operator $U = Q_{F_1} + Q_{F_2} - Q_{F_0}$ acts respectively by $1, 2, 1, 0$. The dimensions are respectively $1, q - 1, 1, q$.

For W use $\mathcal{P}_{q-1} \subset \mathcal{P}_q \subset \mathcal{P}_{2q-1} \subset \mathcal{P}_{2q} = H$. On the four corresponding orthogonal pieces, W acts by $0, 1, 2, 1$, with dimensions $q, 1, q - 1, 1$. This proves (AW7), including the repeated middle subspaces at $q = 1$.

For an explicit intertwiner retain every Gram factor as follows. For any specified ordered independent list $z_0, \dots, z_d$, apply the recursion

$$y_j = z_j - \sum_{i < j} e_i \langle e_i, z_j \rangle_t, \quad a_j = (y_j^* M(t) y_j)^{1/2} > 0, \quad e_j = y_j / a_j.\tag{AW9}$$

The positive real square root fixes its phase. No source density or mass is changed; each a_j is the norm in the original form. Independence proves $y_j \neq 0$ at every step, so this recursion is defined and smooth on the entire parameter interval.

First use $z_j = S^j$, $0 \leq j \leq 2q$, obtaining $p_0, \dots, p_{2q}$. Next use $z_j = \chi S^j$, $0 \leq j \leq q$, obtaining $r_0, \dots, r_q$. These lie in the successive relation spaces in (AW8). Finally let $R_{2q}(t) : E \rightarrow H$ be the original minimum section from AT6 and apply (AW9) to $R_{2q}(t)[1], \dots, R_{2q}(t)[S^{q-1}]$, obtaining $c_0, \dots, c_{q-1}$. The relation orthogonality in AT6 puts this entire list in $F_2^\perp$. Since $J_{2q} R_{2q} = I_E$, the list is independent and spans that q -dimensional complement.

Define T on the first orthonormal basis by

$$\begin{aligned} Tp_j &= c_j & (0 \leq j \leq q-1), \\ Tp_q &= r_0, & Tp_{2q} = r_q, \\ Tp_{q+j} &= r_j & (1 \leq j \leq q-1). \end{aligned} \tag{AW10}$$

The middle range is empty for $q = 1$, with p_q and p_{2q} still distinct. The target list is an orthonormal basis. Thus this linear map is invertible and $T^*MT = M$. The spectral actions already calculated agree on each indicated basis vector, proving $UT = TW$. The recursion, the smooth minimum section and the positive norms prove the asserted smoothness. This completes the construction. $\square$

The exact isometry gives a commutator form of the signed integrand:

$$\mathrm{Tr}((U - W)C) = \mathrm{Tr}(WT^{-1}[C, T]), \quad [C, T] = CT - TC. \tag{AW11}$$

Indeed cyclicity moves T in $\mathrm{Tr}(TWT^{-1}C)$ to the right, and subtraction then gives the displayed product. This T is an isometry of the original source; its cohomology observation is computed, rather than silently assigned an intertwining property. The unchanged arithmetic observation $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} : H \rightarrow Q^{\otimes k}$ satisfies

$$\mathcal{O}T = \sigma_h^{\otimes k} \eta J_{2q} T, \quad \mathcal{O}T - \mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q} (T - I), \quad \ker(\mathcal{O}T) = T^{-1}F_2. \tag{AW12}$$

The last identity follows from both injections in (AW1) and $\ker J_{2q} = F_2$. By (AW10), $T^{-1}F_2$ is exactly $\mathrm{span}\{p_q, \dots, p_{2q}\} = \mathcal{P}_{q-1}^\perp$ in H . At any fixed nonempty support label A , lift every displayed linear map by $(A, x) \mapsto (A, fx)$ and send τ to τ . Consequently the supported-zero inverse image of the lifted $\mathcal{O}T$ at A is the full set $\{A\} \times \mathcal{P}_{q-1}^\perp$. The Taylor unit in (AW1) and the defect in (AW12) remain present.

The spectral bound with every eigenvalue retained

Let $b_1 \leq \dots \leq b_n$ be all eigenvalues, counted with multiplicity, of $D = M(0)^{-1}M(1)$ on its original $M(0)$ Hilbert space. All are positive. Define

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j}, \quad \kappa(D) = b_n/b_1. \tag{AW13}$$

For $q = 1$ the sum is empty and $\mathcal{L}_1(D) = \log(b_3/b_1)$.

Theorem 13.2. *On the unchanged arithmetic source,*

$$|\Delta \mathcal{B}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log \kappa(D). \tag{AW14}$$

In particular the coefficient $2q$ in AT18 can be improved to $2q - 1$, while the first bound retains the full relative spectrum.

Proof. We first prove the needed finite spectral trace inequality. Let C be self-adjoint in a finite Hilbert space with eigenvalues $c_1 \leq \dots \leq c_n$. If P is an orthogonal projection of rank r , let e_j be an orthonormal eigenbasis of C and $d_j = \langle e_j, Pe_j \rangle$. Then $0 \leq d_j \leq 1$ and $\sum_j d_j = r$, since $d_j = \|Pe_j\|^2$ and $\mathrm{Tr} P = r$. Thus $\mathrm{Tr}(PC) = \sum_j c_j d_j$. Its upper bound by $\sum_{j=n-r+1}^n c_j$ follows by subtracting the two expressions: the mass $\sum_{j \leq n-r} d_j$ equals $\sum_{j > n-r} (1 - d_j)$, and every coefficient in the latter group is at least every coefficient in the former. Pairing the same nonnegative mass, or bounding both groups by c_{n-r+1} , proves the inequality. Applying this argument to $-C$ proves the lower bound $\sum_{j=1}^r c_j$. This proof includes repeated eigenvalues and the ranks zero and n .

For an operator Z with spectrum (AW7), its spectral projection A_Z onto eigenvalues 1, 2 has rank $q + 1$, and its spectral projection B_Z onto eigenvalue 2 has rank $q - 1$. The exact decomposition is $Z = A_Z + B_Z$. Applying the proved projection bounds gives

$$\sum_{j=1}^{q+1} c_j + \sum_{j=1}^{q-1} c_j \leq \text{Tr}(ZC) \leq \sum_{j=q+1}^{2q+1} c_j + \sum_{j=q+3}^{2q+1} c_j. \quad (\text{AW15})$$

Each bound is attained for an abstract operator of this spectrum by assigning its eigenspaces to the appropriate eigenvectors of C . This attainment statement is only about the displayed spectral constraint; it assigns no freely chosen orientation to the actual polynomial relation spaces. Both U and W satisfy (AW15). Subtract their bounds and cancel the common c_{q+1} term only after writing both complete sums:

$$|\text{Tr}((U - W)C)| \leq c_{q+2} - c_q + 2 \sum_{j=1}^{q-1} (c_{2q+2-j} - c_j). \quad (\text{AW16})$$

No mutual commutation of U, W, C is used.

For the actual interpolated metric, $C(t) = ((1 - t)I + tD)^{-1}(D - I)$, so its eigenvalues in increasing order are

$$c_j(t) = \frac{b_j - 1}{1 - t + tb_j}. \quad (\text{AW17})$$

The denominator is positive and the derivative with respect to b_j is $(1 - t + tb_j)^{-2} > 0$. Thus the ordering is valid for the entire closed interval. Each $c_j(t)$ is the derivative of $\log(1 - t + tb_j)$ and has integral $\log b_j$. Integrating (AW16) in the exact signed identity (AW6) proves the first inequality of (AW14). Every ratio in (AW13) lies between 1 and b_n/b_1 , so retaining its coefficient gives $1 + 2(q - 1) = 2q - 1$ times $\log \kappa(D)$ for the second bound. This proves the theorem for all $q \geq 1$. $\square$

All bounds retain the literal source masses. If $M(1) = cM(0)$, every $b_j = c$, so (AW13) is zero. Directly each original quotient determinant is multiplied at time t by $(1 - t + tc)^q$, and the four displayed factors in (AW5) cancel. For a fixed invertible source-coordinate arrow F , the transported forms are $M' = F^*MF$, all subspaces have their F^{-1} images, and $D' = F^{-1}DF$. Hence the eigenvalues, (AW6) and (AW13) are unchanged under this explicit map.

The original quartet Gamma theorem, when applied to the actual full quartet packet with displacement $\delta > 0$ and common order m , retains $q = [1 + k(m - 1)](k + 1)^2$ and gives

$$\mathcal{B}_{h,k}^{\text{ar}} \geq 4q \log \left(\frac{\delta k}{2\sqrt{5}} \right) - \mathcal{L}_q(D). \quad (\text{AW18})$$

This is substitution of the complete supplied Gamma theorem into the proved first inequality of (AW14). It proves no uniform estimate for the eigenvalues b_j , and asserts no existence of an off-line packet. The new computed objects are the exact window spectra, their source isometry, its arithmetic observation and commutator, and the strengthened finite bound.

The one-dimensional quotient: exact angle and nonlinear transport

When $q = 1$, the same four endpoints are 0, 1, 1, 2. Their middle factors cancel in the specified ratio, leaving $\mathcal{B}(t) = \log(V_0(t)/V_2(t))$ with both original masses still present. Write $\chi = S - a$. The actual operators now are $U = Q_2$ and $W = I - P_0$. If $\mathcal{N} = (\chi\mathcal{P}_1)^\perp$ and $P_{\mathcal{N}} = I - Q_2$, then $U - W = P_0 - P_{\mathcal{N}}$. Both projections have rank one.

Let $e = 1 \in H$, $z = R_2(t)[1]$, and retain their actual norms $\langle e, e \rangle_t = V_0(t)$ and $\langle z, z \rangle_t = V_2(t)$. The source splitting gives $z = P_{\mathcal{N}}e$, since $J_2e = 1$. Thus $\langle e, z \rangle_t = V_2(t)$, and the angle between the two lines has the exact value

$$\cos^2 \theta(t) = \frac{|\langle e, z \rangle_t|^2}{\langle e, e \rangle_t \langle z, z \rangle_t} = \frac{V_2(t)}{V_0(t)} = e^{-\mathcal{B}(t)}. \quad (\text{AW19})$$

Both Grams are positive, so the cosine is positive. It is strictly less than one for the actual measure (AW3). Otherwise $e \in \mathcal{N}$, implying $\int \chi(S)m_t = 0$ and $\int S\chi(S)m_t = 0$. On the unchanged line $\bar{S} = k - S$, and consequently $|\chi(S)|^2 = ((k - \bar{a}) - S)\chi(S)$. The two equalities would force $\int |\chi(S)|^2 m_t = 0$, contrary to the positive Gram of this nonzero polynomial. Hence $0 < \theta(t) < \pi/2$ and $\mathcal{B}(t) > 0$ on the whole closed interval.

For completeness, take the unit vector $p = e/\sqrt{V_0}$ and write $z/\sqrt{V_2} = \cos \theta p + \sin \theta r$, where r is the resulting unit vector orthogonal to p . The phase is already fixed since $\langle e, z \rangle_t = V_2 > 0$. In this orthonormal pair the exact difference of projections is

$$(P_0 - P_{\mathcal{N}})|_{\text{span}(p,r)} = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & -\sin^2 \theta \end{pmatrix}. \quad (\text{AW20})$$

Its trace is zero and its determinant is $-\sin^2 \theta$. It vanishes on the remaining one-dimensional orthogonal complement. Thus its full eigenvalues are $\sin \theta, 0, -\sin \theta$. Applying the proved rank-one spectral trace bounds to its positive and negative eigenprojections yields

$$|\dot{\mathcal{B}}(t)| \leq \sqrt{1 - e^{-\mathcal{B}(t)}} (c_3(t) - c_1(t)). \quad (\text{AW21})$$

This retains the actual alignment of the two lines.

The function $\Phi(B) = 2 \operatorname{arcosh}(e^{B/2})$ for $B > 0$ has derivative $\Phi'(B) = 1/\sqrt{1 - e^{-B}}$, by differentiating the inverse of the hyperbolic cosine on its positive branch. The preceding strict positivity on the compact interval justifies its smooth composition with $\mathcal{B}(t)$. Divide (AW21) by its positive square-root factor and integrate (AW17). The exact nonlinear endpoint bound is

$$|2 \operatorname{arcosh}(e^{\mathcal{B}(1)/2}) - 2 \operatorname{arcosh}(e^{\mathcal{B}(0)/2})| \leq \log(b_3/b_1). \quad (\text{AW22})$$

Equivalently, with the original values $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/2})$ and $L = \frac{1}{2} \log(b_3/b_1)$, monotonicity of $2 \log \cosh A$ for $A \geq 0$ gives

$$2 \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2 \log \cosh(A_0 + L). \quad (\text{AW23})$$

This last lower bound can be zero although the actual $\mathcal{B}(1)$ is strictly positive, as proved above. Every step concerns the original $q = 1$ quotient and its degree-two source. No extension of that dimension restriction to a quartet quotient is asserted.

Complete proof dependencies

The complete AT calculation accompanies this note, together with its 29 complete source dependencies and independent proof review. Those sources prove the original pushforward, tensor signs, full Taylor-unit injections, theta primitive, positive arithmetic and Gamma moments, and the quartet application. Every new assertion (AW7)–(AW23) is proved above on those unchanged finite objects. A separate independent review supplies the spectral calculation and its edge-case checks.